# mathdash

# Your Guide to Math Contests

*"Mathematics is not about numbers, equations, or algorithms: it is about understanding."*

**By the MathDash Team and Community**

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## Contents

## How To Use This MathDash Guide

This is the official **MathDash Guide** to help you prepare for US-based math competitions, including the MAA's **AMC–AIME–USA(J)MO progression** and **MathCounts**.

We will provide a handout on each topic you may need to know, including walking through some illustrative example problems to show how you can use that technique to solve problems. In order to succeed, we suggest that you:

1. Preview the handout materials at your level (check the rating range) before attempting the problems, ensuring that the relevant materials are fresh on your mind.
2. Clarify anything that is confusing with peers or with a coach using the pre-contest chat
3. Practice solving problems using timed contests on MathDash ← this is where you will spend the bulk of your time. This is "exercise" for your mental muscles
4. Review incorrect solutions to identify gaps in your understanding (aka "upsolve").
5. Chat and discuss your ideas about problems with peers and coaches to refine your intuition.

Every single practice problem should be "solvable" using techniques previously covered at some point in this guide. That doesn't mean you should expect to get a perfect score! Just knowing the techniques is the first step. What we really need to work on together is the actual problem-solving process. We have selected the highest quality problems for you to train the most efficiently, and have also created a fun way to train since you can gain rating and track your progress over time :)

## Some Additional Notes

Whether you are aiming for a perfect score or solving a few problems as a first-time taker, remember that **every bit of progress is valuable** and is putting you ahead of the field.

**This is a LIVING document** that will be updated as you request more modules. Submit any requests for training topics to **coach@mathdash.com**. With your help and feedback, we hope this guide will become one of the best ways to prepare for competition math!

# Sequences and Series

*by sussususus*

https://mathdash.com/contest/sequences-series-problemset

**Disclosure** Typeset by John. Proofs of formulae provided by John.

## Arithmetic Sequences

**Definition 1**

An **arithmetic sequence** is a sequence of numbers where each term is obtained by adding the same number to the previous term. This number is called the common difference, and is denoted by $d$.

For example, the sequence:

$$1, 4, 7, 10, 13, \ldots$$

is an arithmetic sequence with a common difference of $d = 3$, since each term increases by $3$.

Often, we use $a_1, a_2, a_3 \ldots$ to denote the terms of the sequence, and we use $a_n$ to denote the $n$th term (where $n$ is a postive integer).

Note how each term is the previous term plus $d$ and the next term minus $d$. In addition, by repeating this we also have $a_{i+k} = a_i + dk$. When solving problems, if you don't know the common difference, try to find it first and use the above method to get other terms!

Now, we are going to introduce the formula for the first $n$ terms of a arithmetic sequence.

### The $n$th term

**Theorem 1**

If an arithmetic sequence has first term $a_1$ and common difference $d$, then the formula for the $n$th term is:

$$a_n = a_1 + (n-1)d. \tag{1}$$

*Proof.* For each new term, you add $d$ to the initial term. To get to the $n$th term, you add from the first term $n-1$ times. Each time you add you add $d$, so the formula is the initial term plus $n-1$ times $d$. □

**Example 1**

Find the 100th term of the arithmetic sequence $1, 4, 7, 10, 13, \ldots$

**Solution.** $298$.

**Example 2**

How many terms are there in the following? $3, 7, 11, \ldots, 23$

**Solution.** We need to find the $n$ such that the $n$th term is $23$. The common difference is $4$ and $a_1$ is $3$, and $a_n = 23$ so we have

$$23 = 3 + (n-1)4.$$

Solving for that gives $n = 6$.

### Sum of an Arithmetic Series

Suppose we want to find the sum of the first $n$ terms of an arithmetic series.

**Theorem 2**

If $a_1$ is the first term, $a_n$ is the $n$th term (or the final term in our sum), and $n$ is the number of terms, then the formula for the sum of the first $n$ terms is

$$S_n = \frac{n(a_1 + a_n)}{2}. \tag{2}$$

*Proof.* The sum can be written as

$$S_n = a_1 + \underbrace{(a_1 + d)}_{a_2} + \underbrace{(a_1 + 2d)}_{a_3} + \cdots + \underbrace{(a_1 + (n-1)d))}_{a_n}$$

by using our $n$th term formula. It could also be written as

$$S_n = a_n + \underbrace{(a_n - d)}_{a_{n-1}} + \underbrace{(a_n - 2d)}_{a_{n-2}} + \cdots + \underbrace{(a_n - (n-1)d))}_{a_1}.$$

by noticing how subtracting $x$ $d$s from a term gets us the value of the $x$th term before that because the difference between all of the terms are $d$ apart. Adding these together, we note how the $d$, $2d$, $\ldots$, $(n-1)d$ cancels out. We get

$$a_1 \cdot n + a_n \cdot n = 2S_n.$$

Dividing both sides by $2$ and factoring out the $n$ gives the formula. □

You can also prove this by grouping the first number and the last number together, the second and the second last number together, the third and the third last number together, and so on, and noticing their sum are the same. Some very trivial algebra is needed to sort out the middle term for odd $n$.

We can use (1) to rewrite (2) as

$$S_n = \frac{n(2a_1 + (n-1)d)}{2}. \tag{3}$$

It's worth remembering both (2) and (3). Depending on what information you're given, either one can be more useful than the other.

Perhaps the most useful special case of this formula is the following:

**Theorem 3**

The sum of the first $n$ positive integers is given by

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

**Example 3**

On the first day of training, I do one push-up. Every day after that, I do three more push-ups than the previous day. After 100 days, how many push-ups have I done?

**Solution.** This question is basically asking you to find the sum of the first 100 terms of the arithmetic sequence $1, 4, 7, 10, 13, \ldots$.

Use (3). Substitute $n = 100$, $a_1 = 1$, $d = 3$, and the sum is $\frac{100(2+99\times3)}{2} = 14950$. Quick and easy, right?

## Geometric Sequences

Let's move on to the second type of series.

**Definition 2**

A **geometric sequence** is a sequence of numbers where each term is obtained by multiplying the previous term by a constant factor called the common ratio $r$.

For example, the sequence:

$$1, 2, 4, 8, 16, \ldots$$

is a geometric sequence with a common ratio of $r = 2$, since each term is the previous term multiplied by $2$.

Now, we are going to introduce the formula for the first $n$ terms of a geometric sequence.

### The $n$th term

**Theorem 4**

If a geometric sequence has first term $a_1$ and common ratio $r$, then the formula for the $n$th term is:

$$a_n = a_1 \cdot r^{n-1}.$$

This is trivial by repeatedly multiplying from the first term.

**Example 4**

Find the 10th term of the geometric sequence $1458, 486, 162, 54, \ldots$

**Solution.** The numbers are getting smaller! But that doesn't change what we need to do. The first term is $1458$, and the common ratio is $\frac{1}{3}$ by dividing the second term by the first to undo multiplication, so the $10$th term is $1458 \cdot (\frac{1}{3})^9 = \frac{2}{27}$.

Often when you solve problems, you want to find the common ratio first.

**Example 5** (by John)

The second term of a geometric sequence is $5$ and the fourth is $3$. Given the first term is positive, find the fifth term.

**Solution.** The third term is the second term times $r$ and the fourth term is the third times $r$, so the fourth term is the second term times $r^2$. The ratio between the fourth term and the second term is $\frac{3}{5}$ so $\frac{3}{5} = r^2$. Since the first term is positive, and the second term, which is the first term times $r$, is positive, $r$ must be positive so $r = +\sqrt{\frac{3}{5}} = \frac{\sqrt{3}}{\sqrt{5}} = \frac{\sqrt{15}}{5}$. Now we multiply the fourth term by $r$ to get the fifth term, so we get $3 \cdot \frac{\sqrt{15}}{5} = \boxed{\dfrac{3\sqrt{15}}{5}}$.

### Sum of a Geometric Series

**Theorem 5**

The sum of the first $n$ terms of a geometric series is given by

$$S_n = \frac{a_1(1 - r^n)}{1 - r}. \tag{4}$$

*Proof.* The desired sum is

$$S_n = a_1 + a_1 r + a_1 r^2 + \cdots + a_1 r^{n-1}. \tag{5}$$

Multiplying both sides by $r$, we get

$$rS_n = a_1 r + a_1 r^2 + a_1 r^3 + \cdots + a_1 r^n. \tag{6}$$

Subtract (6) from (5), we get

$$S_n - rS_n = (a_1 + a_1 r + a_1 r^2 + \cdots + a_1 r^{n-1}) - (a_1 r + a_1 r^2 + \cdots + a_1 r^n)$$

which cancels out and simplfies to

$$S_n - rS_n = a_1 - a_1 r^n.$$

Factor out $S_n$ on the left and dividing both sides by $1-r$, we get our formula. □

**Theorem 6**

If $-1 < r < 1$, the sum of an *infinite* geometric series will actually be a finite number. The formula for this is

$$S_\infty = \frac{a}{1-r} \tag{7}$$

where $a$ is the first term.

This can be proved by taking the limit of (4) as $n$ goes to infinity; as $n$ goes to infinity, if $|r| < 1$, $r^n$ approaches $0$ since it gets smaller and smaller as you multiply it by a number $< 1$. Therefore, the numerator approaches $a_1$ and that is basically this formula.

Here's an example using the formulas:

**Example 6**

The first term of a geometric series is $8$ and the sum of the infinitely many terms of the geometric series is $16$. Find the common ratio.

**Solution.** By substituting the values in this question into (7), we get

$$\frac{8}{1-r} = 16.$$

Solving the equation, we get $r = \frac{1}{2}$. Done!

## Try the problems

Note: The dashes might have other problem types not introduced here. The problem set has harder problems than those in this chapter, so don't be afraid to ask our awesome MathDash community for help!

**Try the Problems**

https://mathdash.com/contest/sequences-series-problemset

https://mathdash.com/contest-group/sequences-series-1

https://mathdash.com/contest-group/sequences-series-2

https://mathdash.com/contest-group/sequences-series-3

https://mathdash.com/contest-group/sequences-series-4

https://mathdash.com/contest-group/sequences-series-5

# Tricks about Squares and Cubes

*by John Z.*[1]*, testsolved by Krithik*

## Square of a Binomial and Difference of Squares

**Theorem 1** (Square of a Binomial)

$$(x+y)^2 = x^2 + 2xy + y^2$$
$$(x-y)^2 = x^2 - 2xy + y^2$$

**Example 1**

Given $xy = 10$, $x + y = 20$, and $x > y$, find $x$ and $y$.

**Solution.** Note that if you subtract $(x-y)^2$ from $(x+y)^2$ using the right-hand side of theorem 1, we get $(x+y)^2 - (x-y)^2 = 4xy$. We use this to find $x-y$ from $x+y$ and $xy$, and then solve the system involving $x-y$ and $x+y$.

We substitute $x+y=20$ and $xy=10$ and then rearrange the equation:

$$20^2 - (x-y)^2 = 4(10) \implies (x-y)^2 = 360 \implies x-y = \sqrt{360} = 6\sqrt{10}.$$

Note that it's positive since $x > y$. Now, solving the system $x+y=20$ and $x-y=6\sqrt{10}$, we get

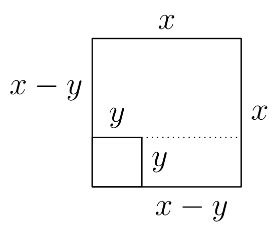
$$\boxed{x = 10 + 3\sqrt{10}, \quad y = 10 - 3\sqrt{10}}. \quad \blacksquare$$

This is probably the most commonly used formula.

**Theorem 2** (Difference of Squares)

$$x^2 - y^2 = (x+y)(x-y)$$

***Optional:*** *Graphic Proof of Theorem 2.* It is trivial by expanding the right-hand side, but there's a more graphical way:




[1]Thanks to Steve for reminding me to add something like example 7 and doing it for me, even though he made some typos. In addition, thanks for Krithik for corrections of various errors.

Note that $x^2 - y^2$ is the area of the larger square minus the smaller one, which, if we want an alternative representation, could be broken down into 2 parts, as shown by the dashed line. The upper rectangle's area is $(x-y) \cdot x$ and the lower rectangle's area is $y \cdot (x-y)$. Adding these areas together and factoring out $x-y$ gives $(x+y)(x-y)$. $\square$

**Example 2**

Calculate $\sqrt{10-\sqrt{51}} + \sqrt{10+\sqrt{51}}$.

**Solution.** First, let the 2 terms be $a$ and $b$ respectively. Note that if we calculate $ab$ it is easy because after we combine the square roots, we can apply the difference of squares formula. Note that $ab$ also applies in the expansion of $(a+b)^2$, so what's missing now is $a^2+b^2$ which is also trivial to calculate. Adding these together and taking the square root gives the desired result.

So let's do it:

$$ab = \sqrt{10-\sqrt{51}}\sqrt{10+\sqrt{51}} = \sqrt{(10-\sqrt{51})(10+\sqrt{51})} = \sqrt{10^2 - \sqrt{51}^2} = \sqrt{49} = 7$$
$$a^2 + b^2 = 10 - \sqrt{51} + 10 + \sqrt{51} = 20$$
$$a+b = \sqrt{(a+b)^2} = \sqrt{a^2+2ab+b^2} = \sqrt{20+14} = \boxed{\sqrt{34}}. \quad \blacksquare$$

**Example 3**

Find the number of the prime factors of $2^{32}-1$.

**Solution.** Note that $2^{32}-1 = 2^{32}-1^{32} = (2^{16})^2 - (1^{16})^2 = (2^{16}+1^{16})(2^{16}-1^{16}) = (2^{16}+1)(2^{16}-1)$, and then you can apply this in the second term in the result repeatedly, until it becomes $(2+1)(2-1)$. This means the factorization is

$$(2^{16}+1)(2^8+1)(2^4+1)(2^2+1)(2+1)(2-1) = 65537 \cdot 257 \cdot 17 \cdot 5 \cdot 3 \cdot 1$$

After checking that those are indeed prime and ignoring the 1, our answer is $\boxed{5}$. $\blacksquare$

*Remark.* If you want to cheat by writing a program to do this with C++ don't forget `long long`.

Note how it's impossible to do this if we just bash and calculate $2^{32}-1$ directly, which is around $4e9$.

**Example 4**

Given that $x + \frac{1}{x} = 3$, find $x^2 + \frac{1}{x^2}$.

**Solution.** Note how $x \cdot \frac{1}{x} = 1$, so we try squaring $x + \frac{1}{x}$, and then subtracting some stuff to get the expression we want.

$$\left(x + \frac{1}{x}\right)^2 = x^2 + 2x\frac{1}{x} + \frac{1}{x^2} = x^2 + \frac{1}{x^2} + 2 = 9$$

It follows that the answer is $\boxed{7}$. $\blacksquare$

**Example 5**

If $a+b=1$ and $a^2+b^2=3$, find $a^4+b^4$.

**Solution.** Since we have $a+b$ and $a^2+b^2$, we square the former and subtract the latter because $(a+b)^2 = (a^2+b^2)+2ab \implies 1 = 3+2ab$ and $ab = -1$. Squaring $(a^2+b^2)$ gets us $a^4+2a^2b^2+b^4 = a^4+b^4+2(ab)^2 = a^4+b^4+2$, and since $a^2+b^2=3$, that is equal to $9$, giving us $a^4+b^4 = \boxed{7}$. ■

**Definition 1** (Factoring)

In the rare case that you don't know, factoring means to write something as a product of a bunch of simpler expressions that cannot be written as products.

**Example 6**

Factor $4x^2-9$.

**Solution.** By difference of squares, we get $\boxed{(2x-3)(2x+3)}$. ■

Note that there are a lot of ways in which these formulas can be applied. The above are just a few of them.

## Sum and Difference of Cubes Formulas

Those are less commonly used but are very *very* elegant.

**Theorem 3** (Sum and Difference of Cubes)

$$x^3+y^3 = (x+y)(x^2-xy+y^2)$$
$$x^3-y^3 = (x-y)(x^2+xy+y^2)$$

Note, the sign in the middle of the linear term is the same as the sign on the left-hand side.

**Example 7** (*Added by Steve*)

Let $a+b=5$ and $a^2+b^2=15$. Find $a^3+b^3$

**Solution.** We have $(a+b)^2=25$, hence $a^2+2ab+b^2=25$ so $2ab=-10$, and $ab=-5$. Thus, $a^3+b^3 = (a+b)(a^2-ab+b^2) = 5\cdot((a+b)^2-3ab) = 5\cdot(25-3\cdot5) = \boxed{50}$. ■

### A Trick Question

**Example 8**

Factor $a^6 - b^6$.

**Solution.** We can start with the difference of squares:

$$\begin{aligned} & a^6 - b^6 \\ = \; & (a^3)^2 - (b^3)^2 \\ = \; & (a^3 - b^3)(a^3 + b^3) \\ = \; & (a+b)(a^2 - ab + b^2)(a-b)(a^2 + ab + b^2) \end{aligned}$$

Or we can start with the difference of cubes:

$$\begin{aligned} & a^6 - b^6 \\ = \; & (a^2)^3 - (b^2)^3 \\ = \; & (a^2 - b^2)(a^4 + a^2b^2 + b^4) \\ = \; & (a+b)(a-b)(a^4 + a^2b^2 + b^4) \end{aligned}$$

Note how they don't match up. The fact is that $a^4 + a^2b^2 + b^4$ is still factorable, and it's $(a^2 - ab + b^2)(a^2 + ab + b^2)$. If you're not convinced, multiply it out! The answer is therefore $\boxed{(a+b)(a^2 - ab + b^2)(a-b)(a^2 + ab + b^2)}$ from the first approach. Please note that $a^2 \pm ab + b^2$ is not factorable.

## Cube of a Binomial

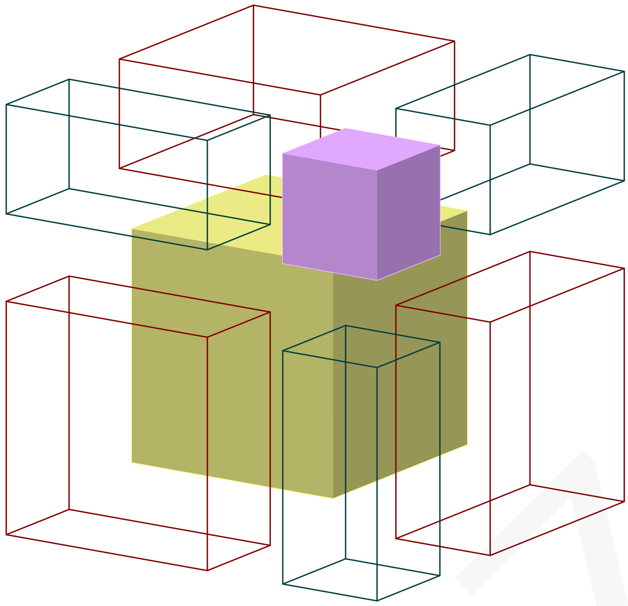
The following is less used but are sometimes still helpful in some problems:

**Theorem 4** (Cube of a Binomial)

$$(x+y)^3 = x^3 + 3x^2y + 3xy^2 + y^3 = (x^3 + y^3) + 3xy(x+y)$$

$$(x-y)^3 = x^3 - 3x^2y + 3xy^2 - y^3 = (x^3 - y^3) - 3xy(x-y)$$

*Proof of sum case without words*[2].



□

**Example 9** (Oxford University Entrance Exam[3])

Given $x + \frac{1}{x} = \sqrt{3}$, find $x^{99} + \frac{1}{x^{99}}$.

**Solution.** Squaring the left side doesn't give anything really useful. However, if we cube it, the resulting expression might be something useful. Sometimes you just have to play around to see what works.

We have

$$\left(x + \frac{1}{x}\right)^3 = x^3 + \frac{1}{x^3} + 3x\frac{1}{x}\left(x + \frac{1}{x}\right) = x^3 + \frac{1}{x^3} + 3(\sqrt{3}) = \sqrt{3}^3 = 3\sqrt{3}.$$

This means that

$$\begin{aligned} x^3 + \frac{1}{x^3} + 3\sqrt{3} &= 3\sqrt{3} \\ x^3 + \frac{1}{x^3} &= 0 \qquad (*) \\ x^6 + 1 &= 0 \\ x^6 &= -1 \end{aligned}$$

Before here it's pretty routine. Now comes the really elegant step: we know that $x^{99} = (x^6)^{16}x^3 = (-1)^{16}x^3 = x^3$. This means that the question is basically asking for $x^3 + \frac{1}{x^3}$. However, that is basically (*), so that means the answer is $\boxed{0}$. ■

Trying to solve this with complex numbers would be a headache. Another remark is that if you can't find anything useful, try squaring and cubing some expressions and it might work out. If you have time, experimenting with stuff never hurts.

[2]Added 2/9/25, Code credit: https://tex.stackexchange.com/questions/644612/ab-whole-cubed-in-latex/644709#644709

[3]From https://www.youtube.com/watch?v=9Syoo5hq8_w, family member on YouTube the other day

## Try the Problems

### Try the Problems & Related Contests

Tricks about Squares #0, 1000

Tricks about Squares #1, 1050 (New)

Tricks about Squares #2, 1100

MATHCOUNTS Chapter Dash 8[a]

Diophantine Equations #6, 1700[b]

MathDash Mini 8[c]

Nested Square Roots #2

TASCC Countdown, 1300

Fun Sum, 1200

[a] Focus on Problem 5, $1 \sim 4$ are trivial, 6 is tricky

[b] *Marginally* related, make sure you know Complete the Square before proceeding, only one step involves stuff in this chapter

[c] Focus on last problem

# Self-Similar Expressions

*by John Z.*

*Note: This isn't really used often. Only a handful of problems require it, and this chapter is largely optional.*

## Self-Similar Expressions

The trick with evaluating self-similar expressions is to find the part that is the same as original, let the expression be $x$, make an equation and solve for $x$.

Let's start with an example:

**Example 1**

Calculate $\boxed{\sqrt{2+\boxed{\sqrt{2+\sqrt{2+\sqrt{2+\sqrt{2+\sqrt{2+\sqrt{2+\cdots}}}}}}}}}$

Note that the square roots just keep nesting, and it will be impossible to evaluate this infinitely by hand. However, note that the magenta box is essentially the original (red box), and if we let $x$ equal to this amount, we can re-write the expression as

$$x = \sqrt{2+x}$$

Squaring both sides:

$$x^2 = 2 + x$$

$$x^2 - x - 2 = 0$$

$$(x-2)(x+1) = 0$$

$$x_1 = 2,\ x_2 = -1$$

Clearly $x$ is positive, so this expression is equal to $\boxed{2}$. ■

This trick works not only on fractions! In fact, here's another similar (pun intended) question that involves fractions instead of square roots:

**Example 2**

Calculate

$$\boxed{\frac{3}{2+\boxed{\frac{3}{2+\frac{3}{2+\frac{3}{2+\frac{3}{2+\frac{3}{\ddots}}}}}}}}$$

Note that if we let this equal $x$, we have

$$\begin{aligned} x &= \frac{3}{2+x} \\ (2+x)(x) &= 3 \\ x^2+2x-3 &= 0 \\ (x+3)(x-1) &= 0 \end{aligned}$$

Clearly this is positive, so $x = \boxed{1}$. ■

**Example 3** (1991 AIME #7)

Find $A^2$, where $A$ is the sum of the absolute values of all roots of the following equation:

$$x = \sqrt{19} + \frac{91}{\sqrt{19}+\frac{91}{\sqrt{19}+\frac{91}{\sqrt{19}+\frac{91}{\sqrt{19}+\frac{91}{x}}}}}.$$

**Solution.** Clearly, if you plug the equation into itself, into itself, and so on, we're going to get something like this:

$$x = \sqrt{19} + \frac{91}{\sqrt{19}+\frac{91}{\sqrt{19}+\frac{91}{\sqrt{19}+\frac{91}{\sqrt{19}+\frac{91}{\sqrt{19}+\frac{91}{\sqrt{19}+\frac{91}{\sqrt{19}+\frac{91}{\sqrt{19}+\ddots}}}}}}}}.$$

This means that

$$x = \sqrt{19} + \frac{91}{x}.$$

So we multiply both sides by $x$ to get $x^2 = x\sqrt{19} + 91$ which means that $x^2 - x\sqrt{19} - 91 = 0$. Plugging in the quadratic formula gives us

$$x_1 = \frac{\sqrt{19} \pm \sqrt{383}}{2} \implies A = |x_1| + |x_2| = \frac{\sqrt{19} + \sqrt{383}}{2} - \frac{\sqrt{19} - \sqrt{383}}{2} = \sqrt{\boxed{383}}. \qquad \square$$

**Example 4**

Calculate $\sqrt{2 + \sqrt{2^2 + \sqrt{2^4 + \sqrt{2^8 + \cdots}}}}$.

Note that if you multiply $\sqrt{1 + \sqrt{1 + \sqrt{1 + \sqrt{1 + \cdots}}}}$ by $\sqrt{2}$, you get $\sqrt{2 + 2\sqrt{1 + \sqrt{1 + \sqrt{1 + \cdots}}}}$. Now the $2$ to the left of the inner square root can be written inside the inner sqrt by multiplying the stuff inside the square root by $2^2$, and that becomes $\sqrt{2 + \sqrt{2^2 + 2^2\sqrt{1 + \sqrt{1 + \cdots}}}}$. Now the $2^2$ can go into the inner-inner square root and become $2^4$ and so on. Therefore,

$$\sqrt{2}\,\boxed{\sqrt{1 + \boxed{\sqrt{1 + \sqrt{1 + \sqrt{1 + \cdots}}}}}} = \sqrt{2 + \sqrt{2^2 + \sqrt{2^4 + \sqrt{2^8 + \cdots}}}}$$

.

Evaluating the red box:

$$\begin{aligned} x &= \sqrt{1 + x} \\ x^2 &= 1 + x \\ x^2 - x - 1 &= 0 \\ x &= \frac{1 \pm \sqrt{1 + 4}}{2} \end{aligned}$$

Clearly this is positive so it's $x = \frac{1+\sqrt{5}}{2}$. Now we know that the original expression is equal to $\boxed{\dfrac{\sqrt{2} + \sqrt{10}}{2}}$. $\blacksquare$

## Summary

**The main idea is that you need to find the sub-expression that is same with the original, let the thing you need to calculate be $x$, and then write an equation and solve for it. Sometimes more sophisticated tricks are needed.**

**Try the Problems**

Algebra Set 1 #6

Nested Square Roots #1

Nested Square Roots #2

# Interesting 3D Problems

*by John Z.*

https://mathdash.com/contest/3d-geometry-mathcounts-chapter-target-level

ALL diagrams prepared with `Asymptote`. No raster graphics is included so the file is super small.

## Volume Formulas

The volume of a prism or cylinder is the area of the base times the height.

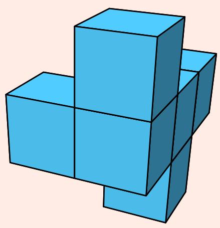
For pyramids or cones, take the volume of the prism or cylinder with the same height and base area and divide by $3$.

*Now try problem **8** in the pset. To find $BCHE$, try to use Pythag to get $FB$ and multiply by $BE$.*

## Surface Area of Weird Shapes

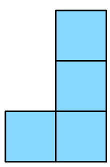
**Example 1**

Find the surface area of this figure below. The side length of each cube is $1$ unit.



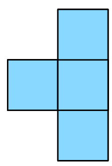
**Solution.** The solution for this kind of stuff, if there's no "groove" faces that can't be seen from the outside at a perspective, is to consider sides from the front and back (which should be the same), left and right (which should be the same), and top and bottom (which should be the same).

Top view:



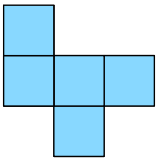
There are $4$ square units in here.

Front view:



There are $4$ square units in here.

Right view:



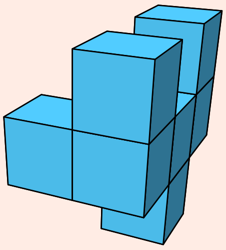
There are $5$ square units in here.

Therefore our total surface area is $4 \cdot 2 + 4 \cdot 2 + 5 \cdot 2 = \boxed{26}$ $\text{units}^2$.

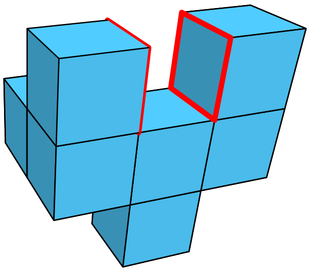
*Now, try Problem* ***1*** *and* ***3*** *in the PSet. For problem 3 is is not made up of a bunch of cubes but the process is similar — get surface area from left + right, top + bottom, front + back. For left & right try not to bash.*

**Example 2**

Find the surface area of this figure below. The side length of each cube is $1$ unit.

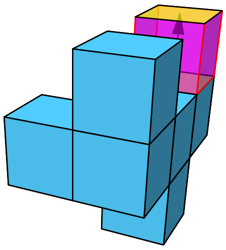


**Solution.** Let us go with our routine method; however, we will soon find that if we count using that method, the faces outlined with red below are the faces that will NOT be counted. Try imagining something spraying paint from the top, left, right, front, and back, and then you'll find that those two faces are not sprayed.



Therefore, in addition to our regular method we need to add $2$. From the front we have $4$ squares; from the right we have $6$ squares; from the top we have $4$ squares, as usual. Muliplying each of those by $2$, adding them together and adding $2$ we get $\boxed{30}$ $\text{units}^2$ as our answer.

**Another way to think about this** This example is basically the same as the last example but with an extra cube at the place with crazy highlighting. You can think of it as the bottom yellow surface begin lifted up, and while dragging the yellow surface up, the four magenta "supporting" surfaces appeared out of thin air.



So the yellow face just moved up without changes in area but the $4$ lateral faces were new. Therefore the surface area of example 2 is just $4$ more than that of example 1. $26 + 4 = 30$ so this is correct.

With that in mind, let us try another problem using this shortcut:

**Example 3**

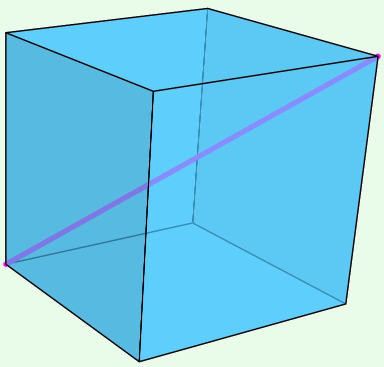
A rectangular prism has a surface area of $80$. Now, a smaller prism with base perimeter $7$ and height $3$ is stacked on the top base of the rectangular prism. Calculate the percent increase in surface area.

**Solution.** The base of the smaller prism that overlaps the rectangular prism's base gets "moved up" to the other base of smaller prism. To support it the lateral surface area of the smaller prism is the part increased. The answer is $\frac{7 \cdot 3}{80} \cdot 100\% = 26.25\%$.

*Now, try Problem **6** in the PSet.*

## Body Diagonal

**Definition 1** (Body Diagonal of a Cube)



In a cube, a segment connecting the two opposite vertices is known as the **body diagonal**, also known as a **space diagonal**. For a cube with side length $s$, the length of the body diagonal is

$s\sqrt{3}$.

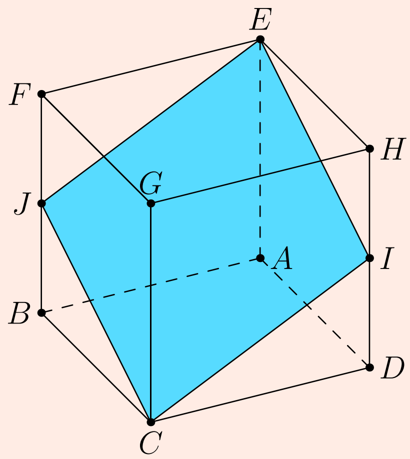
**Example 4**

Find the volume of a cube with body diagonal $5$.

**Solution.** First, we find the side length by dividing $5$ by $\sqrt{3}$ to undo the body diagonal formula, so we get $\frac{5}{\sqrt{3}} = \frac{5\sqrt{3}}{3}$. Now we cube it to get our answer: $\left(\frac{5\sqrt{3}}{3}\right)^2 = \frac{125\cdot 3\sqrt{3}}{27} = \boxed{\frac{125\cdot\sqrt{3}}{9}}$.

**Example 5** (2018 AMC 8 #24)

In the cube $ABCDEFGH$ with opposite vertices $C$ and $E$, $J$ and $I$ are the midpoints of edges $\overline{FB}$ and $\overline{HD}$, respectively. Let $R$ be the ratio of the area of the cross-section $EJCI$ to the area of one of the faces of the cube. What is $R^2$?




*Do this in problem **5** of the problem set before coming here to read the solution.*

**Solution.** $EJCI$ is a rhombus by symmetry. Let side length of the cube be $x$.

$EI$ can be computed in the horizontal cross section containing these points as $x\sqrt{2}$. $EC$ is the body diagonal so it's $x\sqrt{3}$. $\frac{x^2\sqrt{6}}{2}$ is the area of the desired rhombus. The area of the face is $x^2$. The answer is $\frac{6}{4} = \frac{3}{2}$ by simple algebra.

## Scaling

**Example 6**

The surface area of a sphere is three times of that of a sphere of radius $1$. Find the former sphere's volume.

**Solution.** The formula for surface area of a sphere is $4\pi r^2$, so the surface area scales quadratically as the side length. This means that if you scale the side length by a factor of $s$, the surface area will scale by a factor of $s^2$. Since the surface area is scaled by a factor of $3$, the radius is scaled by a side length of $\sqrt{3}$. Therefore the new radius is $\sqrt{3}$ and by the formula $\frac{4}{3}\pi r^3$, the radius is

$$\frac{4}{3}\pi(\sqrt{3})^3 = \frac{4}{3}\pi(3\sqrt{3}) = 4\pi\sqrt{3}.$$

Hence, the volume of the former sphere is $4\pi\sqrt{3}$.

Generally if you scale the shape by a factor of $s$, 2-dimensional measures like surface area will scale by a factor of $s^2$ and 3-dimensional measures like volume will scale by a factor of $s^3$ [4].

*Try problem **10** in the PSet.*

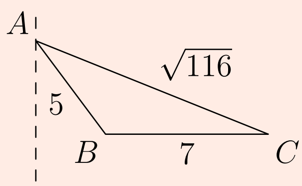
## Solids of Revolution

*In this section we are only covering basic shapes instead of the calculus stuff.*

A **solid of revolution** is a three-dimensional shape created by rotating a two-dimensional region around a specified axis. The solid of revolution of a right triangle around one of its sides is a cone and that of a rectangle around one of its sides is a cylinder. We can add and subtract those things to get the solid of revolution of other shapes.

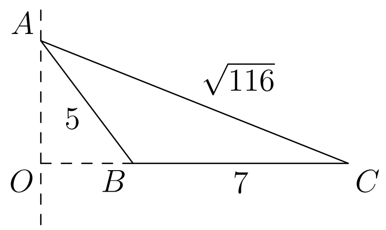
**Example 7**

Find the volume of the solid of revolution of triangle $ABC$ with side lengths $5$, $7$, and $\sqrt{116}$ around the line through $A$ perpendicular to $BC$.




**Solution.** First, realize that the solid of revolution is the solid of revolution of $\triangle AOC$ minus the solid of revolution of $\triangle AOB$.

[4]Except if the shape is a fractal. See https://www.youtube.com/watch?v=gB9n2gHsHN4. A lot of people think fractals are self-similar shapes, while some kids only heard it in a song in *Frozen*.




Now notice that $\triangle AOB$ has a $5$ in it so it might be a $3$–$4$–$5$ triangle. Indeed it is! If $OB = 3$ and $AO = 4$, we can see that $OC^2 + AO^2 = (7+3)^2 + 4^2 = 116 = AC^2$.

This is trivial after. First we find the volume of the solid of revolution of $\triangle AOC$ around $AO$ which is a cone with $OC$ as the radius and $AO$ as the height. That volume is $\frac{1}{3}\pi(7+3)\cdot 4^2 = \frac{160\pi}{3}$. Now we find that of $\triangle AOB$ which is $OB$ as radius and $AO$ as height, so $\frac{1}{3}\pi(3)^2 \cdot 4 = 12\pi$. Subtracting the latter from the former gives $\boxed{\dfrac{124}{\pi}}$ for our volume of the solid of revolution.

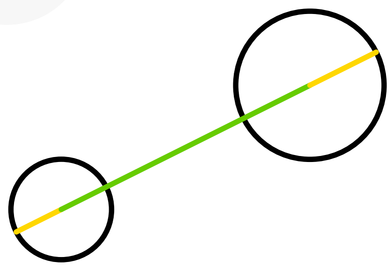
*Now try problem **2** in the pset. The line revolved around is different but the method is the same.*

## Miscellanous Fun Problems

**Example 8**

Find the largest distance between a point on a sphere with center $(-5,1,2)$ and radius $2$ and a sphere with center $(3,2,4)$ and radius $1$?

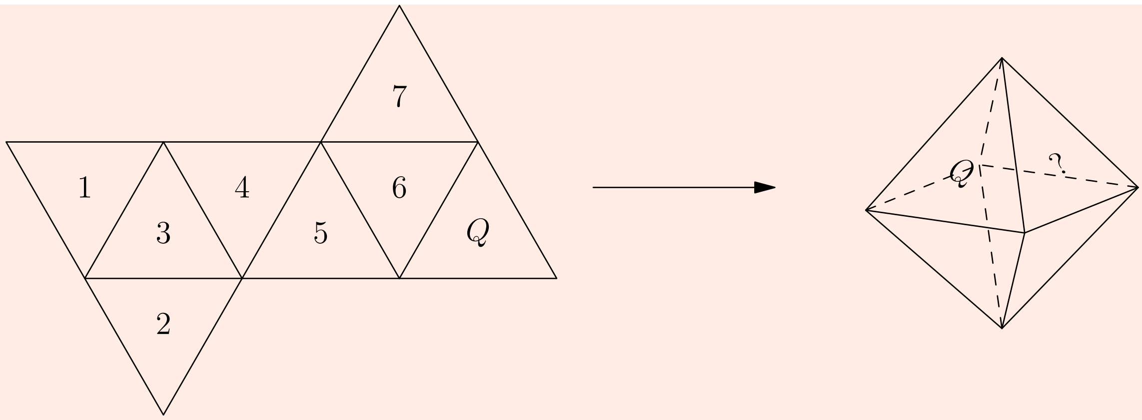
**Solution.** The distance between the centers plus the radii would be the largest. If you're skeptical, here's a demonstration in 2D with different coordinates than the problem:



Now our answer is $\sqrt{8^2+1^2+2^2}+1+2 = 3+\sqrt{69}$.

*Try problem **11** in the problem set.*

**Example 9** (2023 AMC 8 #17)




The net on the left folds to the right. Find the number for $?$.

**Solution.** From imagination we know that $6$ is red and $5$ is yellow. Note that the yellow and the desired surface do not share any corners; however, $5$ shares a corner with all of the faces except $1$ so we know $1$ is the answer.

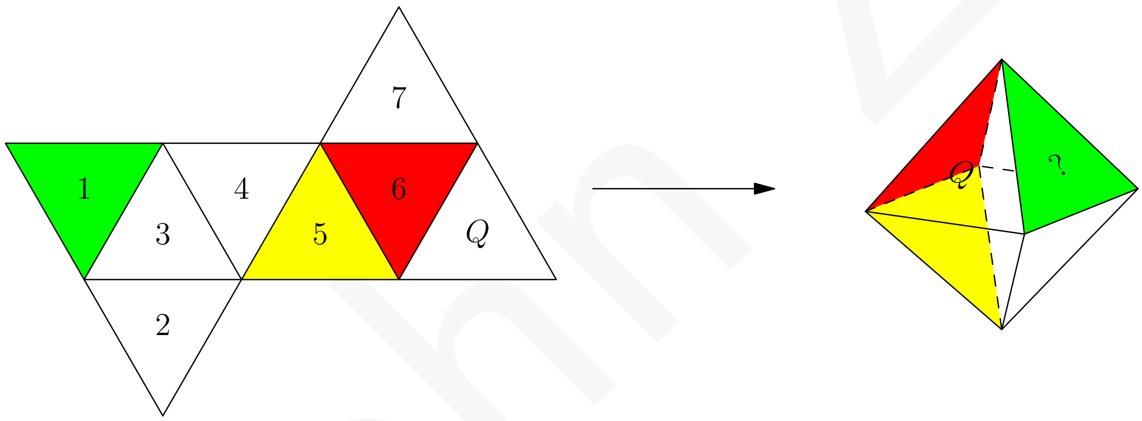
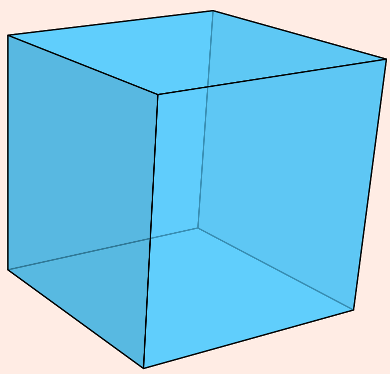


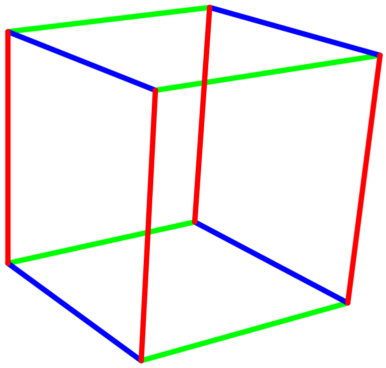

Diagram credits: TheMathGuyD and MRENTHUSIASM at https://artofproblemsolving.com/wiki/index.php?title=2023_AMC_8_Problems/Problem_17.

**Example 10** (2015 AMC 10 #12)

How many pairs of parallel edges does a cube have?



**Solution.** There are $4$ edges in each direction (illustrated with different colors):



Any two of them in each color are parallel and we can add the cases for each of them together. Therefore we get $3\binom{4}{2} = 18$.

**Try the Problems**

**MATHCOUNTS Chapter Target Level 3D Geometry by Akshaj**

# Prime Factors and Divisors

*by MathDash Team*

https://mathdash.com/contest/prime-factors-divisors-handout

**Theorem 1** (Fundamental Theorem of Number Theory)

Every positive integer greater than one can be written as the product of prime numbers in a unique way (except for the order in which the primes are written)

We know that the prime numbers are $2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, \ldots$.

As an example, let's write $60$ as the product of prime numbers:

$$60 = 2 \times 2 \times 3 \times 5 = 2^2 \times 3 \times 5$$

It turns out that this is the **only** way to write $60$ as the product of prime numbers! We call this the **prime factorization** of $60$. In general, each number $N$ can be written in the form

$$2^{\square}3^{\square}5^{\square}7^{\square}11^{\square}13^{\square}17^{\square}19^{\square}\cdots$$

where the boxes are filled with nonnegative integers (that can be zero!).

**Example 1**

Find the prime factorization of $10! = 10 \times 9 \times 8 \times 7 \times 6 \times \cdots \times 2 \times 1$

**Solution:** Multiplying this number out and then trying to find primes that divide it would take a long time! Let's instead do this by finding the prime factorization of each product in the multiplication.

$$\begin{aligned} 10! &= 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 \\ &= (2 \times 5) \times (3^2) \times (2^3) \times 7 \times (2 \times 3) \times 5 \times 2^2 \times 3 \times 2 \times 1 \\ &= \boxed{2^8 \times 3^4 \times 5^2 \times 7} \end{aligned}$$

**Definition 1**

The **divisors** of $N$ are the positive integers $d$ for which $N$ is divisible by $d$. We use the notation $d \mid N$ to mean that $d$ is a divisor of $N$, or $N$ is divisible by $d$, or $d$ divides $N$.

**Example 2**

Find all of the divisors of $60$.

**Solution:** How can we find all of the divisors of a number? Well, we could try each of the $60$ numbers from $1$ through $60$ and see if they evenly divide into $60$. This would take a long time though — so let's search for a quicker solution.

In the case of $N = 60$, one trick we can use is the fact that if $d \mid N$, then $(N/d) \mid N$ as well! (For example, $2$ is a divisor of $60$ — so $60 \div 2 = 30$ is also a divisor of $60$). This is because the divisors come in pairs that multiply to $N = 60$.

All we need to do is find all of the divisors of $60$ that are less than $\sqrt{60}$ — and then calculate $60 \div d$ for each divisor to get the rest of them!

Since $\sqrt{60} \le 8$, we can check $d = 1, 2, \cdots, 8$ and see that $1$, $2$, $3$, $4$, $5$, $6$ are all divisors of $60$ - and then their pairs are $\frac{60}{1}$, $\frac{60}{2}$, $\frac{60}{3}$, $\frac{60}{4}$, $\frac{60}{5}$, $\frac{60}{6}$ — giving us the $10$ divisors

$$\boxed{1,\ 2,\ 3,\ 4,\ 5,\ 6,\ 10,\ 12,\ 15,\ 20,\ 30,\ 60}.$$

How do we know that there are no more? Well, if $d > \sqrt{60}$ and $d$ is a divisor of $60$, then $(60/d)$ is also a divisor of $60$, and $\frac{60}{d} \le \frac{60}{\sqrt{60}} = \sqrt{60}$. So, we would have found $\frac{60}{d}$ in our search already — since we checked all divisors less than $\sqrt{60}$ — and thus have found $d$ because we counted each of those divisors as well as the divisors they are paired with.

**Example 3**

How many divisors does $90000$ have?

If we were to use the method in the previous example, we would have to check $\sqrt{90000} = 300$ numbers to see if they were divisors of $90000$. While better than checking $90000$ values, this is still a lot of numbers to check.

Let's instead try to figure out what the prime factorization of a divisor of $90000$ would look like — maybe this will make them easier to count.

**Theorem 2**

If $a$ is a divisor of $b$, then the exponent of a prime $p$ in the prime factorization of $a$ is less than or equal to the exponent of the same prime $p$ in the prime factorization of $b$

Why is this true? Let's think about the exponent of the prime $p$ in the prime factorization of $\frac{b}{a}$, which is an integer. We know that there are more factors of $p$ in $a$ than $b$ — so the exponent of $p$ in the prime factorization of $\frac{b}{a}$ would be negative! (As an example, you can see that if $p = 3$,

$a = 2 \times 3^3$, and $b = 2^2 \times 3^2$, then $\frac{b}{a}$ would have a negative exponent for $3$). This is not possible, because the prime factorization of any positive integer has only non-negative exponents.

Now — to solve the original problem, let's write $90000 = 9 \times 10^4 = 3^2 \times (2 \times 5)^4 = 2^4 \times 3^2 \times 5^4$

Now that we've prime factorized $90000$, what must the prime factorization of a divisor of $90000$ look like? Well — from Theorem 2, we know that the exponent of $2$ must be less than or equal to $4$, the exponent of $3$ must be less than or equal to $2$, and the exponent of $5$ must be less than or equal to $4$.

Also — since $90000$ has an exponent of $0$ for every prime that isn't $2$, $3$, $5$ — we know that the exponent in a divisor of $90000$ of any prime that isn't $2$, $3$, $5$ must also be less than or equal to $0$!

So, for $d$ a divisor of $90000$, we can write its prime factorization as $d = 2^{\square}3^{\square}5^{\square}7^{\square}11^{\square}\cdots$ and we know:

- the exponent of $2$ is either $0$, $1$, $2$, $3$, $4$
- the exponent of $3$ is either $0$, $1$, $2$, and
- the exponent of $5$ is either $0$, $1$, $2$, $3$, $4$, and
- the exponents of $7$, $11$, $\ldots$ are all $0$.

For each number $d$ of this form, is it necessarily a divisor of $90000$? Well, an integer $d$ is a divisor of $90000$ if there is another integer $k$ such that $d \cdot k = 90000$. For each number of the form $2^{e_0}3^{e_1}5^{e_2}$ with the given conditions on $e_0$,$e_1$,$e_2$, we can let $k = 2^{4-e_0}3^{2-e_1}5^{4-e_2}$, and $d \cdot k$ will be equal to $2^4 3^2 5^4 = 90000$. So, we've figured out a way to write the prime factorization of every divisor of $90000$.

Now all that remains is to count the divisors. There are $5$ choices for the value of $e_0$, $3$ choices for the value of $e_1$, and $5$ choices for the value of $e_2$. Each possible choice gives us a value for $2^{e_0}3^{e_1}5^{e_2}$ that is one of the divisors of $90000$. Since there are $5 \times 3 \times 5 = 75$ ways for us to choose these values, there are $\boxed{75}$ divisors of $90000$.

**Theorem 3** (Number of Divisors)

In general, if $N$ has prime factorization $2^{e_0}3^{e_1}5^{e_2}7^{e_3}\cdots$, then the number of divisors of $N$ is $(e_0 + 1)(e_1 + 1)(e_2 + 1)(e_3 + 1)\cdots$.

This is because there are $e_0 + 1$ ways to choose the exponent of $2$ in the divisor (any integer from $0$ through $e_0$ inclusive), $e_1 + 1$ ways to choose the exponent of $3$, and so on.

**Example 4**

Three consecutive integers multiply to $29760$. What is the middle integer?

**Solution:** Well, a first idea would be to let the integers be $n$, $n+1$, and $n+2$. We get $n(n+1)(n+2) = 29760$. This is pretty difficult to solve though — it is a cubic equation and we don't really have a good way to solve these.

Let's use the fact that $n$ is an integer to help us — perhaps we can try prime factorizing $29760$. We can immediately write $29760 = 10 \cdot 2976$. To prime factorize $2976$, we can start by repeatedly dividing it by $2$ — $2976 = 2 \times 1488 = 4 \times 744 = 8 \times 372 = 16 \times 186 = 32 \times 93$.

Then, we see that $93$ satisfies the divisibility test for $3$ (its digits sum to a multiple of $3$) — so we can divide by $3$ to get $93 = 3 \times 31$.

So, the overall prime factorization is $29760 = 10 \cdot 2976 = 10 \times 2^5 \times 93 = 2^6 \cdot 3 \cdot 5 \cdot 31$.

Now, how do we figure out what $n$ is? Well — we know that $31$ is a divisor of $n(n+1)(n+2)$.

**Theorem 4**

If a prime number $p$ divides $a \times b$, then either $p$ divides $a$ or $p$ divides $b$ (or both). This is because if $p$ didn't divide either $a$ or $b$, then $p$ would not be in the prime factorizations of $a$ or $b$, meaning that their product's prime factorization wouldn't have $p$ in it.

From the above theorem, we know that $31$ is either a divisor of $n$, $n+1$, or $n+2$. Furthermore, since $n(n+1)(n+2) = 29760$, we know that if either of $n$,$n+1$,$n+2$ was a multiple of $31$ larger than $31$, then $n > 60$, which means $n(n+1)(n+2) > n^3 > 216000$ — which is way too big. So, this tells us that one of $n$, $n+1$, $n+2$ must be equal to $31$.

If $n = 31$, then $n+2 = 33$, but $11$ is not in the prime factorization that we obtained — so $33$ cannot be one of the terms. If $n+2 = 31$, then $n = 29$, but $29$ is not in the prime factorization we obtained. So, we must have $n = 30$ — and we can confirm $30 \times 31 \times 32$ has the same prime factorization that we obtained! Thus, our answer, the middle integer, is $\boxed{31}$.

**Theorem 5** (Divisibility Rule for 2)

A number is divisible by $2$ if and only if its last digit is either $0$, $2$, $4$, $6$, $8$

**Theorem 6** (Divisibility Rule for 3)

A number is divisible by $3$ if and only if the sum of its digits is divisible by $3$

**Theorem 7** (Divisibility Rule for 5)

A number is divisible by $5$ if and only if its last digit is either $0$ or $5$

**Theorem 8** (Divisibility Rule for 9)

A number is divisible by $9$ if and only if the sum of its digits is divisible by $9$.

**Theorem 9** (Divisibility Rule for 11)

A number is divisible by $11$ if adding and subtracting the digits in alternating order gives a number divisible by $11$. For example, $3 - 6 + 3 = 0$, and since $0$ is divisible by $11$, $363$ is divisible by $11$.

**Example 5**

How many even perfect square divisors does $1000000$ have?

**Solution:** We will use the ideas from an earlier example — we can prime factorize $1000000 = 10^6 = 2^6 \times 5^6$. Thus, the divisors are $2^{\square}5^{\square}$ where each box is an integer from $0$ through $6$ inclusive. We know that there are $7 \times 7 = 49$ total divisors — the question though is how many of these are even and perfect squares?

For a number $2^{\square}5^{\square}$ to be even, the exponent of two needs to be greater than $0$. For it to be a perfect square, we need both of the exponents to be even, so that they can be evenly divided into two equal numbers. For example, $2^4 5^2$ is a perfect square because it can be written as $(2^2 \times 5) \times (2^2 \times 5)$ — which is because the exponents $4$,$2$ are both even.

So, we need the exponent of $2$ to specifically be an integer from $0$ to $6$ inclusive that is greater than $0$ and even — it can only be $2$, $4$, $6$. And we need the exponent of $5$ to be an integer from $0$ to $6$ inclusive that is even — it can only be $0$, $2$, $4$, $6$.

This gives us a total number of $3 \times 4 = \boxed{12}$ even perfect square divisors of $1000000$.

**Example 6**

A number is in the form of $\overline{12A53B}$, where $A$ and $B$ are not necessarily distinct digits. The number is divisible by $396$. Find it.

**Solution** *(Example and Solution contributed by John)***:** Since there is no straightforward divisibility rule for $396$, we factorize it:

$$396 = 2^2 \cdot 3^2 \cdot 11,$$

which is also $4 \cdot 9 \cdot 11$. Now we have to apply divisibility rules for $4$, $9$, and $11$! A number is divisible by $4$ if its last two digits are divisible by $4$. Since the last two digits are $\overline{3B}$, $B$ must be $2$ or $6$.

- If $B = 2$, then the number will be $\overline{12A532}$. It has to be divisible by $9$, and since $1+2+5+3+2 = 13$ which leaves a remainder of $4$ when divided by $9$, $A$ leaves a remainder of $5$ when divided by $9$, leaving $A = 5$ $(125532)$ the only valid solution. Now we need to check if it's divisible by $11$. Since $1 - 2 + 5 - 5 + 3 - 2 = 0$ which is divisible by $11$, we're good with $125532$.

- If $B = 6$, the number will be $\overline{12A536}$. Since $1 + 2 + 5 + 3 + 6 = 17$ which leaves a remainder of $8$ when divided by $9$, $A$ has to leave a remainder of $1$ when divided by $9$. $A = 1$, giving us $121536$. Now check if it's divisible by $11$. Since $1 - 2 + 1 - 5 + 3 - 6 = -8$ and that is not divisible by $11$, the whole number isn't, leaving this solution bad.

This leaves $\boxed{125532}$ as the only possible number. ■

## Problems w/ Factors

**Definition 2**

Let $\nu_p(a)$ equal the exponent of $p$ in the prime factorization of $a$ for non-zero $a$. Formally,

$$\nu_p(n) = \begin{cases} \max\{k \in \mathbb{N}_0 : p^k \mid n\} & \text{if } n \neq 0 \\ \infty & \text{if } n = 0, \end{cases}$$

where $\mathbb{N}_0$ represents the set of non-negative integers.

The above equation basically means that $\nu_p(n)$ is the largest $k$ such that $p^k$ is a factor of $n$.

Now, try to see if you can make sense of these things:

- For two numbers $a$, $b$, we have $\nu_p(\gcd(a, b)) = \min(\nu_p(a), \nu_p(b))$.
- For two numbers $a$, $b$, we have $\nu_p(\operatorname{lcm}(a, b)) = \max(\nu_p(a), \nu_p(b))$.
- For two numbers $a$, $b$, we have $\nu_p(ab) = \nu_p(a) + \nu_p(b)$.
- For two numbers $a$, $b$, $a \mid b$ is equivalent to $\nu_p(a) \leq \nu_p(b)$ for all $p$.

**Do not memorize them. If you have questions about why they work, ask in the chat.**

**Example 7**

Find the number of factors of $300$ that are also multiples of $12$.

**Solution.** $300 = 2^2 \cdot 3 \cdot 5^2$ and $12 = 2^2 \cdot 3$. We know that $\nu_2(x) = 2$ and $\nu_3(x) = 1$ because in both situations there is an inequality for $\leq$ and $\geq$ of $\nu_2$. For $\nu_5(x)$ you can choose it freely from $0$ to $2$, so there are a total of $\boxed{3}$.

**Example 8**

Find the number of ordered triples $(x, y, z)$ that are all divisors of $600$ and the $\gcd(x, y, z) = 12$.

**Solution.** $600 = 2^3 \cdot 3 \cdot 5^2$ and $12 = 2^2 \cdot 3$. From here we know that $\min(\nu_2(x), \nu_2(y), \nu_2(z)) = 2$ and $\nu_2(x), \nu_2(y), \nu_2(z) \leq 3$. The sequences of $\nu_2$ can be one of the triples where each number is either $2$ or $3$ but not all of them are $3$, so there's $7$ cases for this. For $\nu_3$ we know that for all of those numbers, the value of that is $1$. For $\nu_5$ all of them have to be at most $2$ but *at least* one of them must be $0$. Therefore since we see "at least" we use the principle of inclusion or exclusion, PIE. There can be $9$ cases when one of them is $0$; there are $3$ is two of them are $0$; there are only $1$ if all three are $0$. The cases for this part is $9 \cdot 3 - 3 \cdot 3 + 1 = 19$. Multiplying that by the cases for $\nu_2$ we get $19 \cdot 7 = \boxed{133}$ as our answer.

**Example 9** (2013 CMQQR / 3)

A positive integer $n$ has the property that there are three positive integers $x$, $y$, $z$ such that $\operatorname{lcm}(x, y) = 180$, $\operatorname{lcm}(x, z) = 900$ and $\operatorname{lcm}(y, z) = n$. Determine the number of positive integers $n$ with this property.

**Solution** (By Inferno2332). Obviously $x, y, z$ can't have any prime factors above $5$, otherwise $\operatorname{lcm}(x, y)$ or $\operatorname{lcm}(y, z)$ would be divisible by that prime factor. So let

$$x = 2^{a_1}3^{a_2}5^{a_3}, \quad y = 2^{b_1}3^{b_2}5^{b_3}, \quad z = 2^{c_1}3^{c_2}5^{c_3}.$$

First find the possible values for $\nu_2(n)$. Since

$$\nu_2(180) = \nu_2(900) = 2,$$

we have $\max(a_1, b_1) = 2$ and $\max(a_1, c_1) = 2$. In particular, this means $b_1, c_1 \leq 2$, so $\max(b_1, c_1) = 0, 1, 2$ are all achievable with

$$(a_1, b_1, c_1) = (2, 0, 0),\ (2, 1, 1),\ (2, 2, 2)$$

respectively. So $\nu_2(n)$ can be $0, 1, 2$.

Now move on to $\nu_3(n)$. Since $\nu_3(180) = \nu_3(900) = 2$, we can use the exact same reasoning to find $\nu_3(n)$ can be $0, 1, 2$.

Finally consider $\nu_5(n)$. Since $\nu_5(180) = 1$ and $\nu_5(900) = 2$, we have $\max(a_3, b_3) = 1$ and $\max(a_3, c_3) = 2$. Since the first condition gives $a_3 \leq 1$, the second condition forces $c_3 = 2$. Since $b_3 \leq 1$, the only possibility for $\max(b_3, c_3)$ is $2$. Thus, $\nu_5(n)$ can only be $1$.

To summarize, there are $3$ choices for $\nu_2(n)$, $3$ choices for $\nu_3(n)$, and $1$ choice for $\nu_5(n)$, and thus the answer is $9$.

Thanks to Inferno2332 for writing the solution.

## Try the Problems

Last $2$ contests written by John + testsolved by Samuel & Akshaj. They are relevant to the section by John.

**Try the Prime Factors and Divisors Dashes & Problems**

https://mathdash.com/contest-group/prime-factors-divisors-1
https://mathdash.com/contest-group/prime-factors-divisors-2
https://mathdash.com/contest-group/prime-factors-divisors-3
https://mathdash.com/contest-group/prime-factors-divisors-4
https://mathdash.com/contest-group/prime-factors-divisors-5
https://mathdash.com/contest/divisors-sprint
https://mathdash.com/contest/john-problem-1735000837735
https://mathdash.com/contest/divisor-problem

# Easy Modular Arithmetic

*by sussususus*

https://mathdash.com/contest/mod-arithmetic-problemset/

**Disclosure & Credit** This chapter is authored by sussususus and typeset by John Z. The appropriate level is AMC 8 / MathCounts for this chapter.

This chapter introduces the basics of modular arithmetic. Modular arithmetic is pretty common in math competitions like MATHCOUNTS and the AMC, and it can help you solve many seemingly intractable problems involving remainders!

This is **for beginners**, so there is no advanced content like Fermat's Little Theorem and so on.

Let's begin with an easy warm-up problem.

**Example 1**

Suppose it is midnight (0:00) right now. What time will it be $1000$ hours from now? Give your answer in 24-hour time.

**Solution.** To solve this problem, you can think of the time as a cycle that repeats every $24$ hours. Modular arithmetic helps us simplify this problem by focusing on the remainder after dividing by $24$.

$1000$ divided by $24$ is $41$ remainder $16$, so $1000$ hours is the same as $41$ full days and $16$ leftover hours. Therefore the answer is $16:00$.

In the previous problem, we were working in modulo $24$. This means we only care about the remainders when divided by $24$. When we add $1$ hour to $23$ hours, we go back to $0$.

So in modulo $24$, $25$ is the same as $1$, because they have the same remainder when divided by $24$.

We say that $25$ is congruent to $1$, modulo $24$. And this is written as $25 \equiv 1 \pmod{24}$.

**Example 2**

What integer between $0$ and $4$ is congruent to $6782 \pmod{5}$?

**Solution.** The remainder when $6782$ is divided by $5$ is $2$. So the answer is $2$.

One thing which we can do with a congruence like $6782 \equiv 2 \pmod{5}$ is add the same thing to both sides:

$$6782 + 3 \equiv 2 + 3 \pmod{5}$$

We can do the same thing with subtraction and multiplication.

$$6782 - 3 \equiv 2 - 3 \pmod 5$$
$$6782 \times 3 \equiv 2 \times 3 \pmod 5$$

You can even do it with exponents.

$$6782^3 \equiv 2^3 \pmod 5$$

However, we cannot do it for division. (The rule for division is a bit tricky, and we will not introduce it here. For now just know that it's illegal.)

In other words, when a problem asks for something $\pmod n$, you can reduce each of the numbers in the problem $\pmod n$ first before doing arithmetic operations.

**Example 3**

Find the remainder when $9453 \times 6824 \times 6782 \times 5675341$ is divided by $5$.

**Solution.** We can find that the four numbers in the product are congruent to $3, 4, 2, 1 \pmod 5$, respectively. Thus, the original product is congruent to $4 \times 3 \times 2 \times 1 = 24 \pmod 5$. So the remainder is $4$.

Some problems ask about the **last digit** or **last two digits**. Keep the following in mind:

- The last digit of $n$ is congruent to $n \pmod{10}$.
- The last two digits of $n$ is congruent to $n \pmod{100}$.

**Example 4**

Find the last two digits of $98^{10}$.

**Solution.** Basically we want $98^{10} \pmod{100}$. Now let's try reducing the number $98$. Note that $98 \equiv -2 \pmod{100}$. Yes, negative numbers are allowed and can be very useful.

Now raise both sides to the power of $10$ to get $98^{10} \equiv (-2)^{10} \pmod{100}$. And $(-2)^{10} = 1024 \equiv 24 \pmod{100}$, so the answer is $24$.

Sometimes, it can help to break a large exponent into multiple small exponents. This is particularly helpful if a *power* of the original number is congruent to $1$ or $-1 \pmod m$.

**Example 5**

Find the remainder when $7^{777}$ is divided by $48$.

**Solution.** Unlike the previous problem you can't reduce $7$ any further. So how can we simplify this problem?

The key thing to notice is that $7^2 = 49$. This is congruent to $1 \pmod{48}$. So we can write the expression as

$$7^{777} = \left(7^2\right)^{388} \cdot 7 = 49^{388} \cdot 7$$

Now we can reduce it mod $48$. This simply becomes $1^{388} \cdot 7 \pmod{48}$, so the answer is $7$.

**Try the Problems**

https://mathdash.com/contest/mod-arithmetic-problemset

# MATHCOUNTS

*by MathDash Community*

https://mathdash.com/contests?tag=MATHCOUNTS

**MATHCOUNTS Practice Contests**

**https://mathdash.com/contests?tag=MATHCOUNTS**

# AMC 8

*by MathDash Community*

https://mathdash.com/contests?tag=AMC+8

**AMC 8 Mini Practice Contests**

**https://mathdash.com/league/amc-8-mini**

**AMC 8 Full-length Mock Contests**

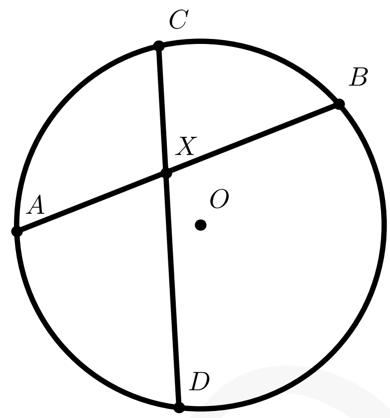
**https://mathdash.com/contests?tag=AMC+8**

# Power of a Point

*by MathDash Team*

https://mathdash.com/maps/map/power-of-a-point

Figure 1: Power of a Point — $AX \cdot BX = CX \cdot DX = R^2 - OX^2$

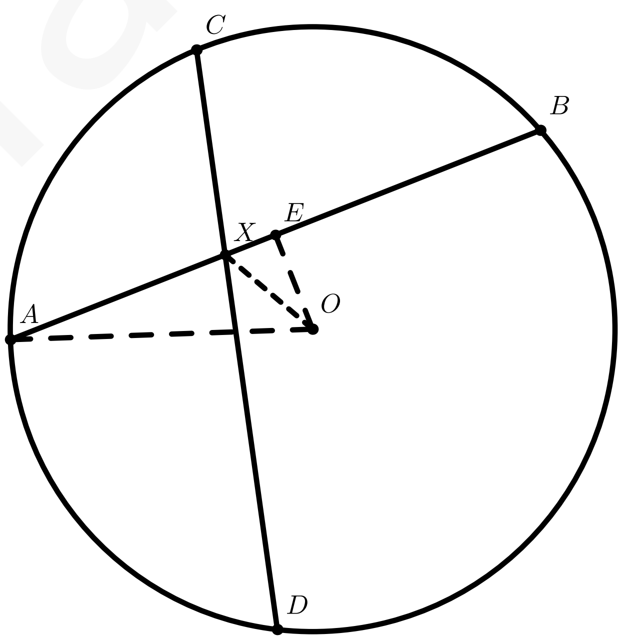



**Theorem 1** (Power of a Point)

Given any circle with center $O$ and radius $R$, and any point $X$ inside that circle, we call $OX^2 - R^2$ the **power** of the point $X$ with respect to the circle.

The theorem states that for any chord $AB$ that goes through $X$, $AX \cdot XB$ is equal to the negative of the power of the point $X$. Since this is constant, this means $AX \cdot XB = CX \cdot XD$.

**Proof:** Let's define $E$ to be the midpoint of chord $AB$, and draw some extra lines



We know that $AE = EB$ by definition, $AO = BO = R$, and $OE = OE$, so $\triangle AEO \cong \triangle BEO$. Thus, $\angle AEO = \angle BEO = 90^\circ$.

We want to find the expression $AX \times XB$. This is equal to $(AE - EX)(BE + EX) = (AE - EX)(AE + EX) = AE^2 - EX^2$. Now, since we have right triangles, let's use the Pythagorean Theorem!

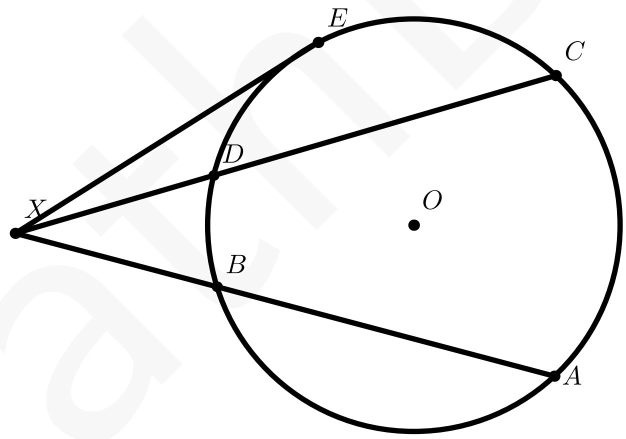
$EX^2 = OX^2 - OE^2$ and $AE^2 = OA^2 - OE^2$. Subtracting these gives that $AX \times XB = AE^2 - EX^2 = OA^2 - OX^2 = R^2 - OX^2$ where $R$ is the radius of the circle. And — since this is the power of point $X$ and only depends on the location of point $X$ and the circle, we know that $AX \cdot XB = CX \cdot XD = R^2 - OX^2$. ■

There is another way to prove that $AX \cdot XB = CX \cdot XD$ using similar triangles! We know that $\angle AXC = \angle BXD$, and $\angle ACX = \angle DBX$ (because they both are half the measure of arc $\overset{\frown}{AD}$) — so by Angle Angle Similarity, we know that $\triangle ACX \sim \triangle DBX$.

This tells us $\dfrac{CX}{AX} = \dfrac{BX}{DX}$ which rearranges to $AX \cdot XB = CX \cdot XD$.

Interestingly, Power of a Point doesn't only hold for points inside the circle — let's look at a case where point $X$ is outside of the circle!

Figure 2: Power of a Point — $XA \cdot XB = XC \cdot XD = OX^2 - R^2$




**Theorem 2** (Power of a Point)

Given any circle with center $O$ and radius $R$, and any point $X$ **inside or outside or on** that circle, we call $OX^2 - R^2$ the **power** of the point $X$ with respect to the circle.

The theorem states that for any chord $AB$ that goes through $X$, $XA \cdot XB$ is equal to the power of the point $X$. Since this is constant, this means $OX^2 - R^2 = XA \cdot XB = XC \cdot XD$.

In addition, we also have that $OX^2 - R^2 = XA \cdot XB = XC \cdot XD = XE^2$.

Where did the $XE^2$ come from, you may ask? Well — consider what happens if we move $C$ and $D$ closer together (maintaining that $CD$ intersects $X$). We still have that $XC \cdot XD = OX^2 - R^2 = XA \cdot XB$ holds as we move $C$ and $D$ closer and closer together. When they meet each other, we still need $CD$ to intersect $X$ — this means that they will meet at point $E$! And so, $XE \cdot XE = OX^2 - R^2 = XA \cdot XB$ giving us our result.

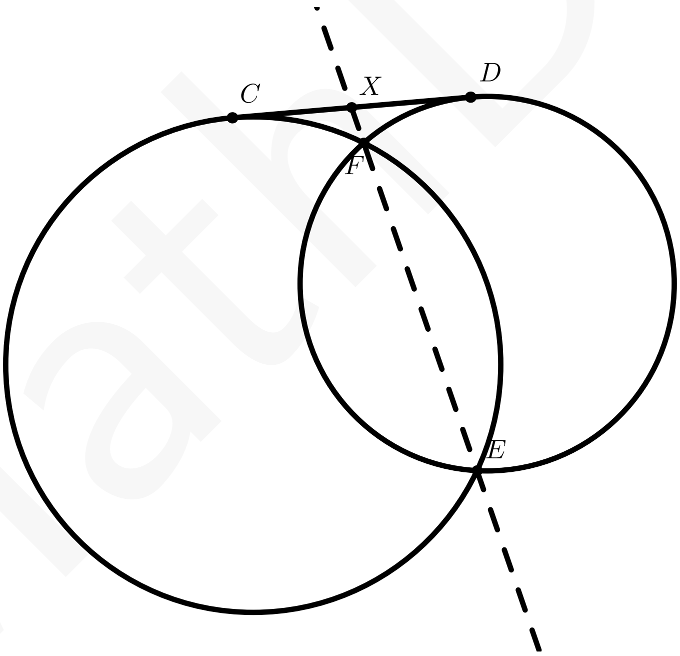
Again, we can also use similar triangles to show $XB \cdot XA = XD \cdot XC = XE^2$. For example, we know $\angle DXB = \angle AXC$, and $\angle XBD = 180 - \angle DBA = \angle XCA$ — where the last equality comes from the fact that opposite angles in a cyclic quadrilateral sum to $180°$.

*As an aside, maybe you will notice that we defined the 'power of a point' to be the negative of our earlier definition. We can keep this consistent using something called* directed lengths *— where we consider $XA = -1 \cdot AX$. If this is the case, then the above theorem holds whether or not $X$ is inside or outside of the circle!*

This is a really powerful theorem that can help us solve a variety of questions.

**Example 1**

Two circles intersect at $F$ and $E$. A line is drawn tangent to both circles at $C$ and $D$, and $EF$ intersects $CD$ at $X$. If $XC = 4$, and the radii of the circles are $10$ and $8$, find $XD$.




**Solution:** Power of a Point on the left circle tells us $XC^2 = XF \cdot XE$. Power of a Point on the right circle also tells us $XD^2 = XF \cdot XE$. So, $XC^2 = XD^2$ — and since $XC = 4$, we must have $XD = \boxed{4}$. The radii of the circles turns out to be a bit of a red herring!

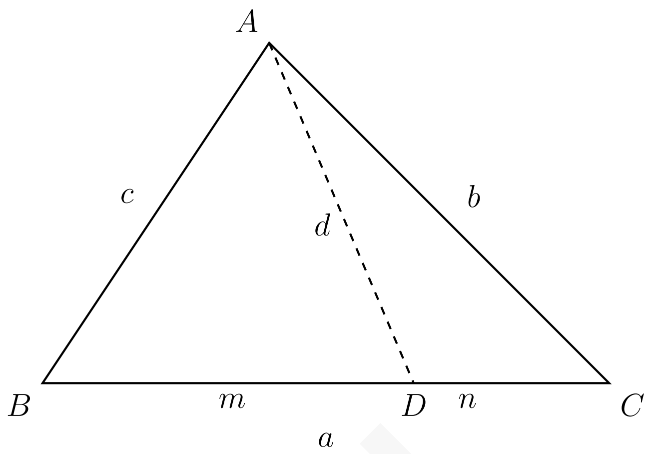
**Try some Power of a Point Problems**

https://mathdash.com/maps/map/power-of-a-point

# Stewart's Theorem

*by Sid, Testsolved by Math645 and Carcool*

https://mathdash.com/maps/map/stewarts




**Theorem 1**

In a triangle cut by any cevian (refer to image above), $man + d^2a = b^2m + c^2n$ or, $man + dad = bmb + cnc$. This can be memorized by the mnemonic "A man and his dad put a bomb in the sink." Note that $m$ is simply $a - n$, and vice versa, so you can solve the equation if you have every other value and either $a$, $m$, or $n$.

**Proof:** Applying Law of Cosines in $\triangle ABD$ at angle $\angle ADB$, and in $\triangle ACD$ at angle $\angle CDA$, we get the following equations:

$$m^2 + d^2 - 2md \cos \angle ADB = c^2$$

$$n^2 + d^2 - 2nd \cos \angle CDA = b^2$$

Notice that $\angle ADB$ and $\angle CDA$ are supplementary, which means that $m\angle ADB = 180^\circ - m\angle CDA$. We can represent the measure of $\angle CDA$ as $\theta$. Since we also know that $\cos\theta = -\cos(180 - \theta)$, we can create the equations:

$$\frac{c^2 - m^2 - d^2}{2md} = \cos\theta$$

$$\frac{n^2 + d^2 - b^2}{2nd} = \cos\theta$$

After substituting, cross multiplying, and simplifying, we get $c^2n + b^2m = m^2n + n^2m + d^2m + d^2n$. Next, we can factor out $(m + n)$ from the terms $m^2n + n^2m$ and $d^2m + d^2n$. Now, we have the equation

$$c^2n + b^2m = (m + n)mn + d^2(m + n).$$

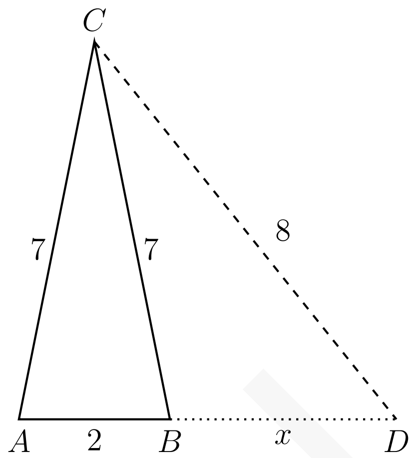
Notice how in the diagram, $m + n$ is equal to $a$ by the Segment Addition Postulate. Therefore, we get our final equation to be

$$\boxed{man + d^2a = b^2m + c^2n}$$

**Example 1**

In $\triangle ABC$, we have $AC = BC = 7$ and $AB = 2$. Suppose that $D$ is a point on line $AB$ such that $B$ lies between $A$ and $D$ and $CD = 8$. What is $BD$? (2005 AMC 12B #6).

**Solution:** First, as we should in any geometry problem, we will construct a quick diagram.




Next, lets use Stewart's Theorem to solve this problem. We get the equation:

$$2(2+x)(x) + 7^2(2+x) = 7^2x + 2(8^2)$$

Simplifying, we get

$$2x^2 + 4x + 98 + 49x = 49x + 128$$
$$2x^2 + 4x - 30 = 0$$
$$x^2 + 4x - 15 = 0$$

By factorizing this equation, we get the solutions to be $3$ and $-5$. Since the length can't be negative, we get that the answer to be $\boxed{3}$.

**Theorem 2**

If $AD$ is an angle bisector, then $d^2 + mn = bc$.

**Theorem 3**

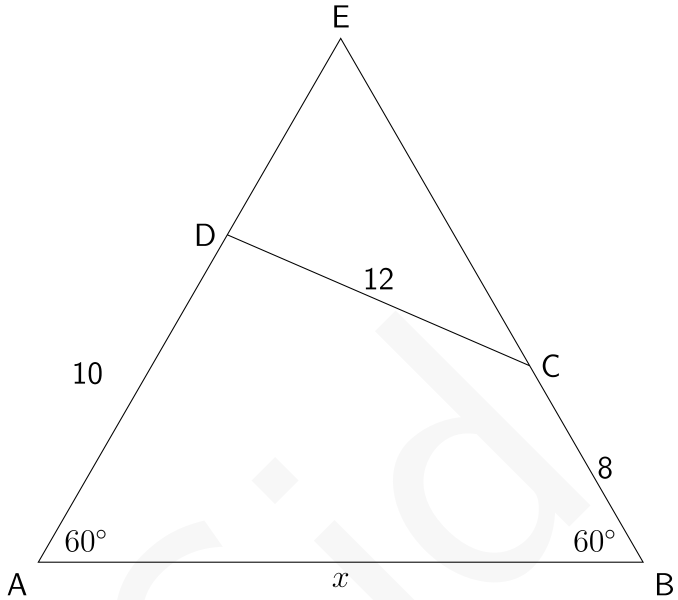
If $AD$ is a median, then $d^2 = \frac{b^2+c^2}{2} - \frac{a^2}{4}$

**Note:** These theorems are not necessary to memorize, as they can easily be derived from Stewart's theorem. However, by memorizing them it makes it much easier to solve problems relating to Stewart's theorem faster.

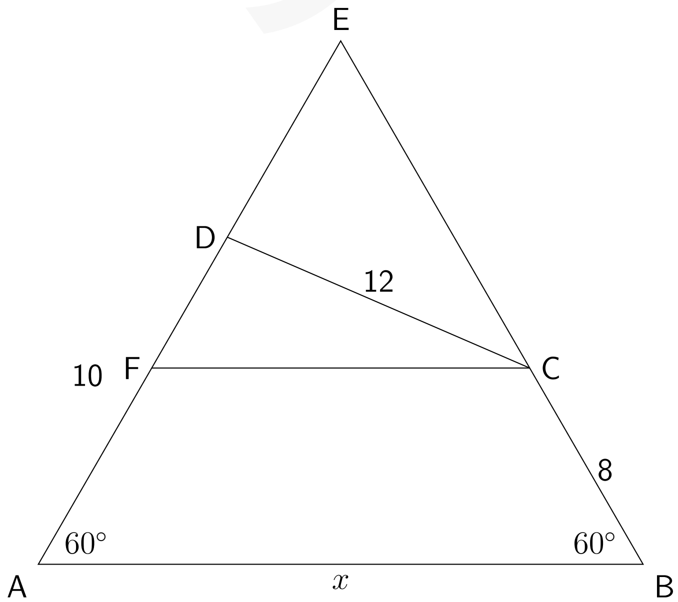
**Example 2**

In quadrilateral $ABCD$, $BC = 8$, $CD = 12$, $AD = 10$, and $m\angle A = m\angle B = 60^\circ$. Given that $AB = p + \sqrt{q}$, where $p$ and $q$ are positive integers, find $p + q$. (2005 AIME I Problem #7).

**Solution:** We start by drawing a diagram. In the following diagram, note that we extend lines $BC$ and $AD$ to intersect at point $E$.




After that, we draw a line from C parallel to line AD, with intersection point F.




Now, we can perform Stewart's Theorem on $\triangle ECF$. Since $FC \parallel AB$, $\triangle EBA \sim \triangle ECF$. This means that $\triangle ECF$ is also equilateral. Also, this means that $AF = CF = 8$ and $DF = 10 - 8 = 2$. Now, applying Stewart's Theorem, where $EF = x$, we get $2x(x-2) + 12^2 x = 2x^2 + x^2(x-2)$. We simplify to get $x^3 - 2x^2 - 140x = 0$, or $x^2 - 2x - 140$. Finally, using the quadratic formula, and discarding the negative solution, we get $x = 1 + \sqrt{141}$. Remember, the problem asked for the length of $AB$, which is equal to the length of $EB$. Therefore, the length of $AB$ is $8 + 1 + \sqrt{141}$, or $9 + \sqrt{141}$, making our answer $\boxed{150}$.

As seen, Stewart's Theorem can be used to easily solve many advanced geometry problems.

**Example 3**

In $\triangle ABC$, we have $AB = 1$ and $AC = 2$. Side $\overline{BC}$ and the median from $A$ to $\overline{BC}$ have the same length. What is $BC$? (2002 AMC 12B #23)

Feel free to draw your own diagram. However, this problem is quite simple. Let's refer to the point of intersection with the median and $BC$ as D. We can apply the formula when $AD$ is a median. Since $BC = AD$, we can represent this with the variable $x$. We get

$$x^2 = \frac{1^2 + 2^2}{2} - \frac{x^2}{4}$$

Simplifying, we get $x = \boxed{\sqrt{2}}$.

## More Stewart's Theorem Problems:

**More Stewart's Theorem Problems**

https://mathdash.com/maps/map/stewarts?tags=Geometry

# Modular Arithmetic I

*by Math645*

https://mathdash.com/maps/map/modular-arithmetic

## Operations

We will consider operations with mods, including addition, subtraction, multiplication, division, and exponentiation. The following $3$ theorems, can be proved easily, yet should make intuitive sense so their proofs are left out.

**Theorem 1** (Addition, Multiplication, Subtraction, Exponentiation)

If $a \equiv b$ mod $c$ and $d \equiv e$ mod $c$, then:

$$a + d \equiv b + e \pmod{c}$$
$$ad \equiv be \pmod{c}$$
$$a - d \equiv b - e \pmod{c}$$

In other words, addition, subtraction, multiplication, work as normal if they have the same mod. As you may have noticed, division was left out, so we will now consider division.

**Theorem 2** (Division Case 1)

If we have $a \equiv b \mod c$, and $n$ such that $n$ divides $a$, $b$, and $c$, then $\frac{a}{n} \equiv \frac{b}{n} \mod \frac{c}{n}$

**Theorem 3** (Division Case 2)

If we have $a \equiv b \mod c$, and any $n$ such that $\gcd(n, c) = 1$, then $\frac{a}{n} \equiv \frac{b}{n} \mod c$.

You may be wondering why we don't force that $n \mid a, b$ as how can we have $\frac{a}{n}$ if it isn't directly an integer. This brings us onto modular division, which is trivial by cross multiplying.

**Definition 1** (Modular Division)

Let's call the fraction $\frac{a}{b} \equiv x \mod m$. Then the fraction or $x$ is defined as the value such that $a \equiv bx \mod m$

This should make sense immediately by cross multiplying. This fraction can be obtained by guessing values that are $a$ mod $m$ and seeing if $b$ divides them. For small numbers guessing is the fastest method, yet for bigger numbers, it is best to use a process described in the Linear Diophantine Equations section of the Diophantine Equations Chapter written by Math645. This chapter has a section for modular division, which contains practice problems with modular division, a method to find this fraction, and a proof of why and when this fraction is defined. Please read the Linear Diophantine Equations Chapter of that book before continuing, as the following chapters do require the use of modular division and thus linear diophantine equations.

**Definition 2** (Modular Inverse)

The modular inverse of $b$ mod $m$ is $\frac{1}{b}$ mod $m$.

**Theorem 4** (Wilson's Theorem)

For a prime, $p$, $(p-1)! \equiv -1 \mod p$.

Proof: As shown in the Linear Diophantine Equations Chapter, the fraction $\frac{a}{b}$ mod $m$ is defined for all relatively prime $b$, $m$. So if we let $m = p$ be a prime, then for $b$ and $p$ to not be relatively prime, they must share a factor other than $1$. Yet the only factor $p$ has other than $1$ is $p$. So they must share the factor of $p$ to not be relatively prime. So $p$ must divide $b$ and $b \equiv 0 \mod p$. So, for $b \not\equiv 0 \mod p$, the fraction is defined. We also showed that the fraction has a unique value mod $p$. So if we have $x$ then there is a unique value, $y$ mod $p$ such that $x \equiv \frac{1}{y} \mod p$ or $xy \equiv 1 \mod p$. Now let's list out $1, 2, 3, 4, \ldots, p-1$. We will now attempt to group every pair $x$, $y$ from our previous definition. But wait, what if $x = y$, then we can't group them. Yet we know that if $x = y$, then $x^2 \equiv 1 \mod p$, so $x^2 - 1 \equiv 0 \mod p$, so $(x-1)(x+1) \equiv 0 \mod p$, so $x \equiv 1 \mod p$, or $x \equiv -1 \equiv (p-1) \mod p$. When we group our numbers, we put these numbers on the side. Now we have the numbers, $1$, $p-1$, (GROUP), (GROUP), (GROUP), $\ldots$. This is just a rearrangement of the numbers $1, 2, 3, 4, \ldots p-1$, so the product of all elements is the same. The product of the elements in each group is $1$, since $xy \equiv 1 \mod p$. So the product of all elements in the rearrangement is $1 \cdot (-1) \cdot 1 \cdot 1 \cdot 1 \cdots 1 \equiv -1$. The product in the original list is $(p-1)!$, so $(p-1)! \equiv -1 \mod p$.

## Exponents

We have a theorem to easily calculate numbers taken to large exponents, using a seemingly unrelated function, Euler's Totient Function.

**Definition 3** (Euler's Totient Function)

Euler's Totient Function, denoted as $\phi(n)$ is the number of integers less than or equal to $n$ that are relatively prime (share no common divisor other than 1) to $n$.

Now, we will form the connection between this function and modular arithmetic exponentiation with the following theorem which lets us easily calculate high exponents by reducing the exponent.

> **Theorem 5** (Euler's Totient Theorem Extended)
>
> For relatively prime $a$ and $m$, $a^n \equiv a^{n \bmod \phi(m)} \mod m$

Proof: This is equivalent to saying that $a^{\phi(m)} \equiv a^0 \equiv 1 \mod m$ as then, the exponent is the same as the exponent mod the totient. To prove this new statement, we will use the set of integers relatively prime to $m$ and less than $m$. This set clearly has $\phi(m)$ elements. Let's call one member of this set, $x$. Clearly, $m$ and $x$ share no divisors. Thus, if we multiply each element of the set by $x$, we aren't adding any common divisors of $m$, so each member of the new set is relatively prime to $m$. There are also still $\phi(m)$ elements, and they are all relatively prime to $m$. When taken mod $m$, no 2 are the same by Divsion Case 2 since if they are the same after, they would have to be the same before. This means that when taken mod $m$ have all the values relatively prime to $m$, which is the same as the inital set. Therefore, the initial set and new set are identical mod $m$, just rearranged. Thus, the product of all elements is the same mod $m$. If we call the product of the terms of the original set, $y$, then the product of the terms of the new set is $y \cdot x^{\phi(m)}$. Setting them congruent, and using Division Case 2, to remove the $y$ (since its relatively prime to $m$) and we get the desired result.

Now we know how to reduce exponents, but calculating the totient by checking every value less than $m$ to see which ones are relatively prime, can still take a while. So, the following theorem provides an easy way to calculate the totient. The proof requires combinatorics and is thus left out until a later part of the book.

> **Theorem 6**
>
> If $n$ can be expressed as $p_1^{a_1} \cdot p_2^{a_2} \cdot p_3^{a_3} \cdots p_x^{a_x}$, then $\phi(n) = n \cdot \frac{p_1-1}{p_1} \cdot \frac{p_2-1}{p_2} \cdot \frac{p_3-1}{p_3} \cdots \frac{p_x-1}{p_x}$

> **Example 1**
>
> How many different remainders can result when the $100$th power of an integer is divided by $125$?

Solution: We note that $\phi 125 = 100$, thus for a relatively prime (to the modulo $125$) $x$, $x^{100} \equiv 1 \mod 125$. Thus $1$ is a possible remainder and occurs for all relatively prime numbers. But, what if its not relatively prime. If a number isn't relatively prime to $125$, it shares a factor with $125$, other than $1$. These factors must be $5, 25$, or $125$. So it must have a factor of $5$ (Since all of $5, 25, 125$ have factors of $5$, so no matter which factor it shares, $5$ is always shared). Anything with a factor of $5$ when taken to the $100$th power must have a factor of $5^{100}$. It's clear that every multiple of $5^{100}$ is a multiple of $5^3 = 125$. So when it's not relatively prime, we will always have a multiple of $125$, which must be $0$ mod $125$. So, in total our possible resiudes are $1$ when it's not relatively prime, and $0$ when it is relatively prime, for a total of $\boxed{2}$ possible remainders.

**Example 2**

What are the last two digits of $97^{38}$?

Since we are working in base $10$, the last $2$ digits will just be the number mod $100$, so we need to calculate $97^{38} \mod 100$. We see that $97$ and $100$ are relatively prime and decide to use Euler's Totient Theorem, so $97^{38} \equiv 97^{38 \mod 40} \mod 100$. $38$ is already as simplified as it can get mod $40$, so what should we do next? We might have noticed that $38$ is really close to $40$, so if we subtract $40$ we would get a negative number with a really small absolute value. We want really small absolute values since it would be easier to compute, compared to taking something to the $38$th power. And negative exponents are just fractions which we know how to deal with. So, $97^{38} \equiv 97^{-2} \equiv \frac{1}{97^2}$. Calculating $97^2$ is still hard, so can we simplify it. Note that $97 \equiv -3 \mod 100$, so $97^2 \equiv (-3)^2 \equiv 9$. So $\frac{1}{97^2} \equiv \frac{1}{9}$. This is a lot easier to work with. We want $x \equiv \frac{1}{9} \mod 100$. We cross-multiply to $9x \equiv 1 \mod 100$ to the Diophantine: $9x = 100y + 1$, or $9x - 100y = 1$. We solve with the Euclidean Algorithm: $(9, 100) = (9, 9 \cdot 11 + 1) = (9, 1) = 1$, or in reverse: $1 = 9 - 8 \cdot 1 = 9 - 8 \cdot (100 - 9 \cdot 11) = 9 \cdot 89 - 8 \cdot 100$, so $x \equiv \boxed{89} \mod 100$.

## CRT

Just like there are system of equations, there are also systems of modular equations that can be solved with CRT or the Chinese Remainder Theorem. Once again, the theorem should intuitive so the proof is left out.

**Theorem 7** (CRT: Chinese Remainder Theorem)

For the modular system of equations:

$$\begin{aligned} x &\equiv y_0 \mod z_0 \\ x &\equiv y_1 \mod z_1 \\ x &\equiv y_2 \mod z_2 \\ x &\equiv y_3 \mod z_3 \\ x &\equiv y_4 \mod z_4 \\ &\vdots \\ x &\equiv y_n \mod z_n \end{aligned}$$

If there are no contradictions, there exists a single solution for $x$ in mod $\operatorname{lcm}(z_1, z_2, z_3, z_4, \ldots, z_n)$

An example of a contradiction is saying that $n \equiv 1 \mod 4$, which means that $n$ must be odd, but then saying $n \equiv 4 \mod 6$, which means that $n$ must be even.

**Example 3**

For a positive integer $p$, define the positive integer $n$ to be $p$-safe if $n$ differs in absolute value by more than $2$ from all multiples of $p$. For example, the set of $10$-safe numbers is $\{3, 4, 5, 6, 7, 13, 14, 15, 16, 17, 23, \ldots\}$. Find the number of positive integers less than or equal to $10{,}000$ which are simultaneously $7$-safe, $11$-safe, and $13$-safe. (Source: 2012 AIME II Problem 12)

Solution: Note that this is basically saying that $n \equiv 3, 4 \mod 7$ and $n \equiv 3, 4, 5, 6, 7, 8 \mod 11$ and $n \equiv 3, 4, 5, 6, 7, 8, 9, 10 \mod 13$. Note that for each value of mod $7$, mod $11$, and mod $13$, we pick we will have a unique system of $3$ modular equations. Well are there any contradictions? We see that since $7$, $11$, $13$ are all relatively prime, there can't be any contradictions, as they share no factors, so no factors must be divisible and can't be divisible at the same time. Thus since there are no contradictions, for each value of mod $7$, mod $11$, and mod $13$, we will have a unique solution mod $\operatorname{lcm}(7, 11, 13) = 1001$. There are $2$ ways to pick for mod $7$, $6$ ways for mod $11$, and $8$ ways for mod $13$, so we can do this in $2 \cdot 6 \cdot 8 = 96$ ways. Since we have mod $1001$, this will always produce $10$ values less than or equal to $10{,}010$. So we have a total of $96 \cdot 10 = 960$ solutions. However, we want less than or equal to $10000$ not $10010$, so we have to subtract off any values that are greater than $10000$ and less than $10010$ and are 7-safe, 11-safe, and 13-safe. Note that $10010$ is divisible by $7$, $11$, $13$. So $10009 \equiv -1 \equiv 6 \mod 7$, fail. $10008 \equiv -2 \equiv 5 \mod 7$, fail. $10007 \equiv -3 \equiv 4 \mod 7$. It's $11 - 3 \equiv 8 \mod 11$ and $13 - 3 \equiv 10 \mod 13$, pass. $10006 \equiv -4 \equiv 3 \mod 7$. It's $11 - 4 \equiv 7 \mod 11$ and $13 - 4 \equiv 9 \mod 13$, pass. $10005 \equiv -5 \equiv 2 \mod 7$, fail. Similarily, $1004$, $1003$, $1002$, $1001$ all fail mod $7$, so they aren't 7-safe and thus fail. In total we have to subtract off $2$ solutions to get $960 - 2 = \boxed{958}$.

If there aren't contradictions, we know there are solutions by CRT but how do we actually find them? To do this, we must convert them from modular equations to algebraic equations. For example, we turn $x \equiv 3$ mod $5$ to $x = 5y + 3$. Then, we can set the algebraic equations equal to each other. Once setting them equal, we can take everything modulo the lower value to eliminate a variable to get a $1$ variable modular equation, which we can solve just like a regular equation by isolating the variable and dividing. Once we get a modular solution we can convert back into an algebraic equation for our variable, and substitute into our initial algebraic equation to get a new algebraic equation and convert that into a modular equation for our answer.

This may sound confusing so let's try this with a small example right now.

**Example 4**

Solve for $x$ if $x \equiv 4 \mod 7$, and $x \equiv 3 \mod 5$.

Converting to algebraic equations gives us, $x = 7a + 4 = 5b + 3$. We know by CRT, we will get a solution mod $35$, which is just a solution for $a$, mod $5$. Taking everything mod $5$, also eliminates $b$, making the equation just $1$ variable which we can solve with operations. Thus we take $7a + 4 = 5b + 3$ mod $5$ to get $2a + 4 \equiv 3 \mod 5$, Using subtraction operation, $2a \equiv -1$

$\bmod 5$. To solve this modular division problem we turn it into a Linear Diophantine Equation so $2a + 5m = -1$. We solve for 1 and then negate: $2a + 5m = 1$. We run the Euclidean Algorithm to $(2, 5) = (2, 2 \cdot 2 + 1) = (2, 1) = 1$. So $1 = 2 - 1 \cdot 1 = 2 - 1 \cdot (5 - 2 * 2) = 2 \cdot 3 - 5 \cdot 1$. So $a \equiv 3 \mod 5$ gives us 1 so $a \equiv -3 \equiv 2 \mod 5$ gives us $-1$. Although we got a solution for $a$ mod 5, it is still hard to relate the equations of different forms, so we convert our value of $a$ to equation form. $a \equiv 2 \mod 5$ is $a = 5c + 2$. Plugging into $x = 7a + 4$, we obtain $x = 7(5c + 2) + 4 = 35c + 18$. Thus, $x \equiv 18 \mod 35$.

For a system of 3 or higher equations, it is beneficial to consider them 2 at a time. Consider any 2 modular equations, and convert it to 1 modular equation and we have effectively reduced the problem. Doing this enough times will solve it.

**Example 5**

Given that $x$ is 3 mod 18, 7 mod 10, and 12 mod 19, find the sum of the two smallest positive integer values of $x$

Solution: It will be easy to find the two smallest integer values of $x$ when we got a modular solution for $x$. Recall to solve 3 equations, we solve any 2 of them and convert it into 1 modular equation, and then we only have 2 modular equations in total to solve. We pick the first 2 equations to solve first, so $x = 18y + 3 = 10z + 7$. Taking everything mod 10, we have $8y + 3 \equiv 7 \mod 10$, subtracting 3 with the subtraction operation we have $8y \equiv 4 \mod 10$. Dividing by 2 with Division Case 1 we have $4y \equiv 2 \mod 5$. Dividing by 2 with Division Case 2 we have $2y \equiv 1 \mod 5$. We can quickly guess $y = 3$ works so $y \equiv 3 \mod 5$. We also did the exact same division in the previous problem. Thus $y = 5n+3$. Plugging into $x = 18y+3$, we obtain $x = 18(5n+3)+3 = 90n+57$. We don't need to convert into modular form since to solve the next system, we convert back into equation form. Now we have $x = 90n + 57 = 19a + 12$. Taking everything mod 19, we have $14n \equiv 12 \mod 19$. Using Division Case 2, we divide by 2 first to get $7n \equiv 6 \mod 19$. Now we solve this Diophantine Equation. So, $7n-19m = 6$. We solve for 1 and multiply by 6. $(7, 19) = (7, 7\cdot2+5) = (7, 5) = (5\cdot1+2, 5) = (2, 5)$. Wait, we could go on all the way until $= 1$, but remember the only reason we keep going down is to get simpler ways to equate something to 1, since we don't know how to do it with the bigger numbers until we work backwards from smaller numbers. In the previous problem though, we already knew how to equate $(2, 5)$ to 1 when we did the division in the previous problem. We recall $1 = 2 \cdot 3 - 5 \cdot 1$. Now we can just write numbers in terms of previous numbers of our Euclidean Algorithm. $1 = 2\cdot3-5\cdot1 = (7-5\cdot1)\cdot3-5\cdot1 = 7\cdot3-5\cdot4 = 7\cdot3-(19-2\cdot7)\cdot4 = 7\cdot11-19\cdot4$. Thus $n \equiv 11 \mod 19$ produces 1 as our result to our Linear Diophantine Equation, so $n \equiv 11*6 \equiv 66 \equiv 9 \mod 19$ produces 6 which is what we needed. Plugging in $n = 19p + 9$ into our equation for $x$, we have $x = 90n + 57 = 90(19p + 9) + 57 = 1710p + 867$. Thus the smallest 2 positive values for $x$ are 867, and $867 + 1710$ which have a sum of $867 + 867 + 1710 = \boxed{3444}$.

Euler's Totient Theorem can be very useful to reduce exponents, however there is a very restricting clause that the base has to be relatively prime to the mod. So often times when we have a number that isn't relatively prime to the mod, we break up the mod into several parts. There will be parts that are relatively prime to the mod which are easy to calculate with Euler's Totient Theorem. There will be parts that aren't relatively prime to the mod which are easy to calculate in general. These parts can be combined with CRT. Also if numbers are big and computation heavy with the totient theorem, CRT can break it into smaller more manageable parts that aren't computation heavy (even if they were relatively prime before being broken up).

**Example 6**

Find the last two digits of $1032^{1032}$. (Source: 2009 HMMT Guts Round)

Recall last 2 digits in base 10 is just mod $10^2 = 100$. So we need $1032^{1032} \mod 100$. We can simplify the base instantly to get $32^{1032} \mod 100$. We can't use Euler's Totient Theorem since 32 and 100 aren't relatively prime. They share the factor of 4. So we decide to break 4 out of 100. So we want $32^{1032} \mod 4$ and $32^{1032} \mod \frac{100}{4} = 25$, and then use CRT to get our solution mod $\operatorname{lcm}(4, 25) = 100$ which is what we need. We solve starting with the relatively prime part so mod 25. $32^{1032} \equiv 7^{1032} \equiv 7^{1032 \bmod \phi(25)=20} \equiv 7^{12} \equiv (7^2)^6 \equiv 49^6 \equiv (-1)^6 \equiv 1 \pmod{25}$. Now the part where they share a factor is a lot simpler. $32^{1032} \mod 4$. Note that 4 divides 32. Thus 4 divides $32^{1032}$, and thus this part is just $32^{1032} \equiv 0 \pmod 4$. We have transformed this problem into a CRT problem where, $x \equiv 1 \pmod{25}$ and $x \equiv 0 \pmod 4$. Solving we have $x = 25y + 1 = 4z$. Taking mod 4, we have $y + 1 \equiv 0 \pmod 4$. Isolating $y$, we get $y \equiv -1 \equiv 3 \pmod 4$. Plugging in $y = 4a + 3$ we have $x = 25y + 1 = 25(4a + 3) + 1 = 100a + 76$. Thus $x \equiv \boxed{76} \pmod{100}$

**Example 7**

Let $a_1, a_2, \ldots, a_{2018}$ be a strictly increasing sequence of positive integers such that

$$a_1 + a_2 + \cdots + a_{2018} = 2018^{2018}.$$

What is the remainder when $a_1^3 + a_2^3 + \cdots + a_{2018}^3$ is divided by 6? (Source: 2018 AMC 10B Problem 16)

Solution: We need to find $a_1^3 + a_2^3 + \cdots + a_{2018}^3 \mod 6$ given the value of $a_1 + a_2 + \cdots + a_{2018} = 2018^{2018}$. The cubes seem to be really annoying so we maybe try to cube the entire thing we have. Yet that produces too many unwanted turns. We try some experiments. $1^3 \equiv 1 \pmod 6$. $2^3 \equiv 8 \equiv 2 \pmod 6$. $3^3 \equiv 27 \equiv 3$, $4^3 \equiv 64 \equiv 4$, $5^3 \equiv 125 \equiv 5$, $0^3 \equiv 0$. We see that $a^3 \equiv a \pmod 6$, is always true. (Can be proved by subtracting $a$ from both sides and algebraic factoring). Thus what we need to find: $a_1^3 + a_2^3 + \cdots + a_{2018}^3 \equiv a_1 + a_2 + a_3 \cdots + a_{2018} \equiv 2018^{2018} \pmod 6$. So now this problem is $2018^{2018} \mod 6$. We see that 2018 isn't relatively prime to 6 since they share the factor of 2. So we consider, $2018^{2018} \mod 2$, and $2018^{2018} \mod 3$ and then use CRT to combine into mod 6. We start with the relatively prime part so $2018^{2018} \equiv 2^{2018 \bmod \phi(3)=2} \equiv 2^0 \equiv 1 \mod 3$. Next the shared factor part so $2018^{2018} \mod 2$. We see that 2 divides 2018 so 2 divides $2018^{2018}$ so this is just $0 \mod 2$. We have $x \equiv 1 \mod 3$, and $x \equiv 0 \mod 2$. Its pretty easy to guess that 4 satisfies both of these so $x \equiv \boxed{4} \pmod 6$. We also use CRT for practice so $x = 3y + 1 = 2z$. Taking everything mod 2 gives us $y+! \equiv 0 \pmod 2$ and $y \equiv -1 \equiv 1 \pmod 2$. So $y = 2a + 1$, and plugging into $x = 3y + 1$ gives us $x = 3(2a + 1) + 1 = 6a + 4$. So $x \equiv \boxed{4} \pmod 6$.

**Example 8**

If $f(x) = x^{x^{x^x}}$, find the last two digits of $f(17) + f(18) + f(19) + f(20)$. (Source: 2008 PUMAC)

Solution: We calculate each input on $f$ seperately.

Starting with $f(17) = 17^{17^{17^{17}}}$. We need to find the last two digits or mod $100$. $17^{17^{17^{17}}} \equiv 17^{17^{17^{17}} \bmod \phi(100)=40} \bmod 100$. So now the problem becomes solving $17^{17^{17}} \bmod 40$. This is just $17^{17^{17} \bmod \phi(40)=16}$. So now the problem becomes $17^{17} \bmod 16$. $17^{17} \equiv 1^{17} \equiv 1 \pmod{16}$. So we plug this back in so $17^{17^{17} \bmod \phi(40)=16} \equiv 17^1 \equiv 17 \pmod{40}$. So we plug back into $17^{17^{17^{17}} \bmod \phi(100)=40} \equiv 17^{17} \bmod 100$. This is is so computation heavy, and it will be much less computation heavy if we split it up into smaller modulos and use CRT. Well what modulos do we split it up into. We need $2$ values that have an lcm of $100$, and we want to keep them as small as possible so in general we want their product to be low as well. Note that $lcm \times gcd = product$ or $100 \times gcd = product$. So if we want the product to be low, we need the gcd to be low. The minimum possible value of the gcd is $1$. So the gcd is $1$ (general method to break things up with CRT) and the product is $100$. This is only possible by splitting it into $1$, $100$ (we still have to calculate mod $100$ so this doesn't help) or $4$, $25$. We decide to split into $4$, $25$. Taking mod $4$, we have $17^{17} \equiv 1^{17} \equiv 1 \pmod 4$. Taking mod $25$, we have $17^{17} \equiv 17^{17 \bmod \phi(25)=20} \equiv 17^{17}$. Hmm that didn't help. It seems like our only way is computational. It is much less than calculating this same thing mod $100$. PUMAC gives a lot of time, so we bash this out. $17^{17} \equiv 17 * 17^{16} \equiv 17 * ((-8)^2)^8 \equiv 17 * (14^2)^4 \equiv 17 * ((-4)^2)^2 \equiv 17 * (-9)^2 \equiv 17 * 6 \equiv 2 \bmod 25$. So we have something $2$ mod $25$ and $1$ mod $4$. CRT gives us $x = 25y + 2 = 4z + 1$, and taking mod $4$, we have $y + 2 \equiv 1 \pmod 4$, so $y \equiv -1 \equiv 3 \pmod 4$. Plugging in $y = 4a + 3$, we get $x = 25(4a + 3) + 2 = 100a + 77$. So $f(17) \equiv 77 \pmod{100}$. We note that in the beginning we tried to calculate things mod $100$ which led to bigger numbers in the beginning, but still manageable, until the end we had huge numbers so we split into smaller modulos. If we started by splitting into smaller numbers, we would have smaller numbers in the beginning and smaller numbers throughout. So for next time, we will start by splitting into smaller modulos, even if we may think it is manageable in bigger modulos, and even if we will always have relatively prime numbers.

Now we calculate $f(18) \bmod 100$. We have no choice but to split since we don't have relatively prime numbers. $18$ and $100$ share the factor of $2$, so we remove all the $2$'s from $100$ and we split into $4$ and $25$. We start mod $25$. $18^{18^{18^{18}}} \equiv 18^{18^{18^{18}} \bmod \phi(25)=20}$. Now the problem becomes $18^{18^{18}}$ mod $20$. These aren't relatively prime since theyt share $2$, so we split $20$ into $4$ and $5$. Starting with mod $5$, $18^{18^{18}} \equiv 3^{18^{18} \bmod \phi(5)=4}$. Note that $2$ divides $18$, so $2^{18}$ divides $18^{18}$, so $2^2$ divides $2^{18}$ so $2^2$ divides $18^{18}$, so $18^{18}$ is $0$ mod $4$. So $3^{18^{18} \bmod 4} \equiv 3^0 \equiv 1 \pmod 5$. Now we calculate mod $4$, so $18^{18^{18}}$ by the same logic as earlier is just $0$ mod $4$, since many powers of $2$ divide it. We have a number $0$ mod $4$ and $1$ mod $5$ which we can do CRT on to get $16$ mod $20$. So $18^{18^{18}} \equiv 16 \pmod{20}$. So plugging into $18^{18^{18^{18}} \bmod \phi(25)=20} \equiv 18^{16}$ mod $25$. We have to bash again yet we see its very quick now. $18^{16} \equiv (18^2)^8 \equiv (-1)^8 \equiv 1 \pmod{25}$. So we know $f(18)$ is $1$ mod $25$, and by the same logic as earlier since many powers of $2$ divide it, its $0$ mod $4$. By CRT we will obtain $f(18) \equiv 76 \pmod{100}$.

Now we calculate $f(19) \bmod 100$. We don't have to split it but from our lesson learned while calculating $f(17)$ it is best to split it, and to make the gcd $1$, we split into $4$ and $25$. We start by calculating mod 25: $19^{19^{19^{19}}} \equiv 19^{19^{19^{19}} \bmod \phi(25)=20}$. So now we calculate $19^{19^{19}} \bmod 20$. Aha, this is just $19^{19^{19}} \equiv (-1)^{19^{19}} \equiv (-1)^{\text{odd}} \equiv -1 \pmod{20}$. So plugging into $19^{19^{19^{19}} \bmod \phi(25)=20} \bmod 25$, we obtain $19^{-1} \bmod 25$. This is just $\frac{1}{19} \equiv a \pmod{25}$. So we solve for $a$ in the Diophantine Equation, $19a - 25b = 1$. So, $(19, 25) = (19, 19 \cdot 1 + 6) = (19, 6) = (6 \cdot 13 + 1, 6) = (6, 1) = 1$. In reverse, $1 = 6 - 5 \cdot 1 = 6 - 5 \cdot (19 - 3 \cdot 6) = 6 \cdot 16 - 5 \cdot 19 = (25 - 19) \cdot 16 - 5 \cdot 19 = 25 \cdot 16 - 19 \cdot 21$. So $a \equiv -21 \equiv 4 \bmod 25$, is the solution. Now we have to calculate mod 4. $19^{19^{19^{19}}} \equiv (-1)^{19^{19^{19}}} \equiv (-1)^{\text{odd}} \equiv -1 \equiv 3$. So $f(19) \bmod 100$ is 4 mod 25 and 3 mod 4. We use CRT so $x = 25y + 4 = 4z + 3$. Taking mod 4, we have $y \equiv 3 \pmod 4$, so $y = 4b + 3$. Plugging in we get $x = 25(4b + 3) + 4 = 100b + 79$. So $f(19) \equiv 79 \pmod{00}$.

Now we calculate $f(20) \bmod 100$. We realize we can't split into parts that are relatively prime and parts that aren't since every prime that divides 100 divides 20. So we already have all the parts that aren't relatively prime. We see that 4 divides 20 so it divides 20 to any power (Even to the power of $20^{20^{20}}$). We see that 25 divides $20^2$ so it divides $20^2$ to any power (Even to the power of $\frac{20^{20^{20}}}{2}$, which is an integer), so 100 divides $f(20)$ so there is no remainder when divided and $f(20) \equiv 0$ mod 100. Now we add them all up to get $77 + 76 + 79 + 0 \equiv 232 \equiv \boxed{32}$ mod 100.

## General Modular Arithmetic Practice Problems

**Example 9**

Calculate the remainder when 97! is divided by 101.

Solution: By Wilson's theorem, we know $100! \equiv -1$ mod 101. Dividing by $100 \cdot 99 \cdot 98$, we obtain $97! \equiv \frac{-1}{100 \cdot 99 \cdot 98} \equiv \frac{-1}{-1 \cdot (-2) \cdot (-3)} \equiv \frac{1}{6}$ mod 101. Thus we need to find $\frac{1}{6}$ mod 101, or just the solution for $x$ in to $6x - 101y = 1$. We continue, $(6, 101) = (6, 6 \cdot 16 + 5) = (6, 5) = (5 \cdot 1 + 1, 5) = (1, 5) = 1$. Reverse, $1 = 5 - 4 \cdot 1 = 5 - 4 \cdot (6 - 5) = 5 \cdot 5 - 4 \cdot 6 = 5 \cdot (101 - 6 \cdot 16) - 4 \cdot 6 = 5 \cdot 101 - 84 \cdot 6$, so $-84 \equiv \boxed{17} \pmod{101}$.

**Example 10**

Let $x_0, x_1, x_2, \ldots$ be a sequence of numbers, where each $x_k$ is either 0 or 1. For each positive integer $n$, define $S_n = \sum_{k=0}^{n-1} x_k 2^k$ Suppose $7S_n \equiv 1 \pmod{2^n}$ for all $n \geq 1$. What is the value of the sum $x_{2019} + 2x_{2020} + 4x_{2021} + 8x_{2022}$? (Source: 2022 AMC 10B Problem 25)

Solution: Firstly we see that $x_{2019} + 2x_{2020} + 4x_{2021} + 8x_{2022} = \frac{S_{2023} - S_{2019}}{2^{2019}}$. Now we just have to calculate $S_{2023}$ and $S_{2019}$. We start with $S_{2023}$. We know that $7S_{2023} \equiv 1 \pmod{2^{2023}}$, which if we call $S_{2023} = y$ we can convert to $7y - 2^{2023}z = 1$. We want to solve for $y$. We start running the Euclidean Algorithm and see that we need to find the quotient and remainder when $2^{2023}$ is divided by 7, before we can reduce the numbers and go backwards to re-increase them. The reaminder is easy with Euler's Totitent Theorem Extended, $2^{2023} \equiv 2^{2023 \bmod 6} \equiv 2^1 \equiv 2$ mod

7. Now it should also be clear that the quotient is just $\frac{2^{2023}-2}{7}$, so we run the Euclidean Algorithm, $(7, 2^{2023}) = (7, 7\cdot(\frac{2^{2023}-2}{7})+2) = (7, 2) = (3\cdot 2+1, 2) = (1, 2)$. So, $1 = 2-1\cdot 1 = 2-1\cdot(7-3\cdot 2) = 2\cdot 4-7 = (2^{2023}-7\cdot\frac{2^{2023}-2}{7})\cdot 4-7 = 2^{2023}\cdot 4-7\cdot(\frac{2^{2023}\cdot 4-1}{7})$. Thus $y \equiv -(\frac{2^{2023}\cdot 4-1}{7}) \mod 2^{2023}$. We also want to bound $y$ or $S_{2023}$. The smallest possible value is $0$, if all the $x_k$ before it are $0$, and the largest possible value, if all the $x_k$ before it are $1$ is $1+2+4+8+16\cdots+2^{2022} < 2^{2023}$ by sum formulas. Thus, since we have a solution mod $2^{2023}$, our answer is simply the smallest possible positive value of that solution mod $2^{2023}$, by basic bounding since any values less are less than $0$, the minimum, and any values greater are more than $2^{2023}$, the maximum. Our value of $y$ we got was $-(\frac{2^{2023}\cdot 4-1}{7})$, which is negative, so we add $2^{2023}$, to hopefully make it positive, and we get $\frac{2^{2023}\cdot 3+1}{7}$ this is positive, more than the minimum, and less than the maximum, and adding or subtracting $2^{2023}$ will make it more than the max or less than the min so this value is $S_{2023} = \frac{2^{2023}\cdot 3+1}{7}$. We proceed with calculating $S_{2019}$. We have that $7S_{2019} \equiv 1 \mod 2^{2019}$. Which by the same logic as earlier, converts to $7b - 2^{2019}c = 1$ where $b = S_{2019}$. First we calculate the remainder and quotient with Euler's Totient Theorem Extended, and we get $2^{2019} \equiv 2^3 \equiv 8 \equiv 1 \mod 7$. So the remainder is $1$, and the quotient is $\frac{2^{2019}-1}{7}$. With Euclidean Algorithm, we obtain $(7, 2^{2019}) = (7, 7\cdot\frac{2^{2019}-1}{7}+1) = (7, 1) = 1$. Running it in reverse gives us $1 = 7-6\cdot 1 = 7-6\cdot(2^{2019}-7\cdot(\frac{2^{2019}-1}{7})) = 7\cdot(1+6\cdot\frac{2^{2019}-1}{7})-6\cdot 2^{2019}$. Thus $S_{2019} \equiv 1+6\cdot\frac{2^{2019}-1}{7} \equiv \frac{6\cdot 2^{2019}+1}{7} \mod 2^{2019}$. By the same bounding argument as last time, this is the solution. Now $S_{2023}-S_{2019} = \frac{2^{2023}\cdot 3+1-2^{2019}\cdot 6-1}{7} = \frac{2^{2019}\cdot 2^4\cdot 3-2^{2019}\cdot 6}{7} = \frac{2^{2019}\cdot 42}{7} = 2^{2019}\cdot 6$. Dividing by $2^{2019}$ from our expression earlier gives us a final answer of $\boxed{6}$

**Try it Yourself!**

https://mathdash.com/maps/map/modular-arithmetic

https://mathdash.com/maps/map/euler's-totient-theorem

https://mathdash.com/contest/Modular-Arithmetic-I-All-in-One-Problem/

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# Cyclic Polynomials

*by Aaron*

## Introduction

Thanks to John Z. for helping with LaTeX and clarifying some confusing parts!

**Definition 1**

A symmetric polynomial is a polynomial that remains unchanged under the exchange of any pair of variables. In other words, swapping any two variables in the polynomial does not affect its form. For example, $x+y+z$ and $x^2(y+z)+y^2(z+x)+z^2(x+y)$ are symmetric polynomials since switching any two pairs of variables leaves the polynomial unchanged.

**Definition 2**

A cyclic polynomial is a polynomial that remains unchanged under cyclic permutations of its variables, which means that if you rearrange the variables in a circular fashion, the polynomial expression stays the same; essentially, it exhibits rotational symmetry among its variables. For example, $x^2y+y^2z+z^2x$ is cyclic, but is not symmetric. (Do you see why?)

All symmetric polynomials are cyclic; however, not all cyclic polynomials are symmetric.

**Theorem 1**

The factorization of a cyclic polynomial must also be cyclic. In addition, the addition, subtraction, multiplication, and division of two cyclic polynomials always result in a cyclic polynomial.

**Theorem 2**

If the polynomial equation $P(x)=a_nx^n+a_{n-1}x^{n-1}+a_{n-2}x^{n-2}+\cdots+a_1x+a_0=0$ has root $x=r$, then $(x-r)$ is a factor of $P(x)$.

**Theorem 3**

For a two-variable polynomial equation $P(x, y) = 0$, if $x = -y$ is a solution, then $(x + y)$ is a factor of $P(x, y)$.

**Theorem 4**

For a three-variable polynomial equation $P(x, y, z) = 0$, if $x = -y - z$ is a solution for $x$, then $(x + y + z)$ is a factor of $P(x, y, z)$.

Sometimes, cyclic polynomials can be lengthy and difficult to write out completely. Because of this, we have a shorter notation for cyclic polynomials, for example:

$$P(x, y, z) = x(y - z) + y(z - x) + z(x - y) = \sum_{\text{cyc}} x(y - z)$$

## Factoring

Let's try some examples. Be sure to attempt the questions on your own before looking at the solution.

**Example 1**

Factor $(x + y + z)^5 - x^5 - y^5 - z^5$.

**Solution.** Since the expression is cyclic, we can start by identifying some simple factors.

If we set $x = -y$, the expression simplifies to $z^5 - x^5 + x^5 - z^5 = 0$. Therefore, by Theorem 3, $(x + y)$ is a factor, which implies that $(y + z)$ and $(z + x)$ are also factors.

We know that $(x+y+z)^5 - x^5 - y^5 - z^5$ is a $5^{\text{th}}$-degree polynomial so what's left is a second-degree polynomial. In addition, by Theorem 1, we know that the rest of the stuff must be cyclic, and the only factor to achieve those two things is in the form of $a(x^2+y^2+z^2)+b(xy+yz+zx)$. Therefore, the expression $(x + y + z)^5 - x^5 - y^5 - z^5$ can be factored as:

$$(x + y)(y + z)(z + x)(a(x^2 + y^2 + z^2) + b(xy + yz + zx)).$$

By substituting $x = y = 1$ and $z = 0$, we find that $30 = 2(2a + b)$. Similarly, for $x = 2$, $y = 1$, and $z = 0$, we have $210 = 6(5a + 2b)$. Solving these equations, we find that $a = b = 5$.

Thus, the final answer is:

$$\boxed{5(x + y)(y + z)(z + x)(x^2 + y^2 + z^2 + xy + yz + zx).}$$

**Example 2**

Factor $(a+b+c)^3-(b+c-a)^3-(a+c-b)^3-(a+b-c)^3$.

**Solution.** The expression is cyclic, so we aim to identify some simple factors.

If we set $a=0$, the value becomes $(b+c)^3-(b+c)^3-(c-b)^3-(b-c)^3=0$. This indicates that the expression has a factor of $a$, which implies it also has a factor of $abc$.

Since it is a cubic expression and we already have a degree of $3$ in $abc$, the complete factorization must take the form $kabc$. When we set $a=b=c=1$, we find that $k=24$. Therefore, the final result is $\boxed{24abc}$.

**Example 3**

Factor $(x+y)^5-x^5-y^5$.

**Solution.** If $x+y=0$, then the expression evaluates to 0 because $(x+y)^5=0^5=0$ and $x^5+y^5$ has a factor of $x+y$, making it $0$. Similarly, if $x=0$, the expression is also 0. This indicates that $xy(x+y)$ is a factor of the expression.

The remaining part must be quadratic. Therefore, we can express it as $xy(x+y)(ax^2+ay^2+bxy)$.

To find the values of $a$ and $b$, we can use the following conditions:

1. If $x=y=1$, substituting gives us $30=2(2a+b)$.
2. If $x=2$ and $y=-1$, we find $-30=-2(5a-2b)$.

Solving these equations leads to $a=5$ and $b=5$.

Thus, the final expression is $\boxed{5xy(x+y)(x^2+y^2+xy)}$.

**Example 4**

Factor $a^3+b^3+c^3-a(b-c)^2-b(c-a)^2-c(a-b)^2-4abc$.

**Solution.** If $a=b+c$, the expression becomes

$$(b+c)^3+b^3+c^3-(b+c)(b-c)^2-b^3-c^3-4(b+c)bc=0.$$

Thus $(b+c-a)$ is a factor, and hence $(a+c-b)$ and $(a+b-c)$ are also factors.

Since it is a 3rd power expression, we must have: $k(b+c-a)(a+c-b)(a+b-c)$. By checking the coefficient of $a^3$, $k=-1$. Hence the answer is: $\boxed{-(b+c-a)(a+c-b)(a+b-c)}$.

**Try the Problems**

https://mathdash.com/contest/john-problem-1737261586390

https://mathdash.com/contest/ifps

# Complex Numbers

*by Sid*

https://mathdash.com/maps/aime-complex-numbers-and-polynomials

**Definition 1** (Complex Numbers)

Complex numbers are numbers that can be expressed as $a + bi$, where $i = \sqrt{-1}$. $a$ is the real part, and $bi$ is the imaginary part. Note that $i^2 = -1$, $i^3 = -i$, and $i^4 = 1$, and after that it repeats the cycle with $i^5 = i$.

**Theorem 1** (Multiplying Complex Numbers)

$(a + bi)(c + di) = (ac - bd) + (bc + ad)i$

**Definition 2** (Complex Conjugate)

One of the most useful properties of complex numbers is the complex conjugate, represented as $\overline{z}$, where $z$ is the original complex number. The complex conjugate can be calculated as $\overline{z} = a - bi$.

**Definition 3** (Complex Plane)

Complex numbers can be graphed on the complex plane, where the value of $a$ is plotted on the x-axis, and the value of $b$ is plotted on the y-axis. The magnitude of a complex number, represented as $|z|$, is the distance from $(0, 0)$, the origin to $(a, b)$, the complex number.

**Theorem 2**

$|z| = \sqrt{a^2 + b^2} = \sqrt{z\overline{z}}$

**Theorem 3** (Fundamental Theory of Algebra)

Any non zero, single-variable, degree $n$ polynomial has exactly $n$ complex roots, counting multiplicity.

**Theorem 4**

Let $F(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$, where all $a_n$ are real numbers. If $z = a + bi$ is a root of $F(x)$, then $\overline{z} = a - bi$ is also a root.

**Example 1**

Let $z$ be a complex number, given that $z^2 = 12 + 16i$. Find $z$, given that the real part of $z$ is positive.

**Solution:** Let $z = a + bi$. We know that $a^2 - b^2 + 2abi = 12 + 16i$. Since $a^2 - b^2$ must be real, and $2abi$ must be imaginary, we can create two separate two-variable equations:

$$2abi = 16i$$
$$a^2 - b^2 = 12$$

For the first equation, we can divide by $2i$ from both sides to get $ab = 8$. From here, we can solve the two variable equation, but it is easy enough to see that $a = 4$, and $b = 2$, or $a = -4$ and $b = -2$ satisfy the equations. We know that the real part must be positive by the problem conditions, so our answer is $\boxed{z = 4 + 2i}$.

**Example 2**

Let $N$ be the number of complex numbers $z$ such that $|z| = 1$, and $z^{720} - z^{120}$ is a real number. Find the remainder when $N$ is divided by $1000$. (2018 AIME I #6)

**Solution:** Since the result is a real number, you know that the $z^{720} - z^{120} = \overline{z}^{720} - \overline{z}^{120}$. Additionally, we know that the magnitude of $z$ is $1$, so $z\overline{z} = 1^2 = 1$, and $z = \frac{1}{\overline{z}}$. Now, we substitute it back in, getting

$$z^{720} - z^{120} = \frac{1}{z^{720}} - \frac{1}{z^{120}}$$

Multiply by $z^{720}$ so all the exponents are positive, and we realize that we have a single variable polynomial with degree $1440$, meaning there should be $1\boxed{440}$ solutions.

**Definition 4** (Argument)

The argument ($\theta$) of a complex number is represented as the angle the complex number makes with the positive real axis (x-axis) and the line from the origin to the complex number.

**Definition 5** (Polar Form)

The polar form is represented with the magnitude and the argument.

**Theorem 5** (Argument)

$\tan\theta = \frac{b}{a}$

**Theorem 6**

$z = a + bi = r(\cos\theta + i\sin\theta) = r\,\mathrm{cis}\,\theta$, where $\mathrm{cis}\,\theta$ is just short form for $\cos\theta + i\sin\theta$.

**Theorem 7** (Euler's Formula)

$r\,\mathrm{cis}\,\theta = re^{i\theta}$. This is the root of arguably the most famous equation in maths, $e^{i\pi} = -1$.

**Theorem 8** (Roots of Unity)

For $n \geq 2$ and $r \neq 0$, we have that $z^n = r$ will have exactly $n$ distinct solutions of the form $e^{2k\pi i/n}$ for $k = 1, 2, 3, \ldots, n$. Geometrically, the solutions can be expressed on a circle with radius $r^{1/n}$ as a regular $n$-gon (or a line for $n = 2$) with radius.

**Theorem 9** (De Moivre's Theorem)

If $z = re^{i\theta}$, and $n$ is real, then $z^n = r^n e^{ni\theta}$

**Example 3** (De Moivre's Theorem)

Find $(3 + 3\sqrt{3}i)^{16}$.

**Solution:** First, we calculate the magnitude to be $\sqrt{36} = 6$. Now, we convert to polar form. We see that that $b/a = \sqrt{3}$, meaning that the angle must be $\frac{\pi}{3}$ radians. This means that $z = 6e^{i\pi/3}$. Now, taking this to the 16th power, we get $6^{16}e^{i16\pi/3}$. Converting this to $\mathrm{cis}\,\theta$, we get

$$6^{16}(\cos 16\pi/3 + i\sin 16\pi/3)$$

Now, we evaluate $\cos 16\pi/3 = -0.5$, and $\sin 16\pi/3 = -\frac{\sqrt{3}}{2}$. Now, converting back to $a + bi$, we get

$$\boxed{z^{16} = -\frac{6^{16}}{2} - \frac{6^{16}\sqrt{3}}{2}}$$

**Example 4**

Given that $z_1 + z_2 + z_3 = 0$, and $|z_1| = |z_2| = |z_3| = 1$, evaluate $z_1z_2 + z_2z_3 + z_3z_1$.

**Solution:** We know that the magnitude is equal to $1$, meaning they lie on the unit circle. WLOG, fix $z_1$ to the top. That means that $z_2$ and $z_3$ must have imaginary parts of $-0.5$ so the real parts cancel out too. Since $\sin 330 = \sin 210 = -0.5$, it means that they are both $30°$ from the real axis, meaning they are equally spaced on the unit circle. This means that $z_1, z_2,$ and $z_3$ can be the solutions to the equation $\omega^3 = 1$, with $z = 1$, $z_2 = \omega$, and $z_3 = \omega^2$. Now, we get our product in terms of $\omega$ to be $\omega + \omega^2 + \omega^3$. This is just $z_1 + z_2 + z_3$, meaning our answer is $\boxed{0}$.

**Example 5**

Given that $z$ is a complex number such that $z + \frac{1}{z} = 2\cos 3°$, find the least integer that is greater than $z^{2000} + \frac{1}{z^{2000}}$. (2000 AIME II #9)

**Solution:** We express $z$ to be on the unit circle in the complex plane. Then, $z = 1 * e^{i\theta} = e^{i\theta}$. Note that $\frac{1}{z} = z^{-1} = e^{-i\theta}$. Expressing these in $\operatorname{cis}\theta$ form, and considering that $z + \frac{1}{z} = 2\cos 3$ we get

$$\cos\theta + i\sin\theta + \cos\theta - i\sin\theta = 2\cos 3\cos\theta = \cos 3\theta = 3$$

Now, we evaluate $z^{2000} + z^{-2000}$ using De Moivre's, noting that the $\sin$ terms will cancel out, getting $2\cos 6000 = 2\cos 240 = -1$, meaning that our answer is $\boxed{000}$.

**Practice Problems:** (Solutions at end of chapter)

**Example 6**

The equation $z^6 + z^3 + 1 = 0$ has complex roots with argument $\theta$ between $90°$ and $180°$ in the complex plane. Determine the degree measure of $\theta$. (1984 AIME #8)

**Example 7**

There are $24$ different complex numbers $z$ such that $z^{24} = 1$. For how many of these is $z^6$ a real number? (2017 AMC 12A #17)

**Example 8**

Let $z = a + bi$ be the complex number with $|z| = 5$ and $b > 0$ such that the distance between

$(1+2i)z^3$ and $z^5$ is maximized, and let $z^4 = c+di$. Find $c+d$. (2013 AIME II #6)

**Example 9**

Let $S$ be the set of all polynomials of the form $z^3 + az^2 + bz + c$, where $a$, $b$, and $c$ are integers. Find the number of polynomials in $S$ such that each of its roots $z$ satisfies either $|z| = 20$ or $|z| = 13$. (2013 AIME II #12)

**Try the Problems (MAPS)**

**https://mathdash.com/maps/aime-complex-numbers-and-polynomials**

**Solutions:**

**#6:** Let $y = z^3$. Now, we get the quadratic

$$y^2 + y + 1 = 0$$

Using the quadratic equation, we get the solution to be $y = -\frac{1}{2} \pm \frac{\sqrt{3}i}{2}$. The argument is $120°$ and $240°$ respectively, and they both have a magnitude of $1$. This means that we can write it as $z^3 = \cos 240° + i\sin 240°$ and $z^3 = \cos 120° + i \sin 120°$. Taking both to the power $\frac{1}{3}$, we get $\cos 40° + i\sin 40°$, and $\cos 80° + i \sin 80°$. Now, we know that the argument repeats in intervals of $120°$ for both, since $360/3 = 120$, so the possible roots would be $40°$, $160°$, $280°$, $80°$, $200°$, and $320°$. The only one between $90°$ and $180°$ is $\boxed{160}$.

**#7:** Let $a = z^6$. Since $z^{24} = 1$, $a^4 = 1$, and since $a$ must be real, $a = \pm 1$. This means that $z^6 = 1$, or $z^6 = -1$. By Fundamental Theory of Algebra, we know there are exactly $6$ complex solutions to both these cases, meaning the answer is $\boxed{12}$.

**#8:** We need to maximize $|z^5 - z^3(1+2i)$. Factoring out the $z^3$, we need to maximize $z^2 - (1+2i)$, which is obviously achieved if they are opposite each other. Since $|z| = 5$, $|z^2| = 25$. Currently, $1+2i$ has a magnitude of $\sqrt{5}$, so we have to multiply it by $5\sqrt{5}$ to transform it into the circle with radius $25$ in the complex plane, which means that it becomes $\sqrt{5}(5+10i)$. Since they are opposite, it means that $z^2 = -\sqrt{5}(5+10i)$. Squaring this, we get $-375 + 500i$, meaning that the answer is $\boxed{125}$.

**#9:** We split it into two cases:

**Case 1:** All the roots are real. Since the magnitude is $20$ or $13$, the possible values are $20$, $-20$, $13$, $-13$. We use stars and bars to evaluate that the amount of ways is $\binom{6}{3} = 20$.

**Case 2:** Two of the roots are imaginary and one is real (Note that because of Theorem 4, the two imaginary roots must be complex conjugates). There are $4$ ways to choose the real roots. For the imaginary roots, let one of the imaginary roots be $m + ni$, and the other be $m - ni$, and denote the real number to be $r$. We know that $\sqrt{m^2 + n^2} = 20$ or $13$. Applying Vietta's equations, we get

$$a = -(2m + r)$$
$$b = m^2 + n^2 + 2mr$$
$$c = -(m^2 + n^2)$$

Since $r$ is an integer, we can ignore that. Additionally, since $\sqrt{m^2 + n^2} = 20$ or $13$, we know that $m^2 + n^2$ is an integer too. Now, we see that the only way that will make $a$, $b$, and $c$ all integers is if $2m$ is an integer. Since $|m|$ has to be less than 20 or 13 (depending on what the magnitude is) (since otherwise $n \leq 0$, which doesn't work), that means that if the magnitude is $13$, then the possible values for $m$ will be $0, \pm\frac{1}{2}, \pm\frac{2}{2}, \pm\frac{3}{2}, \cdots \pm \frac{25}{2}$, meaning there are $51$ ways. Similarily, if the magnitude is $20$, there will $79$ ways. This means in total (for this case), there are $130 * 4 = 520$. Adding up the two cases, we get $520 + 20 = \boxed{540}$.

# Note about $\sum$ and $\prod$ Notation

*Mathematicians just love sigma notation ($\sum$) for two reasons. First, it provides a convenient way to express a long or even infinite series. But even more important, it looks really cool and scary, which frightens nonmathematicians into revering mathematicians and paying them more money.*

— *Caculus II for Dummies, p. 51*

*$\sum$-notation and $\prod$-notation are used later in this book. In case you don't know, here's what they mean.*

**Definition 1** (Capital Sigma Notation)

$\sum_{k=n}^{m} f(k)$ means to take the sum where the terms are given by plugging in different $k$s into $f(k)$, starting from $k = n$ to $k = m$, with an increment of $1$. Note that the part that follows the summation does not have to be a function; it can simply be an expression involving $k$. $k$ can be whatever letter you want, it just has to match the letter in the expression later. It's called a **dummy variable**.

This notation can be used to represent identities concisely. For example,

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k$$

(binomial theorem, theorem 1 in Generating Functions)

If the upper bound is infinity we use $\infty$. An upper bound of infinity basically means the limit of the sum as $n$ gets arbitrarily large. The following means that $ar^0 + ar^1 + ar^2 + \cdots = \frac{a}{1-r}$:

$$\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}, \quad \text{for } |r| < 1$$

**Definition 2** (Capital Pi Notation)

$\prod$ works the same way except it's product, not sum.

A few more examples:

$$\sum_{k=3}^{7} k^2 = 3^2 + 4^2 + 5^2 + 6^2 + 7^2$$

$$\sum_{k=1}^{n} k = 1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$$

$$n! = \prod_{k=1}^{n} k$$

**Exercise.** Write the formula in $\sum$ notation: The sum of the first $n$ perfect cubes is equal to the sum of the first $n$ positive integers squared.

# Linearity of Expectation

*by MathDash Team*

https://mathdash.com/maps/map/linearity-of-expectation

**Definition 1** (Expected Value)

Let $E[A]$ denote the expected value of the random variable $A$. This is defined as the sum, over all possible values $x$ of the random variable $A$, of $x$ multiplied by the probability that the variable is equal to $x$ — or specifically, $E[A] = \sum_x P(A = x) \cdot x$

**Example 1**

A six-sided die is rolled, having sides $1$, $2$, $3$, $4$, $5$, $6$. What is the expected value of the result of the roll?

**Solution:** We can use the definition of expected value — let $A$ be the random variable for the result of the dice roll. We know that the expected value of $A$ is

$E[A] = P(A = 1) \cdot 1 + P(A = 2) \cdot 2 + P(A = 3) \cdot 3 + P(A = 4) \cdot 4 + P(A = 5) \cdot 5 + P(A = 6) \cdot 6$

$P(A = i) = \frac{1}{6}$ for each $i = 1, 2, 3, 4, 5, 6$, so this simplifies to simply $A = \frac{1}{6}(1+2+3+4+5+6) = \frac{21}{6} = 3.5$.

That was easy enough. Now let's look at a very basic example of Linearity of Expectation

**Example 2**

If a random number $a$ has an expected value of $3$, and a random number $b$ has an expected value of $4$, what is the expected value of $a + b$?

Turns out that you can easily guess than the answer would be $3 + 4 = \boxed{7}$. This is precisely what Linearity of Expectation says — for any two random variables $A$, $B$, we have that $E[A + B] = E[A] + E[B]$. More generally,

**Theorem 1** (Linearity of Expectation)

For any random variables $A_1$, $A_2$, $A_3$, $\ldots$, $A_n$,

$$E[A_1 + A_2 + \cdots + A_n] = E[A_1] + E[A_2] + E[A_3] + \cdots + E[A_n].$$

In other words, the expected value of the sum is the sum of the expected values.

Easy enough — why is this even a theorem you may ask — it seems very obvious! Of course the expected value of a sum should be the sum of the expected values. Well, let's take a look at the next problem

**Example 3**

(Classic) $n$ students take a test. Afterwards, the teacher decides to randomly scramble the tests and give each student one of the tests back. What is the expected number of students who get their own test back?

What an impossible looking question — trying to do this naively would mean we would need to count the number of ways to give the tests back such that one student got their own test back, two students, three students, etc. Each of those cases are really tricky problems in their own right.

**Solution:** Let's use the power of **indicator variables** to help us solve this. Let $A_i$ be a random variable defined as follows:

$$A_i = \begin{cases} 1, & \text{if the } i\text{-th student receives their own test} \\ 0, & \text{otherwise} \end{cases}$$

First, I claim that the number of students who get their own test back is $A_1 + A_2 + \cdots + A_n$. Why? This is because $A_i$ will be $1$ for each student that gets their own test back, and $0$ otherwise — so the sum will count the number of students who get their own test back.

Perfect — so we just need to find the expected value of this sum, or

$$E[A_1 + A_2 + \cdots + A_n]$$

Look familiar? We'll use linearity of expectation — this is equal to

$$E[A_1] + E[A_2] + \cdots + E[A_n]$$

What is $E[A_i]$ (for an arbitrary $i$)? Well, the definition of expected value tells us this is $0 \cdot P(A_i = 0) + 1 \cdot P(A_i = 1)$ — so this is just $P(A_i = 1)$, or the probability that the $i$—th student receives their own test. Well — we know what that is — the probability that the $i$-th student receives their own test is $\dfrac{1}{n}$.

So, $E[A_i] = \frac{1}{n}$, and

$$E[A_1] + E[A_2] + \cdots + E[A_n] = \frac{1}{n} + \frac{1}{n} + \cdots + \frac{1}{n} = \boxed{1}$$

In other words, the expected number of students who get their own test back is $1$.

Linearity of Expectation is one of the most beautiful and satisfying tricks to use in contest math! Hopefully you were impressed by the ease at which Linearity of Expectation helped solve that otherwise quite impossible question.

The perhaps surprising part of Linearity of Expectation is that it holds *even when the random variables depend on each other*! In this case, $A_1$ depends on $A_2$ — for example, whether or not person $1$ receives their own test back definitely impacts the probability that person $2$ receives their own test back. However, we can still use Linearity of Expectation here because it holds in general, not only for independent events.

(As a sidenote, for independent random variables $A_1$, $A_2$, $E[A_1A_2] = E[A_1]E[A_2]$ — but this is not true if the random variables depend on each other).

**Example 4**

Ten people of different heights stand in a line in a random order. A person 'looks tall' if they are taller than both the person to the left and to the right of them (or, if they are on the edge, taller than the one person next to them). What is the expected number of people who look tall?

**Solution:** We will use the trick of **indicator variables** yet again! Let $A_i$ be the random variable defined as such:

$$A_i = \begin{cases} 1, & \text{if the } i\text{-th person in line looks tall} \\ 0, & \text{otherwise} \end{cases}$$

Note that the number of people who look tall is $A_1 + A_2 + \cdots + A_{10}$. So we just need to find the expected value of this sum, or

$$E[A_1 + A_2 + \cdots + A_{10}]$$

Which by Linearity of Expectation is equal to

$$E[A_1] + E[A_2] + \cdots + E[A_{10}]$$

Now, how do we find this sum? As in the previous problem, $E[A_i]$ is equal to $0 \cdot P(A_i = 0) + 1 \cdot P(A_i = 1)$ — so this is just $P(A_i = 1)$, or the probability that the $i$-th person in line looks tall.

Let's try to figure out what $P(A_1 = 1)$ is — this is the probability that the first person in line 'looks tall'. Well — each possible random ordering of people can be paired up with another random ordering with the position of the first two people swapped. In exactly one of the two of these, the first person in line 'looks tall' and in the other one, they don't! So, in exactly $\frac{1}{2}$ of the random orderings, the first person in line looks tall. Thus, $P(A_1 = 1) = \frac{1}{2}$. Similarly, $P(A_{10} = 1) = \frac{1}{2}$.

$P(A_2 = 1)$ is a little different — this is the probability that the second person in line 'looks tall'. We can use a similar idea though — we group all of the orderings into groups of $6$ — the group is determined by the ordering of the participants who are not in the $1$, $2$, or $3$ slot. Each group of six is the same ordering except with the order of $1$, $2$, $3$ changed to be one of the $6$ possible orderings. In exactly two of these orderings, the second person in line 'looks tall' — so across all permutations, $\dfrac{1}{3}$ of them have the second person in line 'look tall'. Thus, $P(A_2 = 1) = \frac{1}{3}$, and similarly, $P(A_i = 1) = \frac{1}{3}$ for $i = 2, 3, \cdots, 9$

So, the expected value of the number of people who look tall is $E[A_1 + A_2 + \cdots + A_{10}]$ which is

$$\begin{aligned} E[A_1] + E[A_2] + \cdots + E[A_{10}] &= \frac{1}{2} + \frac{1}{3} + \frac{1}{3} + \cdots + \frac{1}{3} + \frac{1}{2} \\ &= 1 + \frac{8}{3} \\ &= \boxed{\frac{11}{3}}. \end{aligned}$$

You can imagine trying to do this question without linearity of expectation — you'd have to count the number of permutations of $10$ people for which there are $0$ people who look tall, $1$ person who looks tall, $2$ people who look tall, $3$ people who look tall, etc. After performing an incredibly large number of calculations, you would arrive at a final answer of $\frac{11}{3}$ as well. Linearity of Expectation saves us a lot of time!

**Try some Linearity of Expectation Problems**

**https://mathdash.com/maps/map/linearity-of-expectation**

# Recursion

*by Sepehr Golsefidy and Aditya Bisain*

https://mathdash.com/maps/map/recursion

## Introduction

States and Recursion are both fundamental to AMC 10/12 or AIME Problems. They can be a helpful tool to get rid of large amounts of casework, which can often lead to silly mistakes. But how exactly do these tools work?

**Recursion** The idea of recursion is that we have a sequence, say, $a_n$, where $n$ represents the nth term, and we want to be able to express $a_n$ in terms of the previous terms of this sequence, such as $a_{n-1}$, $a_{n-2}$, etc. A very famous example of a this is the Fibonacci sequence, where the nth term of this sequence is denoted by $F_n$, and $F_n$ is defined by the sum of the previous two terms. So, $F_n = F_{n-1} + F_{n-2}$. We call this a recurrence relation.

Let's start with a basic problem.

**Example 1**

Determine the maximum number of regions formed by $n$ lines in the plane.

We can guess by checking small cases that the pattern adds $1$, $2$, $3$, and so on, so the final answer should be $\frac{(n)(n+1)}{2} + 1$, but how are we going to prove it?

## Solution

Let's play around with this. If we have $0$ lines, we have $1$ region. With $1$ line, this splits old region up into $2$ regions. When we add another line, first, notice that we never want lines to be parallel because we want to maximize the number of regions formed. If we draw a 2nd line that is parallel to the first one, we only create 3 regions. If we have it intersect the first line, then it creates 4 regions.

At this point you might've guessed that the answer is $2^n$, where $n$ represents the number of lines we draw. But once we add a third line, we see that the maximum regions we can form is 7...

The idea is that when we add the third line, it can intersect at most the two other lines. The 2 intersections splits up the third line into three separate chunks, and each of these chunks splits up an existing region into 2 regions. So in general, if we have $n$ lines, and we add one more line, this new line will have at most $n$ intersection points, splitting the line up into $n+1$ segments, forming $n+1$ new regions in addition to the previously existing regions.

So, our recurrence relation is $a_{n+1} = a_n + (n+1)$, where $a_n$ represents the maximum number if regions with $n$ lines.

Now lets try to get a closed form for $a_n$. We have that $a_n = n + a_{n-1}$. By our recurrence relation, we can plug in $a_{n-1} = n - 1 + a_{n-2}$, which yields $a_n = n + (n+1) + a_{n-2}$. Plugging in $a_{n-2} = n-2+a_{n-3}$, which gives $a_n = n+(n-1)+(n-2)+a_{n-3}$. At this point, we start to see the triangular number pattern. From here, we can see that $a_n = n + (n-1) + (n-2) + \cdots + 1 + a_0 = \frac{n(n+1)}{2} + 1$, and we are done. (Note: the $+1$ at the end comes from the fact that $a_0 = 1$, which represents that there is 1 region when we draw 0 lines).

The process of recursion, as seen in the previous solution, has two major steps: forming a relation and solving it. Let's try another example of this.

**Example 2**

How many sequences of length $n$ are there consisting of digits in the set $0, 1$ such that there are no two consecutive 0's in this sequence?

## Solution

**Forming the Recurrence:** Note that there are two cases: the sequence ends with a $0$ or a $1$. If the length of the sequences with length $i$ are denoted with $a_i$ if the sequence ends with $0$, and $b_i$ if otherwise. Notice that if this sequence ends in a zero, the previous digit must be 1. As a result, $b_i = a_{i-1}$. If the sequence ends in a 1, the previous digit might be either a 1 or a 0. Thus, $a_i = b_{i-1} + a_{i-1}$.

**Solving:** We try to make the entire thing in terms of one sequence. In this case, it's simple to set $b_{i-1} = a_{i-2}$, and so $a_i = a_{i-1} + a_{i-2}$. Now, if we're looking at the possibility for the $n$ digit sequence, we must find $a_n + b_n = a_n + a_{n-1} = a_{n+1}$. Note that since $a_i$ is defined as the Fibonacci Sequence, we get $a_{n+1} = F_{n+1}$, where $F_{n+1}$ is the $n+1$th term in the sequence. This is our final answer.

We'll cover one more example and then go into our next topic.

**Example 3** (AIME II 2015 P10)

Call a permutation $a_1$, $a_2$, $\ldots$, $a_n$ of the integers $1$, $2$, $\ldots$, $n$ quasi-increasing if $a_k \le a_{k+1} + 2$ for each $1 \le k \le n-1$. For example, $53421$ and $14253$ are quasi-increasing permutations of the integers $1$, $2$, $3$, $4$, $5$, but $45123$ is not. Find the number of quasi-increasing permutations of the integers 1, 2, . . . , 7.

## Solution

We will solve this for $n$ integers.

Lemma: If we call the number of permutations possible for $i$ integers $a_i$, the number of permutations possible for $i+1$ integers is $a_{i+1} = 3a_i$.

Proof: How many ways are there to place the $i+1$ after the $i$ terms? Only before the $i-1$, $i$, and right at the end. Thus, our result is proven.

Before we move to states, I would like to cover a small bit of the general formula of a recursive sequence.

## Characteristic Polynomials

**Definition 1**

If a recurrence equation

$$x_n = c_1x_{n-1} + c_2x_{n-2} + \cdots + c_kx_{n-k}, \qquad (*)$$

exists, define the characteristic polynomial of this recurrence be $f(x) = x^n - c_1x^{n-1} + c_2x^{n-2} + \cdots + c_k$.

Why is this useful? This will be shown soon.

**Example 4**

If a recurrence $G$ is defined with $G_n = c_1G_{n-1}+c_2G_{n-2}+\cdots+c_kG_{n-k}$, define the characteristic polynomial of this recurrence be $f(x) = x^n - c_1x^{n-1} + c_2x^{n-2} + \cdots + c_k$. Then the term $G_n$ can be written as $a_1({r_1}^n) + \cdots + a_n({r_n}^n)$ if $r_i$'s are the roots to our characteristic polynomial, and if the characteristic polynomial has distinct roots.

## Solution

Define generating function $H(x)$ such that $H(x) = G_0 + G_1(x) + \ldots + G_i(x^i) + \ldots$. Here comes the genius step: consider $\sum_{j=1}^{k} c_jH(x)x^{j-1}$. If we observe the $x^{k+i}$ (where $i$ is an integer greater than or equal to $-1$) term in the expansion, we observe that the sum over this is $c_kG_{i+1} + c_{k-1}G_{i+2} + \ldots +$

$c_1G_{i+k}$. But amazingly, this simplifies to $G_{k+i+1}$. But this is the coefficient we get if we divide $\frac{H(x)}{x}$. As a direct result, $\sum_{j=1}^{k} c_jH(x)x^{j-1} = \frac{H(x)+R(x)}{x}$, where $R(x)$ is a polynomial of degree less than or equal to $k-1$.

Through this, we multiply by $x$ on both sides, subtract by $H(x)$ on both sides, and factor $H(x)$ to get $H(x) = \frac{R(x)}{c_kx^k+c_{k-1}x^{k-1}+\cdots+c_1x-1}$. But the denominator is just a flip of the characteristic polynomial! So the roots of the denominator are $\frac{1}{r_i}$ for $i$ ranging from $1$ to $n$. Since the roots of the characteristic polynomial are distinct, we write the partial fraction decomposition of $H(x)$ as above: it can be written as $\sum_{m=1}^{k} \frac{n_m}{1-r_mx}$. We are using the fact that the degree of $R(x)$ is less than or equal to $k-1$ here, and that the denominators divide the other denominator. Now, we use the geometric series formula and compare with the coefficients of $H(x)$. As a result, $G_n = a_1{r_1}^n + a_2{r_2}^n + \cdots + a_k{r_k}^n$, and we have proved our final result.

I implore you to find another result but if the characteristic polynomial has a root with multiplicity more than 1.

This may seem complicated, but it gets more simple through simple numerical examples.

**Example 5**

Prove Binet's formula, which states that $F_n = \frac{(\frac{1+\sqrt{5}}{2})^n - (\frac{1-\sqrt{5}}{2})^n}{\sqrt{5}}$.

## Solution

Fibonacci's characteristic polynomial is $x^2 - x - 1 = 0$. The roots are $\frac{1+\sqrt{5}}{2}$, $\frac{1-\sqrt{5}}{2}$. Now, we look for our coefficients such that $a(\frac{1+\sqrt{5}}{2})^n + b(\frac{1-\sqrt{5}}{2})^n$, and so $a+b=0$, $a+a\sqrt{5}+b-b\sqrt{5}=2$. Solving this and plugging back in, we get the final result.

**Example 6**

(2018 USAJMO P1) For each positive integer $n$, find the number of $n$-digit positive integers that satisfy both of the following conditions:

- no two consecutive digits are equal, and
- the last digit is a prime.

## Solution

Look at the second digit from the left, and call the amount of number of $n$ digit numbers that satisfy this property $a_n$. If it is 0, then the number of $n-2$ digit numbers times 9 is the total possibility. If not, it is $n-1$ digit numbers times 8. Thus, our recurrence is $a_n = 8a_{n-1}+9a_{n-2}$. The characteristic polynomial is $x^2 - 8x - 9$, so we must find constants $a$ and $b$ such that $a(9)^n + b(-1)^n$. Finding the cases where $n = 1$ and $n = 2$, we find that $9a - b = 4$, $81a + b = 32$. Add the two expressions up, find $a$ and $b$, and finish.

**Try some Recursion Problems (MAPS)**

**https://mathdash.com/maps/map/recursion**

# Generating Functions

*by MathDash Team*

https://mathdash.com/maps/map/generating-functions

Generating functions are a powerful tool in combinatorics that allow us to encode sequences of numbers as coefficients in a power series. By manipulating these power series, we can solve some of the most difficult combinatorics and algebra problems in competition math!

**Example 1**

There are three baskets on the ground: one has 2 purple eggs, one has 2 green eggs, and one has 3 white eggs. Eggs of the same color are indistinguishable. In how many ways can I choose 4 eggs from the baskets?

## Solution

Let's express the conditions as polynomials. This may seem crazy at first, but bear with me! Since we can choose either $0$, $1$, or $2$ eggs from the first basket (purple eggs), we can represent this as:

$$x^0 + x^1 + x^2$$

Similarly, we can represent the choices for the green eggs:

$$x^0 + x^1 + x^2$$

For the white eggs, since we can choose $0$, $1$, $2$, or $3$ eggs, we represent this as:

$$x^0 + x^1 + x^2 + x^3$$

Now, consider what happens when we multiply these polynomials together:

$$\underbrace{(x^0 + x^1 + x^2)}_{\text{purple eggs}} \cdot \underbrace{(x^0 + x^1 + x^2)}_{\text{green eggs}} \cdot \underbrace{(x^0 + x^1 + x^2 + x^3)}_{\text{white eggs}}$$

Now let term in the resulting polynomial correspond to choosing one term from each of the original polynomials! For instance, if we choose $x^0$ from the first factor, $x^1$ from the second, and $x^2$ from the third, this would correspond to choosing $0$ purple eggs, $1$ green egg, and $2$ white eggs.

So, this $7^{\text{th}}$-degree monstrosity of a polynomial in fact is the sum of many individual terms each representing a way to pick eggs from our three baskets!

The problem is asking for the number of ways to choose $4$ eggs from the basket. Conveniently, each term representing one of these ways evaluates to $x^4$! So, we only need to find the coefficient of $x^4$ in the resulting polynomial.

Expanding the product gives

$$x^7 + 3x^6 + 6x^5 + 8x^4 + 8x^3 + 6x^2 + 3x + 1$$

So, it turns out that the coefficient of $x^4$ is $8$, meaning there are $\boxed{8}$ ways to choose $4$ eggs.

Furthermore, this wild method of solving this question also tells us that there is $1$ way to choose $7$ eggs, $3$ ways to choose $6$ eggs, $6$ ways to choose $5$ eggs, $8$ ways to choose $3$ eggs, $6$ ways to choose $2$ eggs, $3$ ways to choose $1$ egg, and $1$ way to choose $0$ eggs! (We can see this by simply looking at the coefficients of the polynomial).

**Example 2**

Find a general formula for the Fibonacci sequence, satisfying the recurrence $f_0 = 0$, $f_1 = 1$, and $f_n = f_{n-1} + f_{n-2}$ for $n \geq 2$.

## Solution

Let's tackle the Fibonacci sequence using generating functions! Generating functions are not just for combinatorics; they shine in solving recurrence relations too.

First, let's define the generating function $F(x)$ for the Fibonacci sequence $\{f_n\}$ as:

$$\begin{aligned} F(x) &= f_0 + f_1 x + f_2 x^2 + f_3 x^3 + f_4 x^4 + \cdots && (1) \\ &= 0 + 1x + 1x^2 + 2x^3 + 3x^4 + 5x^5 + \cdots && (2) \end{aligned}$$

In general, this is how we will define the generating function of any sequence! You might be confused about what $x$ "means" here. It's just a generic input to a function — the function itself is what we really care about!

Now, given the recurrence relation $f_n = f_{n-1} + f_{n-2}$, we can express this in terms of $F(x)$ as follows:

Start by writing the recurrence for $n \geq 2$:

$$f_n - f_{n-1} - f_{n-2} = 0$$

Multiply both sides by $x^n$ and sum from $n = 2$ to $\infty$:

$$\sum_{n=2}^{\infty} (f_n - f_{n-1} - f_{n-2}) x^n = 0$$

Now we can be a little clever in simplifying this! Note that

$$\sum_{n=2}^{\infty} f_n x^n$$

is the original generating function without the first two terms. Also,

$$\sum_{n=2}^{\infty} f_{n-1} x^n = \sum_{n=1}^{\infty} f_n x^{n+1} = x \sum_{n=1}^{\infty} f_n x^n$$

is $x$ multiplied by the original generating function missing the first term.

This simplifies to:

$$F(x) - f_0 - f_1 x - x(F(x) - f_0) - x^2 F(x) = 0$$

Plugging in $f_0 = 0$ and $f_1 = 1$:

$$F(x) - x - xF(x) - x^2 F(x) = 0$$

Distribute and collect like terms:

$$F(x)(1 - x - x^2) = x$$

Solving for $F(x)$:

$$F(x) = \boxed{\frac{x}{1 - x - x^2}}$$

And voila, we have a nice closed form for the generating function of the Fibonacci sequence!

Now, to find the general formula for $f_n$, we will write $F(x)$ in a form where the coefficients are clearly defined! To do this, we need to express $F(x)$ in a form that allows us to figure out its coefficients. Let's use partial fraction decomposition.

First, find the roots of the denominator $1 - x - x^2 = 0$:

$$x = \frac{-(-1) \pm \sqrt{1+4}}{2} = \frac{1 \pm \sqrt{5}}{2}$$

Let $\phi = \frac{1+\sqrt{5}}{2}$ (the golden ratio) and $\psi = \frac{1-\sqrt{5}}{2}$.

Thus, we can write:

$$F(x) = \frac{x}{(1 - \phi x)(1 - \psi x)}$$

Using partial fractions:

$$F(x) = \frac{1}{(\psi - \phi)} \left( \frac{1}{1 - \psi x} - \frac{1}{1 - \phi x} \right)$$

Recognizing the sum of geometric series:

$$F(x) = \frac{1}{-\sqrt{5}} \left( \sum_{n=0}^{\infty} \psi^n x^n - \sum_{n=0}^{\infty} \phi^n x^n \right)$$

Therefore, the coefficient of $x^n$ in $F(x)$ is:

$$f_n = \frac{\psi^n - \phi^n}{-\sqrt{5}} = \frac{\phi^n - \psi^n}{\sqrt{5}}$$

Now, remember that $f_n$ was originally defined to be the $n$-th Fibonacci number! So, now we have a general formula for the $n$-th Fibonacci number:

$$f_n = \frac{(1+\sqrt{5})^n - (1-\sqrt{5})^n}{2^n \sqrt{5}} = \frac{\phi^n - \psi^n}{\sqrt{5}}$$

$$\boxed{f_n = \frac{\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n}{\sqrt{5}}}$$

This elegant formula allows us to compute the $n$-th Fibonacci number directly without computing all the previous terms! And the power of generating functions let us come up with this.

Here are some common generating functions — maybe it could be exciting to try proving them yourself!

**Theorem 1** (Binomial Theorem)

For any non-negative integer $n$,

$$(1+x)^n = \sum_{k=0}^{n} \binom{n}{k} x^k.$$

**Theorem 2** (Negative Binomial Theorem)

For any real number $\alpha$ and $|x| < 1$,

$$(1-x)^{-\alpha} = \sum_{k=0}^{\infty} \binom{\alpha+k-1}{k} x^k.$$

**Theorem 3** (Generating Function for Combinations)

For fixed $k$,

$$\sum_{n=0}^{\infty} \binom{n}{k} x^n = \frac{x^k}{(1-x)^{k+1}}.$$

**Theorem 4** (Generating Function for Fibonacci Numbers)

Given the Fibonacci sequence $\{F_n\}$ where $F_0 = 0$, $F_1 = 1$, and $F_n = F_{n-1} + F_{n-2}$ for $n \geq 2$,

$$\sum_{n=0}^{\infty} F_n x^n = \frac{x}{1-x-x^2}.$$

**Theorem 5** (Generating Function for Partitions)

The generating function for the number of partitions of $n$ is given by:

$$\sum_{n=0}^{\infty} p(n) x^n = \prod_{k=1}^{\infty} \frac{1}{1-x^k}.$$

**Theorem 6** (Geometric Series)

For any real number $r$ and $|x| < 1$,

$$\sum_{k=0}^{\infty} r^k x^k = \frac{1}{1 - rx}.$$

**Theorem 7** (Catalan Numbers Generating Function)

The generating function for Catalan numbers $C_n$ is:

$$\sum_{n=0}^{\infty} C_n x^n = \frac{1 - \sqrt{1-4x}}{2x}.$$

**Example 3**

Let $n$ be a positive integer. Show that

$$F_n = \binom{n-1}{0} + \binom{n-2}{1} + \binom{n-3}{2} + \cdots$$

where $F_n$ denotes the $n$th Fibonacci number

## Solution

Well, let $a_n = F_n$ and let $b_n = \binom{n-1}{0} + \binom{n-2}{1} + \binom{n-3}{2} + \cdots$. If we can show that

$$a_0 + a_1 x + a_2 x^2 + \cdots = b_0 + b_1 x + b_2 x^2 + \cdots$$

, then we are done since the polynomials being equal (as polynomials) means that each coefficient is equal!

We found the generating function of the Fibonacci sequence — it is $\frac{x}{1-x-x^2}$. Let's find the generating function of $b_i$!

Note that

$$b_n = \sum_{i=0}^{\infty} \binom{n-1-i}{i}$$

So,

$$\begin{aligned}
\sum_{n=0}^{\infty} b_n x^n &= \sum_{n=0}^{\infty} \sum_{i=0}^{\infty} \binom{n-1-i}{i} x^n \\
&= \sum_{i=0}^{\infty} \sum_{n=0}^{\infty} \binom{n-1-i}{i} x^n \\
&= \sum_{i=0}^{\infty} x^{1+i} \sum_{n=0}^{\infty} \binom{n-1-i}{i} x^{n-1-i}
\end{aligned}$$

Now, we are going to use Theorem 3 to get that this is equal to

$$\begin{aligned}\sum_{i=0}^{\infty} x^{i+1} \cdot \frac{x^i}{(1-x)^{i+1}} &= \sum_{i=0}^{\infty} \left(\frac{x^2}{1-x}\right)^i \cdot \frac{x}{1-x} \\ &= \frac{x}{1-x} \cdot \frac{1}{1-\frac{x^2}{1-x}} \\ &= \frac{x}{1-x-x^2}\end{aligned}$$

So, we have shown that these two sequences are equal, and thus, the result is proven! ■

**Try some Generating Functions Problems**

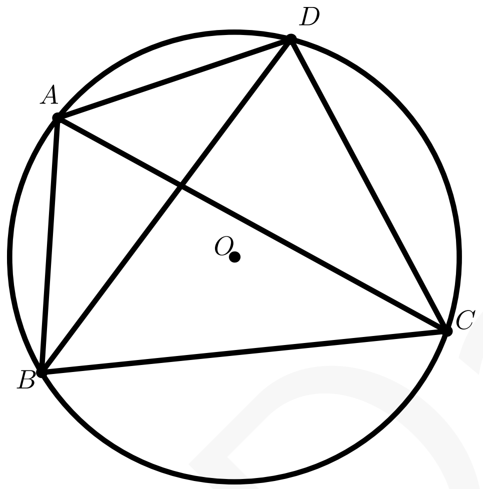
https://mathdash.com/maps/map/generating-functions

# Ptolemy's Theorem

*by MathDash Team*

https://mathdash.com/maps/map/ptolemy

Figure 3: Ptolemy — $AB \cdot CD + AD \cdot BC = AC \cdot BD$



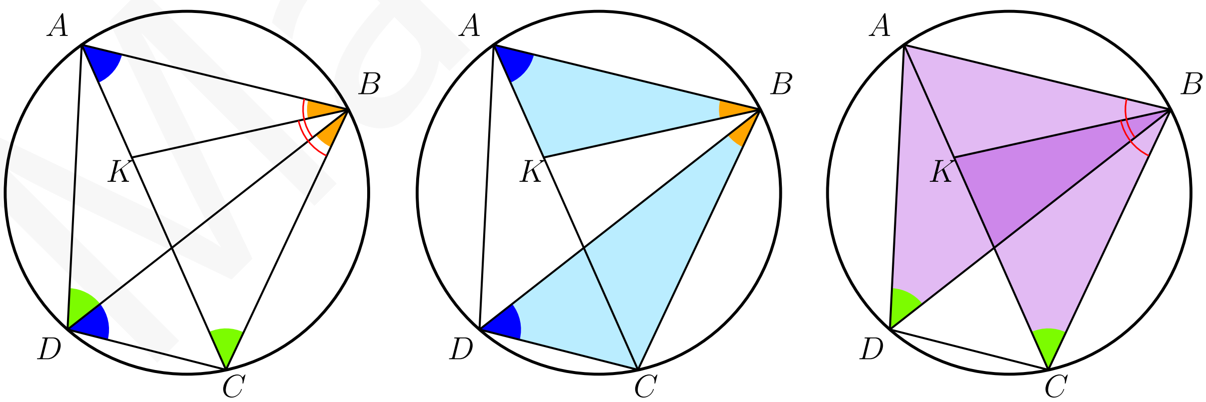



**Theorem 1** (Ptolemy's Theorem)

Given four points on a circle in the order $A$, $B$, $C$, $D$,

$$AB \cdot CD + AD \cdot BC = AC \cdot BD$$

**Proof:**




Source: Wikipedia

Let's construct a point $K$ on $AC$ such that $\angle ABK = \angle DBC$. Note that this makes $\triangle AKB$ is similar to $\triangle BDC$ because of angle-angle similarity (since $\angle BAK = \angle BDC$).

So, $\frac{AB}{AK} = \frac{BD}{DC}$. Cross-multiplying gives us $AB \cdot CD = AK \cdot BD$. Nice — we've been able to re-write one term in the Ptolemy's Theorem equation. Let's do it again!

Since $\angle ABK = \angle DBC$, adding $\angle KBD$ to both tells us $\angle ABD = \angle KBC$. Coupling this with the fact that $\angle BDA = \angle BCK$, we see that $\triangle ABD$ is similar to $\triangle KBC$. This tells us $\frac{AD}{BD} = \frac{KC}{BC}$ or $AD \cdot BC = BD \cdot KC$.

Now, using these two results we get $AB \cdot CD + AD \cdot BC = AK \cdot BD + KC \cdot BD = BD(AK + KC) = BD \cdot AC$, and we've proved the result!

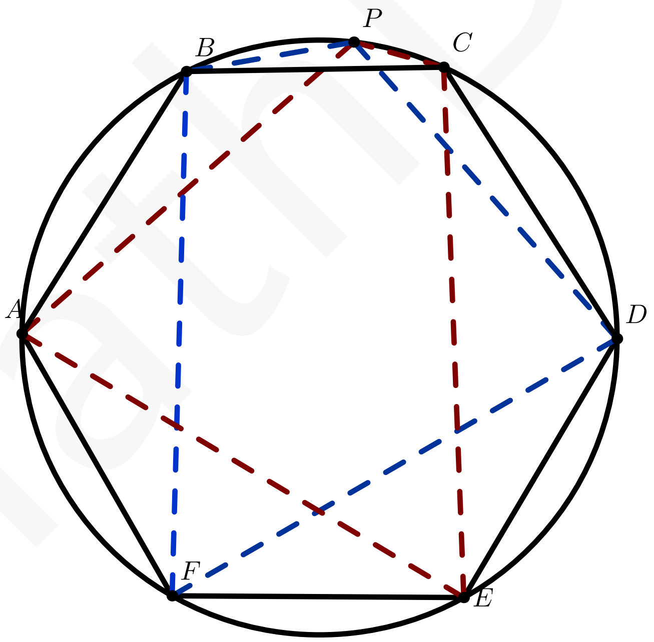
Here's a generalization if you are interested:

**Theorem 2** (Ptolemy's Inequality)

If $A$, $B$, $C$, $D$ are any four points, then $AB \cdot CD + AD \cdot BC \geq AC \cdot BD$, with equality holding if and only if $A$, $B$, $C$, $D$ lie on a circle

**Example 1**

A regular hexagon $ABCDEF$ is inscribed in a circle. If $P$ is on the minor arc $BC$, and $PE = 10$, $PF = 11$, find $PA + PB + PC + PD$.




**Solution:** Let's use Ptolemy on some of the cyclic quadrilaterals in this diagram! If we use Ptolemy's on the red quadrilateral $PCEA$, we get $PA \cdot EC + PC \cdot AE = PE \cdot AC$. Note though that $\triangle ACE$ is an equilateral triangle (being the alternate vertices on a regular hexagon)! So, $AC = CE = AE$. Dividing by this side length gives us $PA + PC = PE$.

Similarly, Ptolemy on the blue quadrilateral gives us $PB \cdot DF + PD \cdot BF = BD \cdot PF$. Since $BD = DF = FB$, dividing by this gives $PB + PD = PF$.

Now adding these together tells us $PA + PB + PC + PD = PE + PF = \boxed{21}$.

**Try some Ptolemy Theorem Problems**

https://mathdash.com/maps/map/ptolemy

# Other Cyclic Quad Theorems

*by Samuel*

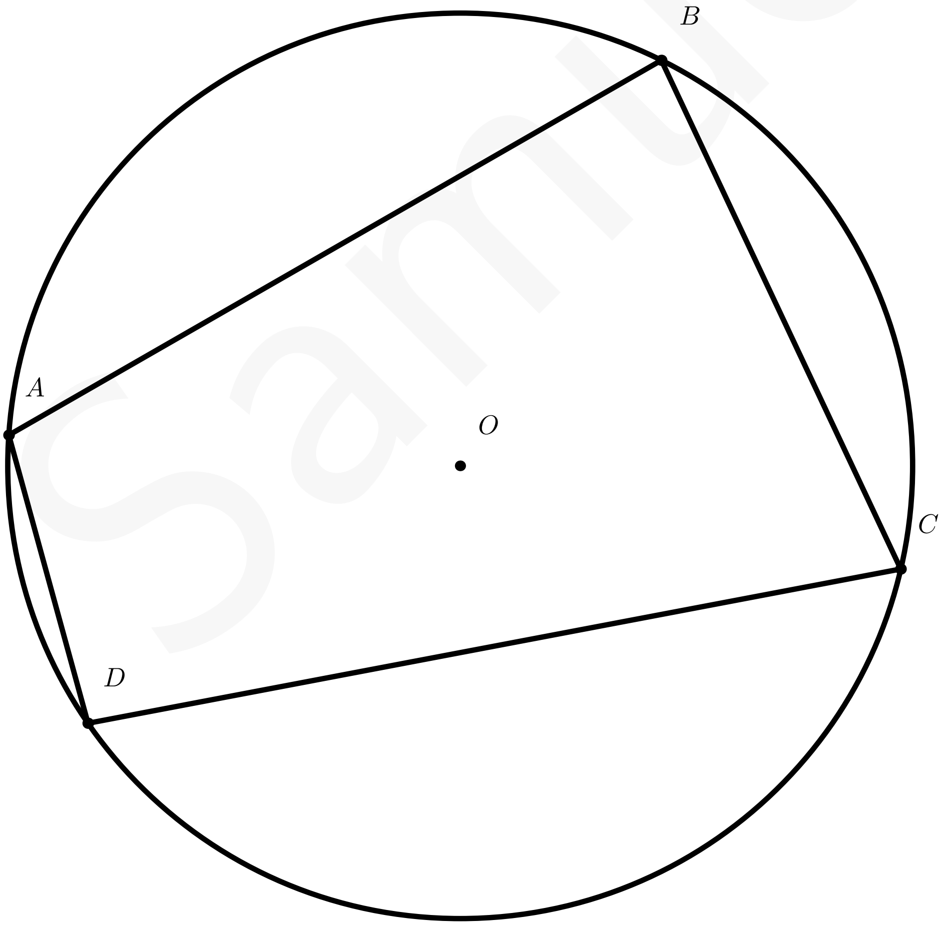
https://mathdash.com/maps/map/cyclic-quads-1

### A brief intro of cyclic quadrilaterals

By now, you probably know Ptolemy's theorem, which only works on cyclic quadrilaterals. However, there are several other theorems and formulas that apply specifically to cyclic quadrilaterals because of their circumscribing circle. The following theorem is probably the easiest and most simple one to prove.

**Theorem 1**

The sum of opposite angles in cyclic quadrilaterals is always $180°$.




*Proof.* Consider the following diagram. Notice how $\angle ABC$ bounds the arc $ADC$. Therefore, we can see that $m\angle ABC = \frac{mADC}{2}$ from our arc length theorems. Also notice that the same applies to $m\angle ADC = \frac{mABC}{2}$. Therefore, $m\angle ABC + m\angle ABC = \frac{mABC+mADC}{2} = \frac{360}{2} = 180°$. This applies to all angles in the cyclic quadrilateral. It can also be proven that the converse of this theorem is true. □

**Example 1**

Given that two angles in a cyclic quadrilateral measure $120°$ and $50°$, determine the degree measures of all other angles.

Since opposite angles must be supplementary, and $120+50 \neq 180$, the given angles are not opposite. Therefore, we can determine that one of the missing angles is $180 - 120 = 60°$, and the other is $180 - 50 = 130°$.

**Example 2**

Let $ABCD$ be a cyclic quadrilateral. The side lengths of $ABCD$ are distinct integers less than $15$ such that $BC \cdot CD = AB \cdot DA$. What is the largest possible value of $BD$?

**Solution.** By Law of Cosines,
$BD^2 = AD^2 + AB^2 - 2(AD)(AB)\cos(\angle BAD)$
$BD^2 = CD^2 + BC^2 - 2(CD)(BC)\cos(\angle BCD)$.
Since $ABCD$ is cyclic, we have $m\angle BAD + m\angle BCD = 180°$, and $2(AD)(AB)\cos\angle BAD = -2(CD)(BC)\cos\angle BCD$. Add these up to get $BD^2 = AD^2 + AB^2 + CD^2 + BC^2$.
Now, try to make $AD(AB) = CD(BC)$. If $AB$ is $14$, then $BC$ or $CD$ will be equal to 7. WLOG $BC = 7$, then $CD = 6$, $DA = 12$. The maximum value of $2BD^2 = 14^2 + 7^2 + 6^2 + 12^2 = 425$, so the maximum value of $BD = \boxed{\sqrt{\frac{425}{2}}}$.

**Theorem 2**

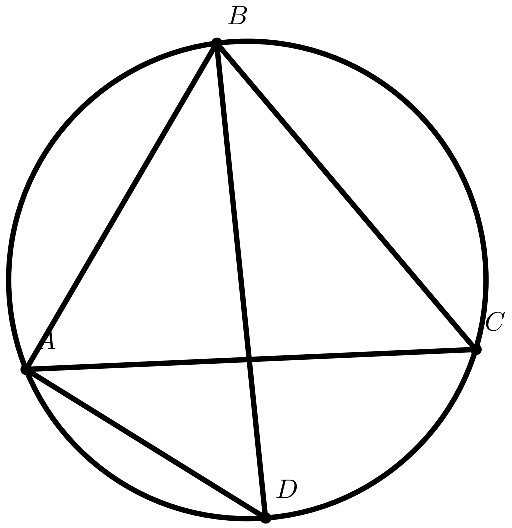
In cyclic quadrilateral $ABCD$, $\angle ABD = \angle ACD$, $\angle BCA = \angle BDA$, $\angle BAC = \angle BDC$, $\angle CAD = \angle CBD$.

There isn't really any proof required. This is simply true because the angles we claim to be equal bound the same arc in the circle! However, this idea is very handy to keep in mind!

**Example 3**

Consider $\triangle ABC$ with $m\angle CAB = 45°$ and $m\angle BCA = 60°$. Let $\overline{BD}$ be the angle bisector of $\angle ABC$, where $D$ is the point of intersection between $\overline{AD}$ and the circumcircle of $\triangle ABC$. Let $E$ be the point opposite of $D$ on the circumcircle of $\triangle ABC$. Determine the measure of $\angle EAC$ (TJ "Fake" Holiday CMIMC TST Problem 3).

Consider the following diagram:




**Solution.** Since $\overline{BD}$ is the angle bisector of $\angle ABC$, we have that $m\angle ABD = m\angle CBD = (75/2)^\circ = 37.5^\circ$. Also, $m\angle CBD = m\angle CAD$, so $m\angle CAD = 37.5^\circ$, and it bounds an $75^\circ$ arc. Since $E$ is on the opposite side of $D$ on the circumcircle, the arc bounded by $DE$ must be a $180^\circ$ arc. So, the measure of $\angle EAC$ will be $\frac{180-75}{2}$, or $\boxed{52.5}$.

**Theorem 3**

The area of a cyclic quadrilateral is $\sqrt{(s-a)(s-b)(s-c)(s-d)}$, where $a$, $b$, $c$, $d$ are the side lenghts of the quadrilateral, and $s$ is the semiperimeter($\frac{a+b+c+d}{2}$). This is also known as Brahmagupta's Formula for the area of a Cyclic Quadrilateral.

The proof is extremely messy, but you can check it out here: Brahmagupta's Formula Proof.

Also, Brahmagupta's Formula is a special case of Bretschneider's formula, when the cosine of the sum of two opposite angles in a cyclic quadrilateral become zero(a nice application of theorem 1!)

**Example 4**

Quadrilateral $ABCD$ with side lengths $AB = 7$, $BC = 24$, $CD = 20$, $DA = 15$ is inscribed in a circle. The area interior to the circle but exterior to the quadrilateral can be written in the form $\frac{a\pi - b}{c}$, where $a$, $b$, and $c$ are positive integers such that $a$ and $c$ have no common prime factor. What is $a + b + c$? (2023 AMC 10A Problem 15)

**Solution.** It is pretty easy to see that the diameter of this circle is $25$ (you can verify this by using the fact that 7–24–25 and 15–20–25 are both Pythagorean Triples, and then applying the converse of Thales's Theorem). Therefore, the area of the circumcircle is $\frac{625\pi}{4}$. Since the quadrilateral is inscribed in a circle, it is cyclic. Brahmagupta's formula tells us that the area of quadrilateral is $\sqrt{(33-7)(33-24)(33-20)(33-15)} = 234$. So, $\frac{625}{4}\pi - 234 = \frac{625\pi - 936}{4}$, and our answer is $\boxed{1565}$.

**Theorem 4**

All perpendicular bisectors of a cyclic quadrilateral are concurrent, and intersect at the circumcenter of the cyclic quadrilateral. The converse is also true.

*Proof.* Imagine each side as an individual chord in the circle. It is known that the perpendicular bisector of the chord passes through the center of the circle. Since this applies to all four sides, all perpendicular bisectors of a cyclic quadrilateral are concurrent at the circumcenter of the circle. $\square$

Some more example problems(tricky!):

**Example 5**

Triangle $ABC$ is an isosceles right triangle with $AB = AC = 3$. Let $M$ be the midpoint of hypotenuse $\overline{BC}$. Points $I$ and $E$ lie on sides $\overline{AC}$ and $\overline{AB}$, respectively, so that $AI > AE$ and $AIME$ is a cyclic quadrilateral. Given that triangle $EMI$ has area $2$, the length $CI$ can be written as $\frac{a-\sqrt{b}}{c}$, where $a$, $b$, and $c$ are positive integers and $b$ is not divisible by the square of any prime. What is the value of $a+b+c$?(2018 AMC 12A Problem 20).

**Example 6**

Let $ABCD$ be a cyclic quadrilateral with $AB = 4$, $BC = 5$, $CD = 6$, and $DA = 7$. Let $A_1$ and $C_1$ be the feet of the perpendiculars from $A$ and $C$, respectively, to line $BD$, and let $B_1$ and $D_1$ be the feet of the perpendiculars from $B$ and $D$, respectively, to line $AC$. The perimeter of $A_1B_1C_1D_1$ is $\frac{m}{n}$, where $m$ and $n$ are relatively prime positive integers. Find $m+n$.(2021 AIME I Problem 11)

**Example 7**

Let $ABC$ be an acute, scalene triangle, and let $M$, $N$, and $P$ be the midpoints of $\overline{BC}$, $\overline{CA}$, and $\overline{AB}$, respectively. Let the perpendicular bisectors of $\overline{AB}$ and $\overline{AC}$ intersect ray $AM$ in points $D$ and $E$ respectively, and let lines $BD$ and $CE$ intersect in point $F$, inside of triangle $ABC$. Prove that points $A$, $N$, $F$, and $P$ all lie on one circle(USAMO 2008).

**Try the problems!**

https://mathdash.com/maps/map/cyclic-quads-1

# Incenter-Excenter

*by MathDash Team*

https://mathdash.com/maps/map/incenter-excenter

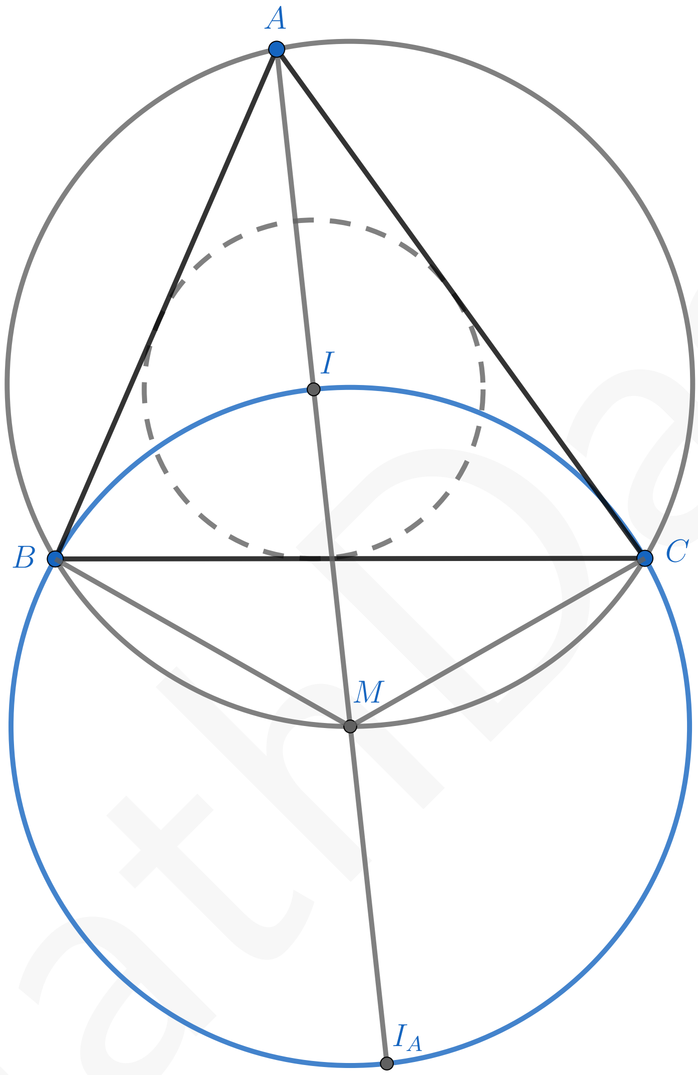



Figure 4: Fact 5: $BM = IM = CM = I_AM$



> **Theorem 1** (Incenter-Excenter Theorem / Fact 5)
>
> Let $I$ be the incenter of $\triangle ABC$, $I_A$ the $A$-excenter, and $M$ the midpoint of the arc $BC$ on the circumcircle of $\triangle ABC$. Then, $A$, $I$, $M$, $I_A$ are collinear (on the angle-bisector of $\angle A$, and $B$, $I$, $C$, $I_A$ lie on a circle centered at $M$ (so $BM = IM = CM = I_AM$).

**Proof:** Well, if $M$ is the midpoint of the arc $BC$, then $\overline{BM}$ and $\overline{CM}$ have equal angle measures. So, $\angle BAM = \angle CAM$ (both are half of the measures of the arc). Thus, $M$ lies on the angle bisector of $\angle BAC$. We know, by definition that $I$ and $I_A$ lie on this angle bisector as well — (remember $I_A$ is defined as the intersection of the $A$-angle bisector and the $B$, $C$ external angle bisectors) — so we know now $A$, $I$, $M$, $I_A$ are collinear.

We also know from $M$ being the midpoint of the arc $BC$ that $BM = CM$. Furthermore, note that $\angle IBM = \angle IBC + \angle CBM = \frac{1}{2}\angle ABC + \angle CAM = \frac{1}{2}\angle ABC + \frac{1}{2}\angle BAC = 90 - \frac{1}{2}\angle ACB$. Also, angle $\angle BIM = 180 - \angle AIB = \angle BAI + \angle ABI = \frac{1}{2}\angle ABC + \frac{1}{2}\angle BAC = 90 - \frac{1}{2}\angle ACB$. So, $\angle BIM = \angle IBM$ and thus $BM = IM$.

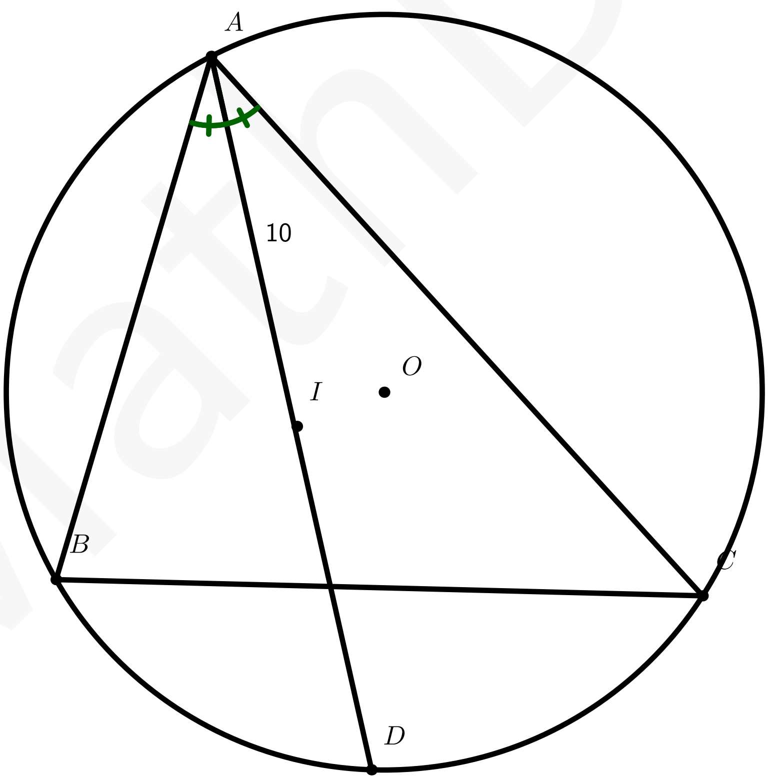
Furthermore, similar angle chasing gives us $\angle MBI_A = \angle CBI_A - \angle CBM = \frac{1}{2}(180 - \angle ABC) - \angle CAM = 90 - \frac{1}{2}\angle ABC - \frac{1}{2}\angle BAC = \frac{1}{2}\angle ACB$, and $\angle MI_AB = \angle ICB = \frac{1}{2}\angle ACB$. So, $MBI_A$ is isosceles and $MB = MI_A$.

Thus, the length equalities are proven as well — $MB = MI_A = MI = MC$.

This theorem by itself is a very common configuration in advanced computational geometry and olympiad geometry! Let's use it to solve a problem

**Example 1** (2013 HMMT Team #6)

Let triangle $ABC$ satisfy $2 \cdot BC = AB + AC$, and have incenter $I$ and circumcircle $\omega$. Let $D$ be the intersection of $AI$ and $\omega$. If $AI = 10$, find $DI$.




**Solution:** First, we can immediately tell that from the Incenter-Excenter Theorem, $D$, being the intersection of $AI$ with the circumcircle of $ABC$, is the midpoint of arc $BC$! This tells us $BD = CD = ID$ — so we simply need to find what this common length is. Let it be $x$.

Now let's use Ptolemy's Theorem on $ABDC$ — we get $AB \cdot CD + AC \cdot BD = AD \cdot BC$, or $AB \cdot x + AC \cdot x = AD \cdot BC$. Thus, $x(AB + AC) = AD \cdot BC$. Substituting in the given statement $2BC = AB + AC$ gives us $2BC \cdot x = AD \cdot BC$, or $AD = 2x$.

Also note that $AI = AD - ID = 2x - x = x$, so $AI = x = 10$. Thus, $ID = x = \boxed{10}$.

**Try some Incenter Excenter Problems**

https://mathdash.com/maps/map/incenter-excenter

# Introduction to Diophantine Equations

*by Math645*

https://mathdash.com/maps/map/diophantine-equations

**Definition 1** (Diophantine Equation)

A Diophantine Equation is any equation where all we need to find are integer solutions, or the number of integer solutions. Examples of Diophantine Equations include: $3x + 5y = 12$, or $4x^2 - y^2 = 1$

As you can probably tell, Diophantine Equations come in many forms. There is no standard formula or algorithm for Diophantine Equations, as there is for quadratics or cubics. So, this chapter will be very different, as there will be very few theorems, and mainly just methods.

## Linear Diophantine Equations

Anything in the form of $ax + by = c$, is a linear Diophantine Equation. To figure out when these have roots, we have a theorem.

**Theorem 1** (Bezout's Lemma Extended)

For integers $a$, $b$, there exist integer solutions for $x$ and $y$ to the equation, $ax + by = c$ if and only if $\gcd(a, b) \mid c$

Proof: Recall that if and only if means we must prove the original statement and its inverse. We start by proving the original statement: For integers $a$, $b$, there exist integer solutions for $x$ and $y$ to the equation, $ax + by = c$ if $\gcd(a, b) \mid c$. Note that if $\gcd(a, b) \mid c$, then $\frac{c}{\gcd(a,b)}$ is an integer. Therefore if we can prove that there are solutions of $(x, y)$ for $ax + by = gcd(a, b)$, we can multiply $x$ and $y$ by the integer, $\frac{c}{\gcd(a,b)}$, and we would get $ax + by = c$. We proceed by considering the set of positive integers that can be expressed as $ax + by$. By the Well Ordering Principle, this set must have a minimum, let's call it $n$. Let's express $a$, in terms of $n$, a quotient $m$, and a remainder $p$ with $a = nm + p$, for $0 \le p < n$. Now we know since $n$ is an element of the set, we express it as $n = ax + by$. Plugging this into $a = nm + p$, we obtain $a = (ax + by)m + p$, rearranging gives us $p = a(1 - nx) + b(-ny)$. We have just expressed $p$ in the form $ax + by$, which means if it is positive, it belongs to the set, but is less than the minimum since $0 \le p < n$. Therefore, $p = 0$. Thus $a = nm$, and $n \mid a$. By symmetry, $n \mid b$, so we can say $n \mid \gcd(a, b)$. Thus $n$ is a common divisor of $a$ and $b$. We also know that $\gcd(a, b) \mid a$, thus $\gcd(a, b) \mid ax$, and similarly $\gcd(a, b) \mid by$, and $\gcd(a, b) \mid ax + by$. Thus since the gcd divides the left side, it must divide the right side. Thus $\gcd(a, b) \mid n \mid \gcd(a, b)$ and so $n = \gcd(a, b)$. Since $n$ belongs to the set, the gcd can be expressed as $ax + by$, and we are done. Now we prove the inverse: If $\gcd(a, b) \nmid c$, then there don't exist integers $x, y$ such that $ax + by = c$. As we proved earlier, the gcd divides the left side of the equation so it must divide the right side, which it doesn't in this case, so there are no solutions.

**Example 1**

Two farmers agree that pigs are worth $300$ dollars and that goats are worth $210$ dollars. When one farmer owes the other money, he pays the debt in pigs or goats, with "change" received in the form of goats or pigs as necessary. (For example, a $390$ dollar debt could be paid with two pigs, with one goat received in change.) What is the amount of the smallest positive debt that can be resolved in this way? (Source: 2006 AMC 10A Problem 22)

Solution: This problem can be reworked to the linear Diophantine Equation, $300x + 210y = c$. Thus with Bezout's Lemma Extended, $30 \mid c$, so the least positive value of $c$ is $\boxed{30}$

Now we try to find solutions to Diophantine Equations.

**Theorem 2**

Once we have $1$ solution to $ax + by = c$, if we increase $x$ by $\frac{kb}{\gcd(a,b)}$, and decrease $y$ by $\frac{ka}{\gcd(a,b)}$ for any integer $k$, then we produce the entire solution set.

Proof: Our sum increases by $\frac{kab}{\gcd(a,b)}$ and then decreases by $\frac{kab}{\gcd(a,b)}$. But, how do we know this will produce all the solutions? We proceed with a proof by contradiction. Let's call our initial solution, $(x_0, y_0)$, and for the sake of contradiction, we claim that there exists a solution $x, y$ such that its value of $x$ is in between two elements of the set $x_0 + \frac{kb}{\gcd(a,b)}$. Let's call the greatest value in the set lower than this value of $x$, $x_1$. Now, $x - x_1 = m$ and clearly $0 < m < \frac{b}{\gcd(a,b)}$. Now we say we change the value of $y$ by $n$. Clearly, $am = bn$. All the divisors of $b$ divide the right side of the equation, so they must divide the left. Thus, all the divisors of $b$ must divide $am$. $a$ takes care of all the divisors of $b$ that are also divisors of $\gcd(a, b)$, thus all that is left are the divisors of $\frac{b}{\gcd(a,b)}$. So $\frac{b}{\gcd(a,b)} \mid m$. Yet, $0 < m < \frac{b}{\gcd(a,b)}$, creating a contradiction, and we have proved that no other solutions exist.

Now, we only need to find $1$ solution, which for small numbers can be obtained by guessing or for larger numbers by applying the Euclidean Algorithm in reverse.

Recall, the Euclidean Algorithm takes two numbers, and constantly takes them modulo one another until their greatest common divisor is reached. Now, instead of taking the values modulo, if we express it as a quotient and a remainder, we form a linear combination. This will let us express it as a linear diophantine equation, when we run it in reverse. If this sounds confusing, the example will clear it up.

**Example 2**

Find a solution to $7x - 24y = 5$

Solution: Once we find values of $x$ and $y$, that make the $7x - 12y = 1$, we can multiply $x$ and $y$ by $5$ to finish. We chose $1$ because it is $\gcd(7, 24)$, which we can create a solution for using the Euclidean Algorithm in reverse. We start, $(7, 24) = (7, 7 \cdot 3 + 3) = (7, 3) = (3 \cdot 2 + 1, 3) = (1, 3) = 1$. Since $3$ is a multiple of $1$, we stop and now go in reverse, expressing 1 in the coefficients before it. We start with the multiple

$$
\begin{gathered}
1 = \\
3 - 2 \cdot 1 = \\
3 - 2 \cdot (7 - 2 \cdot 3) \\
(24 - 3 \cdot 7) - 2 \cdot (7 - 2(24 - 3 \cdot 7)) = \\
24 \cdot 5 - 17 \cdot 7.
\end{gathered}
$$

Since this may be confusing, we will show the process step by step. We start off with the very end of our Euclidean Algorithm chain which is always the gcd, in this case 1. Then, since we have 2 values that are a multiple of each other (which will always happen in the 2nd to last step of the Euclidean Algorithm), we express the gcd, in this case 1, with 3 and 1, pushing us back a step. Now we see we obtained the 1 by expressing 7 as $3 \cdot 2 + 1$ so we can express 1 as $7 - 3 \cdot 2$, pushing us back a step. Now we see we obtained the 3 by expressing 24 as $3 \cdot 7 + 3$ so we can express 3 as $24 - 3 \cdot 7$, pushing us back a step. Finally, we combine like terms for $24$ and $7$, to finish. Thus $24 \cdot 5 - 17 \cdot 7 = 1$, so $24 \cdot 25 - 17 \cdot 35 = 5$, and $\boxed{(-35, 25)}$ is a solution. Using the previous theorem, we also know that $\boxed{(35 + 24k, 25 - 7k)}$ are solutions.

**Example 3**

A class collects 50 dollars to buy flowers for a classmate who is in the hospital. Roses cost 3 dollars each, and carnations cost 2 dollars each. No other flowers are to be used. How many different bouquets could be purchased for exactly 50 dollars? (Source: 2008 AMC 10B Problem 8)

Solution: This is just $3x + 2y = 50$ for non-negative $x, y$. We start with the Euclidean Algorithm: $(3, 2) = (2 \cdot 1 + 1, 2) = (1, 2) = 1$. Now going backwards, $1 = 2 - 1 \cdot 1 = 2 - 1 \cdot (3 - 2) = 2 - 1 \cdot 3 + 1 \cdot 2 = 2 \cdot 2 - 1 \cdot 3$, Thus $x = -1, y = 2$ is a solution if the equation is equal to $1$. We multiply by $50$, and obtain $x = -50, y = 100$ is a solution. Yet they must both be non-negative. We see that we can add $3$ to $y$ and subtract $2$ from $x$ to generate all solutions (by the previous theorem since their gcd is 1). We need to make $x$ positive, so we add the smallest multiple of $2$ greater than or equal to $50$ which is just $50$, so we must subtract $75$ from $y$ to get the solution $x = 0, y = 25$. Now we continue generating as many as we can until one of them become negative. We can't subtract from $x$, so we add to $x$, subtracting from $y$. So, we stop when $y$ becomes negative. Every time we generate a solution, we subtract $3$ from $y$, so our solutions happen by subtracting $0$, subtracting $3$, subtracting $6\ldots$ all the way till subtracting the greatest multiple of $3$ less than or equal to $25$ which is $24$. It's clear the sequence: $0, 3, 6, 9, \ldots, 24$ has $\boxed{9}$ terms.

**Example 4**

Find the number of lattice points that the line $19x + 20y = 1909$ passes through in Quadrant I. (Source: 2008 iTest Problem 18)

Solution: This is just $19x+20y = 1909$ for non-negative $x, y$. We start with the Eculidean Algorithm: $(19, 20) = (19, 19 \cdot 1 + 1) = (19, 1) = 1$. Going backwards, we obtain $1 = 19 - 18 \cdot 1 = 19 - 18 \cdot (20 - 19 \cdot 1) = 19 - 18 \cdot 20 + 19 \cdot 18 = 19 \cdot 19 - 18 \cdot 20$. Thus, $(19, -18)$ is a solution for it to be equal to $1$. We multiply by $1909$, to get the solution $(19 \cdot 1909, -18 \cdot 1909)$. Now we try to make $y$ non-negative by repeatedly adding $19$. However many times we add $19$ to $y$, we must subtract $20$ from $x$. Now we see that if we add $19$, $1909$ times, we add $1909 \cdot 19$ to $y$ and get $1909$, which is $100$ more $19$'s than what we want, which is the smallest value to generate all solutions. So we only add $1809$ $19$'s to bring the $y$ to $1909 - 100 \cdot 19 = 9$. We subtract $20 \cdot 1809$ from $x$ to get the solution $(19 \cdot 1909 - 20 \cdot 1809, 9)$. Now to generate all solutions, we continue adding $19$ to $y$, subtracting $20$ from $x$, until $x$ becomes negative. Note that if we subtract another $100$ $20$'s, we obtain $x = 19 \cdot 1909 - 1909 \cdot 20 = -1909$. This is negative so we need to add back $1920$ since adding $1900$ makes it still negative and $1920$ is the next smallest multiple of $20$. Adding $1920 = 20 \cdot 96$, gives us an $x$ value of $19 \cdot 1909 - 1909 \cdot 20 + 20 \cdot 96 = 19 \cdot 1909 - 1813 \cdot 20$, being the smallest value of $x$. The largest was $19 \cdot 1909 - 1809 \cdot 20$, giving us a total of $\boxed{5}$ non-negative solutions.

**Example 5**

Let $n$ be the number of ways $10$ dollars can be changed into dimes and quarters, with at least one of each coin being used. Then $n$ equals (Source: 1968 AHSME Problem 19)

Solution: This is just positive solutions to $10x+25y = 1000$, or dividing by $5$ gives us $2x+5y = 200$. While we can apply Euclidean Algorithm, as stated earlier another way to find your first solution is to guess. It's clear to see and guess that $(100, 0)$ works, and we have our first solution. To bring it to positive, we add 2 to $y$, and subtract 5 from $x$, to give (95, 2). For practice, we use the Euclidean Algorithm. $(2, 5) = (2, 2 \cdot 2 + 1) = (2, 1) = 1$. Backwards gives, $1 = 2 - 1 = 2 - (5 - 2 \cdot 2) = 2 - 5 + 2 \cdot 2 = 2 \cdot 3 - 5 \cdot 1$ or $(3, -1)$. Multipling by $200$, gives $(600, -200)$. We must add $2$ to $y$, until we reach positive so we add $202$ in total since it's the least multiple of $2$ to bring us to positive. We add $101$ $2$'s to $y$, so we subtract $101$ $5$'s from $x$, and obtain $(95, 2)$, just like by guessing. Now, to keep us in positives, we can subtract $5$ from $x$, a maximum of $18$ times since $19$ will make $x = 0$. So subtracting anything from $0 - 18$ times gives us a total of $19$ options or $\boxed{19}$ solutions.

While linear diophantine equations have many uses, especially for problems like the above practice problems, perhaps the most important use of Linear Diophantine Equations is modular division. Discussed in Modular Arithmetic I by Math645, modular division can be used a ton in advanced

Number Theory Problems. If you don't know what modular division is, reading its section of Modular Arithmetic I is vital.
With this problem, we will uncover many properties of modular division, so even if you know how to solve, read the solution to uncover many modular division properties, and methods.

We consider the modular division equation: $x \equiv \frac{a}{b}$ mod $m$.
Solution: This is just $bx \equiv a$ mod $m$. Before we continue, it is probably best to simplify this as much as possible. We can remove all common factors of $a$ and $b$ using the division operations. Division Operation $1$ works for all factors shared between $a, b,$ and $m$. Then Division Operation $2$ works for all factors shared between $a, b$ and not $m$. We don't use it for factors which aren't shared between $a$ and $b$ since that brings us to modular division which is the problem we are solving. Now we have $dx \equiv c$ mod $n$ where $\gcd(c, d) = 1$ since all common factors were removed with the division operations. This is equivalent to $dx - c = ny$ or $dx - ny = c$. Aha, we have just convered a modular division problem into a linear diophantine equation. Now it is clear to see that the fraction is only defined when there exist solutions to this linear diophantine equation. There exist solutions only if the greatest common divisor of $d$ and $n$ divides $c$. Thus every common divisor of $d$ and $n$ must divide $c$. Yet since $\gcd(c, d) = 1$, the only divisor of $d$ that divides $c$ is $1$. And since every common divisor of $d$ and $n$ must be a divisor of $d$ (obviously) and $c$ (stated earlier), every common divisor must be $1$. So $\gcd(d, n) = 1$ if and only if a solution exists. Now since the gcd is $1$, solutions for $x$ repeat every $\frac{n}{\gcd(d,n)} = n$ so there is only $1$ unique value mod $n$, and this produces all solutions.

**Example 6**

Solve the modular equation: $x \equiv \frac{6}{12}$ mod $9$.

Rookie Mistake: $\frac{6}{12} \equiv \frac{1}{2}$ mod $9$. Fraction simplification isn't a valid operation in modular arithmetic.

Solution: Multiply everything by $12$ (Cross Multiply) to get $12x \equiv 6$ mod $9$. We see that $\gcd(12, 6)$ isn't $1$ so we start removing common factors. The factor of $3$ is shared, and $3$ divides the modulo, $9$, so when removing it by Division Case $1$, we remove it from the modulo too to get $4x \equiv 2$ mod $3$. This is the reason why the rookie mistake of fraction simplification failed. Next, we divide everything by $2$ with Division Case $2$ so we don't have to remove it from the modulus, to get $2x \equiv 1$ mod $3$ so $2x = 1 + 3y$ so $2x - 3y = 1$ It is very easy to guess that $x = 2$ works so the solutions are $x \equiv 2$ mod $3$, yet we solve the Linear Diophantine Equation anyways. $2x - 3y = 1$ so the Euclidean Algorithm gives $(2, 3) = (2, 2 \cdot 1 + 1) = (2, 1) = 1$, and backwards: $1 = 2 - 1 \cdot 1 = 2 - 1 \cdot (3 - 2 \cdot 1) = 2 \cdot 2 - 3 \cdot 1$, so $x = 2$ is a solution and $\boxed{x \equiv 2 \text{ mod } 3}$ is the answer.
This solution path will always work for modular division mentioned in the Modular Arithmetic I chapter by Math645.

## General Diophantine Equations

Now, Diophantine Equations start to become fun, as there is no longer an algorithm or method to solve Diophantine Equations. Not all Diophantine Equations are linear Diophantine Equations, and they can have exponents, products of variables, more variables, elements which aren't integers, and more. For these Diophantine Equations which aren't linear, there are 2 main ways of solving them: Factoring, or taking all sides modulo a value. We start with Factoring.

## Factoring

If we can factor a Diophantine Equation, into a part with products of variables on one side, and a constant on the other side, we can consider integer factors of the constant, and apply those onto the variables since if the variables are integers, expressions in terms of the variables are also integers, and there are a limited number of integer factors of the constant.

**Example 7**

How many non-negative integer solutions for $(x, y)$ exist to the equation: $(x+2)(y+1) = 840$.

Solution: This is already factored, and in the way we want it. So we just consider factors of $840$. For example, $7$ is a factor of $840$. If we assign $7$ to $(x + 2)$, then it's clear that $y + 1 = 120$. Thus, we just got a solution for $(x, y)$ from a factor of $840$. This factor contributed $1$ solution. Similarly each factor will contribute $1$ solution. But, how do we know there don't exist solutions of $(x, y)$ that can't be found by considering factors of $840$. Simply, if $x + 2$ isn't a factor of $840$, then $y + 1$ is a fraction and so is $y$, which isn't possible. Thus we just need to find the number of positive integer factors of $840$, and consider some extreme cases. We can prime factorize it as $2^3 \cdot 3 \cdot 5 \cdot 7$, which as covered in a previous chapter of the MathDash Book has $4 \cdot 2 \cdot 2 \cdot 2 = 32$ factors. However $x$ and $y$ must be non-negative or $x \geq 0$ and $y \geq 0$. Thus, $x + 2 \geq 2$ and $y + 1 \geq 1$. Since the factors are positive integers, there are no constraints on $y+1$, but $x+2$ can still be a positive integer while not being greater than or equal to $2$. This happens only when $x + 2 = 1$ and $1$ is a factor of $840$, so we have to subtract this case since it has a negative value of $x$, but was still counted since $x + 2$ has a positive value and factor of $840$. So the answer is simply $32 - 1 = \boxed{31}$.

Once we can get it to a factored form as the previous equation, it is simple, but how do we get it to the factored form. This requires knowledge of many algebraic factorizations, and at times even nice observations and guessing. We will now share some algebraic factorizations that help in factoring Diophantine Equations, the proof of which requires Algebra so is left out.

**Theorem 3** (Simon's Favorite Factoring Trick)

$ab+xa+yb+n = (a+y)(b+x) - xy + n$ for constants/coefficients $x$, $y$, $n$ and variables $a$, $b$

**Theorem 4** (Difference of squares)

$a^2 - b^2 = (a - b)(a + b)$

We now look at a generalization of the previous theorem. The reason this was still kept despite having a generalization, is because it is very important for competition math, and should be known quick enough without having to apply the generalization.

**Theorem 5** (Difference of powers general case)

$a^n - b^n = (a - b)(a^{n-1} + a^{n-2}b + a^{n-3}b^2 + a^{n-4}b^3 + a^{n-5}b^4 + \ldots a^2b^{n-3} + ab^{n-2} + b^{n-1})$

**Theorem 6** (Sum of odd powers general case)

$a^{2n+1} + b^{2n+1} = (a + b)(a^{2n} - a^{2n-1}b + a^{2n-2}b^2 - \ldots - ab^{2n-1} + b^{2n})$

There doesn't exist a general case for sum of even powers.

**Theorem 7** (Factor Theorem)

If $a$ is a root of $P(x)$ then $x - a$ is a factor of $P(x)$

This means wherever we can calculate roots with Algebra, we can factor into a Diophantine Equation. There are many more factoring theorems, yet these are the most important basics, and should be understood deeply.

**Example 8**

The harmonic mean of two positive integers is the reciprocal of the arithmetic mean of their reciprocals. For how many ordered pairs of positive integers $(x, y)$ with $x < y$ is the harmonic mean of $x$ and $y$ equal to $6^{20}$? (Source: 1996 AIME Problem 8)

Solution: This is just asking us how many integer solutions to $\frac{1}{\frac{\frac{1}{x}+\frac{1}{y}}{2}} = 6^{20}$. This can be simplified to $\frac{2xy}{x+y} = 6^{20}$, and by cross multiplying, dividing everything by $2$, and moving everything to one side, we get $xy - (x) \cdot \frac{6^{20}}{2} - (y) \cdot \frac{6^{20}}{2} = 0$. Then by Simon's Favorite Factoring Trick, we can write this as $(x - \frac{6^{20}}{2})(y - \frac{6^{20}}{2}) - \frac{6^{40}}{4} = 0$, or $(x - \frac{6^{20}}{2})(y - \frac{6^{20}}{2}) = \frac{6^{40}}{4}$. So how many factors does $\frac{6^{40}}{4}$ have? Well $\frac{6^{40}}{4} = 3^{40} \cdot 2^{38}$. This has $41 \cdot 39$ factors. We also need $x < y$, which by symmetry happens half of the time when $x \neq y$. Since $\frac{6^{40}}{4}$ is a perfect square, there is one solution for $x = y$, so we have the number of solutions as $\frac{41 \cdot 39 - 1}{2} = \boxed{799}$.

**Example 9**

The integer solutions to the following equation are $(x_1, y_1)$ and $(x_2, y_2)$. Find $x_1 + x_2 + y_1 + y_2$
$x^2 + x + 1 = y^2 - 1$

Hint: Move everything to one side and use the Quadratic Formula for $x$
Solution: Move everything to the left side and write as a Quadratic in $x$, to get $x^2+x+(2-y^2) = 0$. Note that $x$ must be an integer so it must be rational, so the discriminant of this quadratic equation must be a perfect square, lets call it $m^2$. Note that the discriminant is $4y^2 - 7 = m^2$. Isolating the constant, 7, we have $4y^2 - m^2 = 7$. Using difference of squares, we can factor this into $(2y - m)(2y + m) = 7$. Note that $y$ must be a positive integer and if $m$ is positive $2y + m$ must be positive. If $m$ is negative $2y - m$ must be positive. So since one factor is positive, so is the other. Thus we have $2$ possibilities: $2y + m = 7$ and $2y - m = 1$ or $2y - m = 1$ and $2y + m = 7$. In both cases we have $y = 2$ and $m = \pm 3$. Note that by the Quadratic Formula, $x = \frac{-1 \pm m}{2} = \frac{-1 \pm 3}{2}$. So we have $x = 1, -2$ and for each value, $y = 2$. So our answer is just $1 + 2 + (-2) + 2 = \boxed{3}$.

**Example 10**

How many integer solutions are there to $a^2 + 1 = b^2 + 2024$? (Source: 2024 SIMC 10)

Solution: Factor to $(a - b)(a + b)$=$2023$. $2023$ has $6$ factors. Negative factors work for a total of $\boxed{12}$ solutions.

**Example 11**

Let $f(x)$ be a cubic function with leading coefficient $1$. If $f(y) = f(-y) = f(3y) = 0$ for an integer $y$, find integer solutions to $(x, y)$ for

a) $f(x) = 12$

b) $f(x) = -2$

c) $f(x) = 3$

Hint: Use the Factor Theorem
Solution: We can factor $f(x)$ by the Factor Theorem to get $(x - y)(x + y)(x - 3y)$. We first solve part a. We see that we need it to be equal to $12$, an even result. Thus, one of the factors must be even. We see that the factors differ by an amount of $2y$, an even number, so if one of the factors is even, they all are. Thus if it's even, it must be divisible by $2^3 = 8$. So there are no solutions for it to be equal to $12$. Same thing for it to be equal to $-2$. For it to be equal to $3$, it doesn't need to be even, so it doesn't need to be divisible by $8$. We see that $3$ is prime, so it must be factored as $1 * 1 * 3$, or $1 * -1 * -3$. Now, we see that the factors must differ by the same even amount of $2y$. Thus the first case is impossible since they differ by different amounts of $0, 2$. The second case we

see they differ by $2$, so $2y = \pm 2$. So $y = \pm 1$. If $y = 1$, then $x = 0$. If $y = -1$, then $x = -2$, giving us the solutions, $(0, 1)$ and $(-2, -1)$.

Sometimes, there is no easy factoring trick, and instead you just have to experiment, try new things, manipulate, and even guess factorings.

**Example 12**

Integers $a$, $b$, and $c$ satisfy $ab + c = 100$, $bc + a = 87$, and $ca + b = 60$. What is $ab + bc + ca$? (Source: 2024 AMC10A Problem 23)

Solution: As stayed earlier by trying many different manipulations, expirements, adding equations, multiplying equations, and subtracting equations, you will find that by subtracting the second equation from the first equation, we have $ab + c - bc - a = 13$. This can be factored as $(b - 1)(a - c) = 13$. This gives us $4$ cases since $13$ has $2$ positive integer factors, and $2$ negative integer factors:

Case $1$: $b - 1$=13 and $a - c = 1$. Then, $b = 14$, and $a = c + 1$. Then $ac + b = c^2 + c + 14 = 60$, so $c^2 + c = 46$. This has no integer solutions.

Case 2: $b-1 = -13$ and $a-c = -1$. Then $b = -12$, and $c = a+1$. Then $ac+b = a^2+a-12 = 60$, so $a^2 + a = 72$. $a = 8, -9$ is a solution. Thus, $(8, -12, 9)$ and $(-9, -12, -8)$ work. Plugging into $ab + c = 100$, we see that only $(-9, -12, -8)$ works, so $ab + bc + ac = 108 + 96 + 72 = \boxed{276}$. We don't need to test any other cases since we got something that works and the AMC's only have $1$ correct answer.

**Try it Yourself!**

https://mathdash.com/maps/map/diophantine-equations

https://mathdash.com/contest/Diophantine-Equations-All-in-One-Problem

# Orders

*by Math645, Testsolved by Krithik and Anthony*

**Definition 1** (Order)

We define order as $\text{ord}_m n = x$ if $x$ is the lowest positive integer such that $n^x \equiv 1 \mod m$. If this is the case, we can say that the order of $n$ mod $m$ is $x$.

For example the $\text{ord}_8 3 = 2$ since $3^2 \equiv 1 \pmod 8$ and no lower power of $3$ is $1 \pmod 8$.

**Example 1**

Let's calculate the order of $3$ mod $10$ or $\text{ord}_{10} 3$

**Solution:** We start calculating powers of $3$. $3^1 \equiv 3 \not\equiv 1$, $3^2 \equiv 9 \not\equiv 1$, $3^3 \equiv 27 \equiv 7 \not\equiv 1$, $3^4 \equiv 81 \equiv 1$ mod $10$. Thus $\text{ord}_{10} 3 = 4$

Note that if $\gcd(m, n) > 1$, then there are no solutions to $n^x \equiv 1$ mod $m$, as n to any positive power can never be $1$ mod $m$. So for the rest of this chapter, whenever we have $\text{ord}_m n$, we assume $m$ and $n$ are relatively prime, unless otherwise stated.

There are several theorems using orders that can help us to calculate orders of numbers much easier than brute forcing, and solve problems that seemingly have no connection to orders.

**Theorem 1** (Order Theorem 1)

$x^{(\text{ord}_m x)} \equiv 1 \pmod m$

This builds us onto our next theorem:

**Theorem 2** (Order Theorem 2)

**If** $\text{ord}_m x \mid n$, **then,** $x^n \equiv 1 \pmod m$

**Proof:** Note that if $\text{ord}_m x \mid n$ then $\frac{n}{\text{ord}_m x}$ is an integer. Now we can take both sides of our first theorem, $x^{(\text{ord}_m x)} \equiv 1$ mod $m$ to the power of $\frac{n}{\text{ord}_m x}$ to get $(x^{\text{ord}_m x})^{\left(\frac{n}{\text{ord}_m x}\right)} \equiv 1^{\frac{n}{\text{ord}_m x}}$. The left side is clearly just $x^n$. The right side is $1$ to the power of an integer which is just $1$. Simplifying we get, $x^n \equiv 1 \pmod m$ and we are done.

Our next theorem is very similar, but instead is reversed.

> **Theorem 3** (Order Theorem 3)
>
> If $x^n \equiv 1 \pmod{m}$ then $(\operatorname{ord}_m x) \mid n$

Proof: We do a proof by contradiction. Assume, for sake of contradiction, that $\operatorname{ord}_m x$ doesn't divide $n$.

Then $n \not\equiv 0$ mod $\operatorname{ord}_m x$, so we can express n as $a(\operatorname{ord}_m x) + b$ where $0 < b < \operatorname{ord}_m x$, and $a$ and $b$ are integers.

Plugging this into the given gives us $x^{(a*(\operatorname{ord}_m x)+b)} \equiv 1$. Simplifying the left as follows:

$$x^{(a*\operatorname{ord}_m x+b)} \equiv x^{a*\operatorname{ord}_m x} * x^b \equiv (x^{\operatorname{ord}_m x})^a * x^b \equiv 1^a * x^b \equiv x^b$$

Thus we have $x^b \equiv 1$ mod $m$ where $0 < b < \operatorname{ord}_m x$. This goes against our definition of order since the order is the SMALLEST positive integer with the criteria. Thus the assumption is wrong and our proof by contradiction is complete.

The next theorem is very useful, and helps form connections between problems that don't seem to involve orders, with problems that require order.

> **Theorem 4** (Order Theorem 4)
>
> The exponents of $n$ mod $m$ repeat every $\operatorname{ord}_m n$ and no sooner or later.

Proof: We calculate $n^{(x+\operatorname{ord}_m n)}$ mod $m$ to see $n^{x+\operatorname{ord}_m n} \equiv n^x * n^{\operatorname{ord}_m n} \equiv n^x * 1 \equiv n^x$, and the order is the smallest value for which this is true since the order is the smallest value $a$ such that $n^a \equiv 1$

We continue the theorems section with 2 theorems that helps us calculate the order, and can be used in multiple ways.

> **Theorem 5** (Order Theorem 5)
>
> $(\operatorname{ord}_m b) \mid \phi(m)$

Proof: Simply plug in $n = \phi(m)$ into Order Theorem 3 and use Euler's Totient Theorem. This helps us calculate orders as we only have to check values that divide the totient.

> **Theorem 6** (Order Theorem 6)
>
> If we prime factorize $m$ into $p_1^{a_1} \cdot p_2^{a_2} \cdot p_3^{a_3} \cdots p_n^{a_n}$, then
>
> $$\operatorname{ord}_m b = \mathbf{lcm}(\operatorname{ord}_{(p_1^{a_1})} b,\ \operatorname{ord}_{(p_2^{a_2})} b,\ \operatorname{ord}_{(p_3^{a_3})} b,\ \ldots,\ \operatorname{ord}_{(p_n^{a_n})} b).$$

Proof: First we prove that $\operatorname{ord}_m b$ must be divisible by the order of b modulo $p_x^{a_x}$ for all $x$. Then, we prove it must be the least common multiple of these numbers. Let's consider $p_x^{a_x}$ for any arbitrary $x$. We know that $a_x$ is the largest power of $p_x$ that divides m. Let's denote the order we are looking for, $\operatorname{ord}_m b = y$. Now by Order Theorem 1, we know $b^y \equiv 1 \mod m$. Thus splitting $m$ into it's prime factors, we know $b^y \equiv 1$ mod $p_x^{a_x}$. By Order Theorem 4, we know that this is possible when $y$ is a multiple of $\operatorname{ord}_{(p_x^{a_x})} b$. We also know that when $y$ is a multiple, it will work. Thus the smallest value that is a multiple of each $\operatorname{ord}_{(p_x^{a_x})} b$, is the lcm of all $\operatorname{ord}_{(p_x^{a_x})} b$

Now we can start with some problems:

**Example 2**

Problem 1: Given that $x^{32} \equiv 513 \pmod{514}$, find $\operatorname{ord}_{514} x$

Solution: Located at the end of the chapter with the rest of the solutions.

If you were unable to solve the previous problem, understand the solution before attempting the next one. It uses the same concept, but different numbers to test your understanding of the solution. If you did solve the previous problem, try solving the next one as quick as possible.

**Example 3**

Problem 2: Given that $x^{81} \equiv 719 \pmod{974}$, find $\operatorname{ord}_{974} x$. It is also given that $719^3 \equiv 1$ mod $974$.

Solution: Once again, as with the rest of the solutions to the problems, it's at the end of the chapter.

Now that you should be comfortable with orders lets go into competition-style math problems which don't have orders in the problem statement.

**Example 4**

Problem 3: Find the least odd prime factor of $2019^8 + 1$ (Source: 2019 AIME I Problem 14).

**Example 5** (Source: Olympiad Number Theory Through Challenging Problems)

Problem 4: Let $a > 1$ and $n$ be given positive integers. If $p$ is an odd prime divisor of $a^{2^n} + 1$, prove that $p - 1$ is divisible by $2^{n+1}$.

**Example 6**

Problem 5: Find the smallest prime number $p$ such that $p^2 \mid (n^4+1)$ for some positive integer $n$. (Inspiration: 2024 AIME I Problem 13).

**Example 7**

Problem 6: Find the least positive integer $n$ such that when $3^n$ is written in base $143^2$, its two right-most digits in base $143$ are $01$. (Source: 2018 AIME I Problem 11).

**Example 8**

Problem 7: How many positive integer multiples of $1001$ can be expressed in the form $10^j - 10^i$, where $i$ and $j$ are integers and $0 \leq i < j \leq 99$? (Source: 2001 AIME II Problem 10).

**Example 9**

Problem 8: The values of $y$ that satisfy the following equation can be expressed as $a$ mod $b$. Find $ab - a$:

$$3^{2y} \equiv 3^{y+2} \text{ mod } 221$$

**Try some Order Practice Contests**

Order Problem 1 (1800)
Order Problem 2 (1900)
Order Problem 3 (1900)
Order Problem 4 (2000)
Order Problem 5 (1900, Proof)
Order Problem 6 (2350, Proof)
Number Theory Contest 1 (Practice problems)
Number Theory Contest 2 (Practice problems, including Order)

Further Exercises: If you want more practice problems after the MathDash contests listed above, try 1982 USAMO Problem 4. If you know LTE, try 2021 JMC 10 Problem 25 and try solving the original 2024 AIME I Problem 13, instead of the modified version given here. The solutions, being easy to find online, aren't included. If you want to test your knowledge and calculations of order, you can think of random numbers, find their order, and use Order Calculator to check your work.

## Solutions

**Problem 1:** Note that $513 \equiv -1 \pmod{514}$, so we write our equation as $x^{32} \equiv -1$. Then we square both sides to get $x^{64} \equiv 1$. Thus, applying Order Theorem 3 with $n = 64$ and $m = 514$, the order must divide 64. So the order must be 1, 2, 4, 8, 16, 32, 64. Now we apply Order Theorem 2 with $m = 514$ and $n = 32$. Thus, if the order divides 32, $x^{32} \equiv 1$, which we know is false as given in the problem. Thus, the order doesn't divide 32, but does divide 64, so the order is simply $\boxed{64}$.

**Problem 2:** We cube both sides to get $x^{243} \equiv 1$. Thus, applying Order Theorem 3 with $n = 243$ and $m = 974$, the order must divide 243. So the order must be 1, 3, 9, 27, 81, 243. Now we apply Order Theorem 2 with $m = 974$ and $n = 81$. Thus, if the order divides 81, $x^{81} \equiv 1$, which we know is false as given in the problem. Thus, the order doesn't divide 81, but does divide 243, so the order is simply $\boxed{243}$.

**Problem 3:** Let the least odd prime factor be $p$. Then, $p \mid 2019^8 + 1$ or $2019^8 + 1 \equiv 0 \mod p$. Subtracting 1, we get $2019^8 \equiv -1 \mod p$. Using the same logic as in the previous problems, we obtain $\operatorname{ord}_p 2019 = 16$. Thus using Order Theorem 5, with $m = p$ and $b = 2019$, we obtain $16 \mid \phi(p)$. Since $p$ is prime, $\phi(p) = p - 1$. Thus $16 \mid p - 1$. We test multiples of 16, to see if they are 1 less than a prime to quickly see that the smallest such primes are $17$, $97$, and $113$. Now we test all of them to see if they work, starting from the lowest since as soon as we have the smallest value we are done.

1. Working mod 17, $2019^8 + 1 \equiv 13^8 + 1 \equiv (-4)^8 + 1 \equiv 16^4 + 1 \equiv (-1)^4 + 1 \equiv 1 + 1 \equiv 2 \not\equiv 0$

2. Working mod 97, $2019^8 + 1 \equiv 79^8 + 1 \equiv (-18)^8 + 1 \equiv 324^4 + 1 \equiv 33^4 + 1 \equiv 1089^2 + 1 \equiv 22^2 + 1 \equiv 484 + 1 \equiv 96 + 1 \equiv 97 \equiv 0$. Thus $\boxed{97}$ works.

**Problem 4:** We know $p \mid a^{2^n} + 1$ and since $p$ is odd, $p$ doesn't divide 2, and thus $p$ doesn't divide their difference, $a^{2^n} - 1$. Thus, $a^{2^n} - 1 \not\equiv 0 \mod p$, or $a^{2^n} \not\equiv 1 \mod p$. Now going back to the given, $p \mid a^{2^n} + 1$, so we also have $p \mid (a^{2^n} + 1) * (a^{2^n} - 1)$ which simplifies to $p \mid a^{2^n+1} - 1$. Thus $a^{2^n+1} - 1 \equiv 0 \mod p$ or $a^{2^n+1} \equiv 1 \mod p$. Combining this with $a^{2^n} \not\equiv 1$, and using the logic with problem 1, we see that $\operatorname{ord}_p a = 2^{n+1}$. Applying Order Theorem 5 with $m = p$ and $b = a$, we have $2^{n+1} \mid p - 1$.

**Problem 5:** We can write $p^2 \mid (n^4+1)$ as $n^4+1 \equiv 0 \pmod{p^2}$. Subtracting 1, we obtain $n^4 \equiv -1$. Using the same method as problems 1 and 2, we obtain $\operatorname{ord}_{(p^2)} n = 8$. Plugging $m = p^2$ and $b = n$ into Order Theorem 5, we obtain $8 \mid \phi(p^2)$. Using the definition of totient, we get $8 \mid p(p-1)$. Since

$p$ and $p-1$ are relatively prime, we have $8 \mid p$ or $8 \mid p-1$. If $8 \mid p$, $p$ can't be prime, so this case isn't possible. Thus, $8 \mid p-1$. It is easy to see that the smallest $p$ is $\boxed{17}$.

**Problem 6:** We can write the given statement as $3^n \equiv 1 \pmod{143^2}$. Using Order Theorem 6, we break it up and handle the 2 cases separately. We want the smallest $n$ such that:

1. $3^n \equiv 1 \pmod{11^2}$. This is simply $\operatorname{ord}_{121} 3$, which we can calculate using Order Theorem 5 with $m = 121$ and $b = 3$. So $b \mid \phi(121) = 110$. Thus, $b = 1, 2, 5, 10, 11, 22, 55, 110$. We start with the lowest values and build our way up.

- $3^1 \equiv 3 \not\equiv 1$
- $3^2 \equiv 9 \not\equiv 1$
- $3^5 \equiv 243 \equiv 1 \pmod{121}$

Thus, $n$ must be a multiple of 5.

2. Now, we need the smallest $n$ such that $3^n \equiv 1 \pmod{169}$. Thus, using Order Theorem 5, $n \mid 156$. We also know that $3^n \equiv 1 \pmod{13}$. Trying values, we quickly get that the smallest $n$ for this is 3. Thus, with Order Theorem 2, we have $3 \mid n \mid 156$. This limits $n$ to 3, 6, 12, 39, 78, 156. Once again, starting with the smallest values and working our way up mod 169. Although it is tedious, we can constantly take mod 169 to get small numbers. We get:

- $3^3 \equiv 27 \not\equiv 1$
- $3^6 \equiv 729 \equiv 53 \not\equiv 1$
- $3^{12} \equiv (3^6)^2 \equiv 53^2 \equiv 2809 \equiv 105 \not\equiv 1$
- $3^{39} \equiv 3^{36} \cdot 3^3 \equiv 105^3 \cdot 27 \equiv (-64)^3 \cdot 27 \equiv (-192)^3 \equiv -23^3 \equiv 529$

Thus using Order Theorem 6, we solve the least common multiple of 39 and 5 is $\boxed{195}$.

**Problem 7:** We need values of i and j such that $1001 \mid 10^j - 10^i$. We can rewrite this as $10^j - 10^i \equiv 0$ mod $1001$ or $10^j \equiv 10^i$ mod $1001$. Dividing by $10^i$, we obtain $10^{j-i} \equiv 1$ mod $1001$. Using Order Theorem 4, we know that they will repeat every $\operatorname{ord}_{1001} 10$ times, so we calculate this order.

$10^1 \equiv 10 \not\equiv 1,$

$10^2 \equiv 100 \not\equiv 1,$

$10^3 \equiv 1000 \not\equiv 1$

$10^4 \equiv 10^3 * 10 \equiv 1000 * 10 \equiv -1 * 10 \equiv -10 \equiv 991 \not\equiv 1$

$10^5 \equiv 10^4 * 10 \equiv -10 * 10 \equiv -100 \equiv 901 \not\equiv 1$

$10^6 \equiv 10^5 * 10 \equiv -100 * 10 \equiv -1000 \equiv 1.$

Thus the order is $6$. So the powers of $10$ repeat every $6$. We want $10^{j-i} \equiv 1$ which we just showed happens first when $j - i = 6$, and repeats every $6$, so $6 \mid j - i$, but $j - i > 0$ since $j > i$. If $j - i = 6$, we have $94$ values for $j$ $(7–100)$ and each value has $1$ correspondent value of $i$. If $j - i = 12$, we have $88$ values for $j$ $(13–100)$ and each value corresponds to $1$ value of $i$. Continuing the pattern, we have $94 + 88 + 82 + 76 \cdots + 4$, and using arithmetic series formulas, we have $\frac{16*(94+4)}{2}$, or $\boxed{784}$.

**Problem 8:** We divide both sides by $3^{y+2}$ to get our equation to $3^{y-2} \equiv 1 \bmod 221$. Using Order Theorem 4, it is clear to see that $3^0 \equiv 1$ and they repeat every $\operatorname{ord}_{221} 3$ so the solutions are $y - 2 \equiv 0$ mod $\operatorname{ord}_{221} 3$. Thus, $y \equiv 2 \mod \operatorname{ord}_{221} 3$. We calculate $\operatorname{ord}_{221} 3$ using Orders Theorems 4 and 6. Firstly we factor $221$. We realize it is $4$ less than $225$ so we can use difference of squares to see that $221 = 225 - 4 = (15 - 2)(15 + 2) = 13 * 17$. Thus with Order Theorem 6, we calculate, $\operatorname{lcm}(\operatorname{ord}_{17} 3, \operatorname{ord}_{13} 3)$.

1. To calculate $\operatorname{ord}_{17} 3$, we use Order Theorem 5. We see that since 17 is prime, $\phi(17) = 16$, Thus the order must be 1, 2, 4, 8, or 16. We test values starting from the smallest modulo 17.

$3^1 \equiv 3 \not\equiv 1$.

$3^2 \equiv 9 \not\equiv 1$.

$3^4 \equiv 81 \equiv 13 \not\equiv 1$.

$3^8 \equiv (3^4)^2 \equiv 13^2 \equiv 169 \equiv -1 \equiv 16 \not\equiv 1$. Thus the order must be 16 since it's not 1, 2, 4, 8.

2. To calculate $\operatorname{ord}_{13} 3$, we apply Order Theorem 5. We see that since 13 is prime, $\phi(13) = 12$. Thus the order must be 1, 2, 3, 4, 6, or 12. We test values from the smallest modulo 13.

$3^1 \equiv 3 \not\equiv 1$

$3^2 \equiv 9 \not\equiv 1$

$3^3 \equiv 27 \equiv 1$. Thus $\operatorname{ord}_{13} 3$ is 3 and $\operatorname{ord}_{221} 3 = \operatorname{lcm}(3, 16) = 48$. So from earlier $y \equiv 2$ mod $48$. So $a = 2$, and $b = 48$. $ab - a = 2 * 48 - 2 = 96 - 2 = \boxed{94}$.

# AIME

*by MathDash Community*

https://mathdash.com/contests?tag=AIME

**AIME Mock Contests**

**https://mathdash.com/contests?tag=AIME**

# Functional Equations

*by Timur*

https://mathdash.com/maps/map/functional-equations

## Introduction to Functional Equations

Solving functional equations is a main part in Olympiad Algebra. Now, let's run through techniques used to solve functional equations.

### Concepts

Let $f\colon A \to B$ be a function. The set $A$ is called **domain**, and the set $B$ the **codomain**. Now, let's look at a couple of definitions which will be useful:

**Definition 1**

A function $F\colon A \to B$ is **injective** if $f(x) = f(y) \iff x = y$.

**Definition 2**

A function $F\colon A \to B$ is **surjective** if for all $b \in B$, there exists at least one $x$ such that $f(x) = b$. (Sometimes also called *onto*).

**Definition 3**

A function is **bijective** if it is both injective and surjective.

### Example

**Example 1**

Find all functions $f\colon \mathbb{R} \to \mathbb{R}$ such that

$$f(x) + xf(1-x) = x$$

for all $x \in \mathbb{R}$.

**Solution.** Substituting $x = a$ and $x = 1 - a$, the identities appear to be

$$f(a) + af(1-a) = a \tag{1}$$
$$f(1-a) + (1-a)f(a) = 1 - a. \tag{2}$$

From identity (1), it follows that

$$f(1-a) = 1 - \frac{f(a)}{a}. \tag{3}$$

From identity (2), it follows that

$$f(1-a) = (1-a)(1-f(a)). \tag{4}$$

Equating (3) and (4) and rearranging the terms, we find that

$$af(a) - f(a) + \frac{f(a)}{a} = a$$
$$f(a) = \frac{a^2}{a^2 - a + 1}.$$

Now you may ask in this step *Is it over?* No, we should verify our answer. As following:

$$\begin{aligned} f(x) + xf(1-x) &= \frac{x^2}{x^2 - x + 1} + x\frac{(1-x)^2}{(1-x)^2 - (1-x) + 1} \\ &= \frac{x^2}{x^2 - x + 1} + \frac{x^3 - 2x^2 + x}{x^2 - x + 1} \\ &= x. \end{aligned}$$

□

*Remark.* Now you may ask why should we recheck our answer. The answer is it may contradict to other equations.

*Remark.* Things to remember:

- Keep in mind that most functional equations are identities. In other words, you can plug in anything in the domain if it will help.
- Common things to plug in: Zero, $x = y$, anything that results in canceling an $f$(something) term (such as plugging in $(x, \frac{1}{x})$ in $f(x) + f(y) = (xy - 1)f(x)f(y)$.
- Many expressions you'll see are **cyclic**, i.e. the variables used in cycle. For example, if you see $f(x) + f(\frac{2}{x})$ or $f(x) + f(1-x)$, you can also plug in $x \to \frac{2}{x}$ or $x \to 1 - x$ to get the same expression.

## Resources.

★ Chapters 3,4 of OTIS Excerpts by Evan Chen

1. Introduction to Functional Equation by Evan Chen
2. Functional Equation by Pang-Cheng, Wu
3. 100 FE Problems by Amir Hossein Parvardi

**Try the problems**

**Functional Equations**

https://mathdash.com/maps/map/functional-equations

# Inequalities

*by Krithik*

## Introduction to Inequalities

Inequalities are becoming increasingly common in math olympiads. In this chapter, we discuss the most useful theorems as well as some techniques to solve inequalities.

## Notation

First, we must define some notation.

**Definition 1** (Homogeneity)

Define a function, polynomial, or inequality as **homogeneous** if all its terms have the same degree.

**Definition 2** (Cyclic Sum Notation)

Define $\sum_{\text{cyc}} f(x_1, x_2, \ldots, x_n)$ as

$$\sum_{\sigma} f(x_{\sigma(1)}, x_{\sigma(2)}, \ldots, x_{\sigma(n)}),$$

where $\sigma$ ranges over all cyclic permutations of $(1, 2, \ldots, n)$. A cyclic permutation is one that simply shifts the numbers by some constant. For example, if we had 3 variables, $a$, $b$, and $c$,

$$\sum_{\text{cyc}} a^3 = a^3 + b^3 + c^3$$

and

$$\sum_{\text{cyc}} a^2 b = a^2 b + b^2 c + c^2 a.$$

**Definition 3** (Symmetric Sum Notation)

Define $\sum_{\text{sym}} f(x_1, x_2, \ldots, x_n)$ as

$$\sum_{\sigma} f(x_{\sigma(1)}, x_{\sigma(2)}, \ldots, x_{\sigma(n)}),$$

where $\sigma$ ranges over all permutations of $(1, 2, \ldots, n)$. For example, if we had 3 variables, $a, b$, and $c$,

$$\sum_{\text{sym}} a^3 = 2a^3 + 2b^3 + 2c^3$$

and

$$\sum_{\text{sym}} a^2 b = a^2 b + b^2 c + c^2 a + ab^2 + bc^2 + ca^2.$$

**Definition 4** (Majorizing)

Let sequence $p = (a_1, a_2, \ldots, a_n)$ **majorize** $q = (b_1, b_2, \ldots, b_n)$ if, for all integers $i$ such that $1 \leq i < n$, the sum of the first $i$ numbers in $p$ is greater than the sum of the first $i$ numbers in $q$, and

$$a_1 + a_2 + \cdots + a_n = b_1 + b_2 + \cdots + b_n.$$

This is denoted by $p \succ q$.

## Basic Inequalities

### Trivial Inequality

This is the simplest inequality, yet it can be used to derive almost all the other, more complicated ones.

**Theorem 1** (Trivial Inequality)

For any real number $x$,

$$x^2 \geq 0.$$

Equality holds if and only if $x = 0$.

### AM-GM Inequality

This is perhaps the most widely used inequality.

**Theorem 2** (AM-GM Inequality)

For any positive integer $n$ and positive real numbers $a_1, a_2, \ldots, a_n$,

$$a_1 + a_2 + \cdots + a_n \geq n\sqrt[n]{a_1 a_2 \ldots a_n}.$$

Equality occurs when $a_1 = a_2 = \cdots = a_n$.

## Proof

We shall use a special type of induction called **Cauchy Induction**. To use Cauchy induction, we must prove that

1. AM-GM holds for $n = 2$
2. AM-GM holds for $2n$ given that it holds for $n$
3. AM-GM holds for $n-1$ given it holds for $n$.

1) We know by the trivial inequality that

$$(\sqrt{a_1} - \sqrt{a_2})^2 \geq 0,$$

so

$$a_1 + a_2 - 2\sqrt{a_1 a_2} \geq 0$$

or

$$a_1 + a_2 \geq 2\sqrt{a_1 a_2}.$$

2) Now, assume that

$$a_1 + a_2 + \cdots + a_n \geq n\sqrt[n]{a_1 a_2 \ldots a_n}$$

for all positive real numbers $a_1, a_2, \ldots, a_n$. We can see that if we have numbers $a_1, a_2, \ldots, a_{2n}$ are positive real numbers, then

$$\begin{aligned} a_1 + a_2 + \cdots + a_{2n} &= (a_1 + a_2 + \cdots + a_n) + (a_{n+1} + a_{n+2} + \cdots + a_{2n}) \\ &\geq n\sqrt[n]{a_1 a_2 \ldots a_n} + n\sqrt[n]{a_{n+1} a_{n+2} \ldots a_{2n}} \\ &\geq 2n\sqrt[2n]{a_1 a_2 \ldots a_{2n}}. \end{aligned}$$

3) Assume that

$$a_1 + a_2 + \cdots + a_n \geq n\sqrt[n]{a_1 a_2 \ldots a_n}$$

for all positive real numbers $a_1, a_2, \ldots, a_n$. We can let $a_n = \frac{1}{n-1}(a_1 + a_2 + \cdots + a_{n-1})$ to get that

$$\tfrac{n}{n-1}(a_1 + a_2 + \cdots + a_{n-1}) \geq n\sqrt[n]{a_1 a_2 \ldots a_{n-1}(a_1 + a_2 + \cdots + a_{n-1})/(n-1)}.$$

Raising this to the power of $n$ and simplifying give

$$(a_1 + a_2 + \cdots + a_{n-1})^{n-1} \geq (n-1)^{n-1} a_1 a_2 \ldots a_{n-1}.$$

The result is straightforward after this. ■

**Theorem 3** (Weighted AM-GM)

For any positive real number $n$, positive real numbers $a_1, a_2, \ldots, a_n$, and positive weights $\omega_1, \omega_2, \ldots, \omega_n$ such that $\omega_1 + \omega_2 + \cdots + \omega_n = 1$,

$$a_1\omega_1 + a_2\omega_2 + \cdots + a_n\omega_n \geq a_1^{\omega_1} a_2^{\omega_2} \ldots a_n^{\omega_n}.$$

Equality occurs when $a_1 = a_2 = \cdots = a_n$.

**Theorem 4** (Weighted Power Mean Inequality)

For $n$ positive real numbers $a_i$ and $n$ positive real weights $w_i$ with sum $\sum_{i=1}^{n} \omega_i = 1$, the power mean with exponent $t$, for some real number $x$, is defined by

$$M(t) = \begin{cases} a_1^{\omega_1} a_2^{\omega_2} \dots a_n^{\omega_n} & \text{if } x = 0 \\ (\omega_1 a_1^x + \cdots + \omega_n a_n^x)^{\frac{1}{x}} & \text{otherwise} \end{cases}.$$

For any real numbers $k_1$ and $k_2$, $M(k_1) \geq M(k_2)$ if $k_1 \geq k_2$. Equality occurs when $a_1 = a_2 = \cdots = a_n$.

Weighted AM-GM and Weighted Power Mean are generalizations of AM-GM that are far more powerful but less commonly used. We shall prove these theorems later in this chapter.

**Example 1** (1983 AIME Problem 9)

Find the minimum value of $\frac{9x^2 \sin^2 x + 4}{x \sin x}$ for $0 < x < \pi$.

## Solution

By AM-GM, we get that

$$\frac{9x^2 \sin^2 x + 4}{x \sin x} = 9(x \sin x) + 4/(x \sin x) \geq 2\sqrt{36} = 12.$$

By Intermediate Value Theorem, there is some value of $x$ where $x \sin x = 2/3$, so the answer is $\boxed{012}$.

**Example 2**

Let $a$, $b$, $c$ be positive real numbers such that

$$(a - \tfrac{1}{a}) + (b - \tfrac{1}{b}) + (c - \tfrac{1}{c}) = 0.$$

Prove that $a + b + c \geq 3$.

## Solution

We know that

$$a + b + c = \tfrac{1}{a} + \tfrac{1}{b} + \tfrac{1}{c},$$

so

$$(a+b+c)^2 = (a+b+c)(\tfrac{1}{a}+\tfrac{1}{b}+\tfrac{1}{c}).$$

By Power Mean Inequality,

$$(a+b+c)/3 \geq ((\tfrac{1}{a}+\tfrac{1}{b}+\tfrac{1}{c})/3)^{-1},$$

so

$$(a+b+c)^2 = (a+b+c)(\tfrac{1}{a}+\tfrac{1}{b}+\tfrac{1}{c}) \geq 9,$$

or $a+b+c \geq 3$. ■

**Cauchy-Schwarz Inequality**

**Theorem 5** (Cauchy-Schwarz Inequality)

For any positive integer $n$ and any two sequences of real numbers $a_1, a_2, \ldots, a_n$ and $b_1, b_2, \ldots,$ $b_n$,

$$(a_1^2 + a_2^2 + \cdots + a_n^2)(b_1^2 + b_2^2 + \cdots + b_n^2) \geq (a_1 b_1 + a_2 b_2 + \cdots + a_n b_n)^2.$$

Equality occurs if there is some constant $\lambda$ such that $a_i = \lambda b_i$ for all $i \in [1, n]$.

## Proof

By AM-GM,

$$\sum_{i=1}^{n}\sum_{j=1}^{n} a_i^2 b_j^2 + a_j^2 b_i^2 \geq \sum_{i=1}^{n}\sum_{j=1}^{n} 2a_i b_i a_j b_j.$$

This simplifies to

$$2\left(\sum_{i=1}^{n} a_i^2\right)\left(\sum_{i=1}^{n} b_i^2\right) \geq 2\left(\sum_{i=1}^{n} a_i b_i\right)^2. \blacksquare$$

**Theorem 6** (Titu's Lemma)

For any positive integer $n$ and any two sequences of positive real numbers $a_1, a_2, \ldots, a_n$ and $b_1, b_2, \ldots, b_n$,

$$\frac{a_1^2}{b_1} + \frac{a_2^2}{b_2} + \cdots + \frac{a_n^2}{b_n} \geq \frac{(a_1 + a_2 + \cdots + a_n)^2}{b_1 + b_2 + \cdots + b_n}.$$

## Proof

By Cauchy-Schwarz,

$$\left(\frac{a_1^2}{b_1} + \frac{a_2^2}{b_2} + \cdots + \frac{a_n^2}{b_n}\right)(b_1 + b_2 + \cdots + b_n) \geq (a_1 + a_2 + \cdots + a_n)^2,$$

and the result follows immediately. ■

**Theorem 7** (Hölder's Inequality)

Let $\{\{a_{ij}\}_{i=1}^{m}\}_{j=1}^{n}$ be a doubly indexed sequence of nonnegative real numbers and let $\omega_1, \omega_2, \ldots,$ $\omega_n$ be positive real weights.

$$(a_{11}+a_{21}+\cdots+a_{m1})^{\omega_1}(a_{12}+a_{22}+\cdots+a_{m2})^{\omega_2}\ldots(a_{1n}+a_{2n}+\cdots+a_{mn})^{\omega_n}$$

$$\geq a_{11}^{\omega_1}a_{12}^{\omega_2}\ldots a_{1n}^{\omega_n}+a_{21}^{\omega_1}a_{22}^{\omega_2}\ldots a_{2n}^{\omega_n}+\cdots+a_{m1}^{\omega_1}a_{m2}^{\omega_2}\ldots a_{mn}^{\omega_n}$$

The proof of Hölder's inequality is very difficult and will not be presented in this chapter.

**Example 3**

Let $a$, $b$, $c$ be positive real numbers such that

$$(a-\tfrac{1}{a})+(b-\tfrac{1}{b})+(c-\tfrac{1}{c})=0.$$

Prove that $a+b+c\geq 3$.

## Solution

This is the exact same question we encountered earlier, but let's prove it with Cauchy-Schwarz. We know that

$$a+b+c=\tfrac{1}{a}+\tfrac{1}{b}+\tfrac{1}{c},$$

so

$$(a+b+c)^2=(a+b+c)(\tfrac{1}{a}+\tfrac{1}{b}+\tfrac{1}{c}).$$

By Cauchy-Schwarz,

$$(a+b+c)(\tfrac{1}{a}+\tfrac{1}{b}+\tfrac{1}{c})=(1+1+1)^2=3^2.$$

Therefore, $a+b+c\geq 3$. ■

In many advanced problems, we can use Hölder's inequality to simplify the expressions.

**Example 4** (2001 IMO Problem 2)

Let $a$, $b$, $c$ be positive real numbers. Prove that $\frac{a}{\sqrt{a^2+8bc}} + \frac{b}{\sqrt{b^2+8ca}} + \frac{c}{\sqrt{c^2+8ab}} \geq 1$.

## Proof

By Hölder's inequality,

$$\left(\sum_{\text{cyc}} \frac{a}{\sqrt{a^2+8bc}}\right)\left(\sum_{\text{cyc}} \frac{a}{\sqrt{a^2+8bc}}\right)\left(\sum_{\text{cyc}} a(a^2+8bc)\right) \geq (a+b+c)^3.$$

Then, after simplifying, we get that we just need to prove that

$$(a+b)(b+c)(c+a) \geq 8abc.$$

AM-GM can then complete the proof. ■

### Muirhead & Schur

These inequalities are extremely useful when dealing with multi-variable homogeneous inequalities. However, we must first define a notation that we will use often when applying Muirhead.

**Definition 5**

Define

$$[(a_1, a_2, \ldots, a_n)] = \sum_{\text{sym}} x_1^{a_1} x_2^{a_2} \ldots x_n^{a_n}.$$

Now, we may state Muirhead's Inequality.

**Theorem 8** (Muirhead's Inequality)

Let $p$ and $q$ be sequences of real numbers such that $p \succ q$. Then,

$$[p] \geq [q].$$

Not all homogeneous inequalities can be solved with Muirhead, though. This is where Schur's inequality can be helpful. Schur's inequality only works on 3 variables, however.

**Theorem 9** (Schur's Inequality)

For all nonnegative $a$, $b$, $c$ and positive $r$,

$$\sum_{\text{cyc}} a^r(a-b)(a-c) \geq 0.$$

We will prove this result later in the chapter. It is worth memorizing some special cases of Schur's inequality.

**Theorem 10** (Schur's Inequality, $r = 1$)

For all nonnegative $a$,$b$,$c$,

$$a^3 + b^3 + c^3 + 3abc \geq \sum_{\text{sym}} a^2 b$$

**Theorem 11** (Schur's Inequality, $r = 2$)

For all nonnegative $a$,$b$,$c$,

$$a^4 + b^4 + c^4 + abc(a+b+c) \geq \sum_{\text{sym}} a^3 b$$

## Calculus Inequalities

Technically, no high school olympiads ever require calculus, so this chapter is optional. However, calculus can be used to solve difficult inequalities faster. The proofs of these inequalities are extremely advanced and will not be presented.

### Convexity/Concavity and Monotonicity

We must define some new terminology before introducing these inequalities.

**Definition 6** (Convexity)

Call a function $f$ **convex** if $f''(x) \geq 0$.

**Definition 7** (Concavity)

Call a function $f$ **concave** if $f''(x) \leq 0$.

**Definition 8** (Monotonicity)

Call a function $f$ **monotonically increasing** if $f'(x) > 0$ for all $x$ and **monotonically decreasing** if $f'(x) < 0$ for all $x$. A function is **monotonic** if it is either always monotonically increasing or always monotonically decreasing.

### Jensen's Inequality

This is the most powerful inequality in this chapter.

**Theorem 12** (Jensen's Inequality)

If $f$ is a convex function for real numbers $x_1, x_2, \ldots, x_n$, and $\omega_1, \omega_2, \ldots, \omega_n$ are positive weights such that $\omega_1 + \omega_2 + \cdots + \omega_n = 1$,

$$\omega_1 f(x_1) + \omega_2 f(x_2) + \cdots + \omega_n f(x_n) \geq f(\omega_1 x_1 + \omega_2 x_2 + \cdots + \omega_n x_n).$$

If $f$ is concave, then

$$\omega_1 f(x_1) + \omega_2 f(x_2) + \cdots + \omega_n f(x_n) \leq f(\omega_1 x_1 + \omega_2 x_2 + \cdots + \omega_n x_n).$$

**Example 5** (Weighted AM-GM)

Prove theorem 3.

## Solution

Let $f(x) = \ln(x)$. We can see that $f'(x) = x^{-1}$, so

$$f''(x) = -x^{-2} < 0$$

for positive $x$. Therefore, $f$ is concave, so by Jensen's inequality,

$$\omega_1 \ln(a_1) + \omega_2 \ln(a_1) + \cdots + \omega_n \ln(a_n) \leq \ln(\omega_1 a_1 + \omega_2 a_2 + \cdots + \omega_n a_n).$$

Taking $e$ to the power of both sides gives the desired inequality. ■

**Example 6** (Weighted Power Mean)

Prove theorem 4.

### Solution

We shall break this into $3$ parts.

**PART 1:** For $t > 0$, $M(t) \geq M(0) \geq M(-t)$.
This is trivial by Weighted AM-GM.

**PART 2:** For $k_1 > k_2 > 0$, $M(k_1) \geq M(k_2)$.
Let $f(x) = x^{k_2/k_1}$.

$$f''(x) = \tfrac{k_2}{k_1} \cdot \tfrac{k_2 - k_1}{k_1} x^{k_2/k_1 - 2} < 0$$

for $x > 0$, so $f(x)$ is concave. Therefore, by Jensen's Inequality,

$$(\omega_1 a_1^{k_1} + \cdots + \omega_n a_n^{k_1})^{\frac{k_2}{k_1}} \geq \omega_1 a_1^{k_2} + \cdots + \omega_n a_n^{k_2},$$

proving the inequality.

**PART 3:** $0 > k_1 > k_2$.
Let $f(x) = x^{k_2/k_1}$.

$$f''(x) = \tfrac{k_2}{k_1} \cdot \tfrac{k_2 - k_1}{k_1} x^{k_2/k_1 - 2} > 0$$

for $x > 0$, so $f(x)$ is convex. Therefore, by Jensen's Inequality,

$$(\omega_1 a_1^{k_1} + \cdots + \omega_n a_n^{k_1})^{\frac{k_2}{k_1}} \leq \omega_1 a_1^{k_2} + \cdots + \omega_n a_n^{k_2},$$

but since $k_1$ is negative, this is equivalent to what we wanted to prove, completing the proof. ■

### Karamata's Inequality

This inequality is very powerful. It can usually help in smoothing variables, a technique we will learn later in this chapter.

**Theorem 13** (Karamata)

Let $a_1, a_2, \ldots, a_n$ and $b_1, b_2, \ldots, b_n$ be real number sequences such that $(a_1, a_2, \ldots, a_n) \succ (b_1, b_2, \ldots, b_n)$, and let $f$ be a convex function.

$$f(a_1) + f(a_2) + \cdots + f(a_n) \geq f(b_1) + f(b_2) + \cdots + f(b_n).$$

If $f$ is concave, then

$$f(a_1) + f(a_2) + \cdots + f(a_n) \leq f(b_1) + f(b_2) + \cdots + f(b_n).$$

## More Techniques

### Smoothing

This is when we manipulate the variables to bring the two sides of the inequality together until we get something manageable. To use smoothing, we can prove that, after bringing two variables closer, the two sides get closer, or by using Karamata's inequality.

**Example 7** (1999 USAMO Problem 4)

Let $a_1, a_2, \ldots, a_n$ be real numbers such that

$$a_1 + a_2 + \cdots + a_n \geq n \qquad \text{and} \qquad a_1^2 + a_2^2 + \cdots + a_n^2 \geq n^2$$

and $n > 3$. Prove that $\max(a_1, a_2, \ldots, a_n) \geq 2$.

## Proof

Assume FTSOC that $a_1, a_2, \ldots, a_n < 2$. If all of the numbers are nonnegative, then

$$a_1^2 + \cdots + a_n^2 < 4n \leq n^2,$$

which gives a contradiction. Hence, at least one term is negative. Then, we can perform the following algorithm: if there are two terms $-x$ and $-y$ that are negative, we may replace them with $-(x+y)$ and $0$ and all of the conditions still hold. Using this, we can reduce the sequence to only one negative number, $-k$. Then, we may use smoothing to convert all the nonnegative terms to $2$. We know that

$$n \leq a_1 + \cdots + a_n < 2(n-1) - k,$$

so $k < n - 2$. Then, $a_1^2 + \cdots + a_n^2 < 4(n-1) + (n-2)^2 = n^2$, giving a contradiction. ■

### Ordering

**Definition 9**

Ordering is the technique where we order the variables in an inequality and use this order to solve a problem.

**Example 8**

Prove Theorem 9.

## Proof

Assume WLOG that $a \geq b \geq c$.

$$\sum_{\text{cyc}} a^r(a-b)(a-c) = c^r(c-b)(c-a) + (a-b)(a^r(a-c) - b^r(b-c)).$$

We can see that $c^r(c-b)(c-a)$, $a-b$, and $a^r(a-c) - b^r(b-c)$ are all positive, which proves the inequality. ■

### Substitution

We can often use substitution to help simplify the inequality. This can help extremely if we are given some initial condition. Some common ones include the following.

**Theorem 14**

Let $a$, $b$, $c$ be positive real numbers. If $abc = k$, then we can substitute $a = x/y\sqrt[3]{k}$, $b = y/z\sqrt[3]{k}$, $c = z/x\sqrt[3]{k}$.

## Proof

$$abc = \tfrac{x}{y}\sqrt[3]{k}\tfrac{y}{z}\sqrt[3]{k}\tfrac{z}{x}\sqrt[3]{k} = k. \blacksquare$$

**Theorem 15**

Let $a$, $b$, $c$ be positive real numbers. If $ab+bc+ca+2abc = 1$, then we can substitute $a = \frac{x}{y+z}$, $b = \frac{y}{x+z}$, and $c = \frac{z}{x+y}$.

## Proof

We can see that

$$\begin{aligned} ab + bc + ca + 2abc &= \sum_{\text{cyc}} \frac{xy}{(x+z)(y+z)} + \frac{2xyz}{(x+y)(y+z)(z+x)} \\ &= \frac{1}{(x+y)(y+z)(z+x)} \left( \sum_{\text{cyc}} xy(x+y) + 2xyz \right) \\ &= 1. \blacksquare \end{aligned}$$

**Theorem 16**

Let $a$, $b$, $c$ be positive real numbers. If $a+b+c+2=abc$, then we can substitute $a=\frac{y+z}{x}$, $b=\frac{x+z}{y}$, and $c=\frac{x+y}{z}$.

## Proof

This is exactly the same as the previous substitution, but $a$, $b$, $c$ are replaced by $1/a$, $1/b$, $1/c$. ■

### Homogenizing

If the problem gives a condition and asks to prove a non-homogeneous inequality, we can use this condition to homogenize the inequality, and then solve the inequality without the condition.

**Example 9**

Given that, for $x, y, z > 0$, $xyz = 3$, prove that $(x+y+z)\sqrt[3]{9} \leq x^3+y^3+z^3$.

## Proof

$\sqrt[3]{9} = (xyz)^{\frac{2}{3}}$, so the inequality we seek reduces to

$$(x+y+z)(xyz)^{\frac{2}{3}} \leq x^3+y^3+z^3,$$

which is true by Muirhead's inequality. ■

**Example 10** (AoPS, krithikrokcs)

Prove that for all positive real numbers $a$, $b$, $c$ such that $abc = 1$,

$$\sum_{\text{cyc}} \frac{a}{a+b+ab^3} \geq 1.$$

## Proof

We can use substitution. Let $a = x/y$, $b = y/z$, and $c = z/x$. We can see that, by substituting, simplifying, and using Titu's lemma, we can obtain

$$\begin{aligned}
\sum_{\text{cyc}} \frac{a}{a+b+ab^3} &= \sum_{\text{cyc}} \frac{x/y}{x/y + y/z + xy^2/z^3} \\
&= \sum_{\text{cyc}} \frac{x^2z^4}{x^2z^4 + xy^2z^3 + x^2y^3z} \\
&\geq \frac{(\sum_{\text{cyc}} xz^2)^2}{\sum_{\text{cyc}} x^2z^4 + 2\sum_{\text{cyc}} xy^2z^3} \\
&= 1. \blacksquare
\end{aligned}$$

### Tangent Line Trick (Calculus)

Given a function that we want to bound, we can find the line tangent to a point on the graph of this function using calculus. We usually use the point that gives the equality case of our inequality. Often, this tangent line either gives an upper bound or lower bound of the function.

**Example 11** (AoPS, krithikrokcs)

Let $a$, $b$, $m$, and $n$ be real numbers such that $m, n > 0$ and $a, b > 1$ and $a^m = b^n$. Furthermore, let $p = \sqrt{n \ln(a) \ln(b)/m}$ and $q = \sqrt{m \ln(a) \ln(b)/n}$ Prove that

$$a^p + b^q \geq ap + bq \geq 2a + 2b - 2e \geq 2e(p+q) - 2e.$$

## Proof

Let $e^k = a^m = b^n$. We can take the natural log of this to get that $k = m \ln a = n \ln b = \sqrt{mn \ln a \ln b}$, giving $p = k/m = \ln a$ and $q = k/n = \ln b$. Therefore, we need to prove that

$$a^{\ln a} + b^{\ln b} \geq a \ln a + b \ln b \geq 2a + 2b - 2e \geq 2e(\ln a + \ln b) - 2e.$$

Let $f(x) = x^{\ln x}$, $g(x) = x \ln x$, $h(x) = 2x - e$, and $j(x) = 2e \ln x - e$. Now, we need to prove that

$$f(a) + f(b) \geq g(a) + g(b) \geq h(a) + h(b) \geq j(a) + j(b).$$

Now, we shall prove that for all numbers $x > 1$ (although this is true for all positive real numbers),

$$f(x) \geq g(x) \geq h(x) \geq j(x).$$

First of all, it is easy to see that $j'(x) = 2e/x$ and $j''(x) = -2e/x^2 \leq 0$, so $j(x)$ is concave. Therefore, any tangent line to $j(x)$ will be above the graph, and since $j'(e) = 2e/e = 2$ and $j(e) = h(e)$, $h(x)$ is tangent to $j(x)$. Therefore, $h(x) \geq j(x)$.

Now, we can see that $g'(x) = \ln x + 1$ and $g''(x) = 1/x \geq 0$, so $g(x)$ is convex and its tangent line is below it. Therefore, since $g'(e) = \ln e + 1 = 2$ and $g(e) = h(e)$ $h(x)$ is tangent to $g(x)$, and $g(x) \geq h(x)$.

Because $\ln(x)$ is concave and $y = x - 1$ is tangent to it, $x - 1 \geq \ln(x)$. However, it is obvious that $x^2 - x \geq x - 1$. Therefore, $x^2 - x \geq \ln(x)$, so $x^2 \geq x + \ln x$. We can substitute $\ln(x)$ to get $(\ln x)^2 \geq \ln(x) + \ln(\ln x)$, so $e^{(\ln x)^2} \geq e^{\ln x} e^{\ln(\ln(x))}$ or $x^{\ln x} \geq x \ln x$. Therefore, $f(x) \geq g(x)$. ■

It is left as an (easy) exercise to the reader to prove that the equality case happens if and only if $a = b = e$ and $m = n = 1$, and in this case,

$$a^p + b^q = ap + bq = 2a + 2b - 2e = 2e\,(p + q) - 2e = 2e.$$

**Example 12** (2017 USAMO Problem 6)

Find the minimum possible value of the following given that $a+b+c+d = 4$ and $a, b, c, d \geq 0$:

$$\sum_{\text{cyc}} \frac{a}{b^3 + 4}$$

## Proof

After testing this function, we can see that $(2, 2, 0, 0)$ appears to yield the maximum value of $\frac{2}{3}$. To prove this, we can see that the tangent line to

$$f(x) = \frac{1}{x^3 + 4}$$

at $x = 2$ is $\dfrac{1}{4} - \dfrac{x}{12}$. We want a lower bound on this function, so we must prove that

$$\frac{1}{x^3 + 4} \geq \frac{1}{4} - \frac{x}{12}.$$

Expanding and simplifying turns this into proving that

$$x^4 - 3x^3 + 4x = x(x + 1)(x - 2)^2 \geq 0$$

for $x \geq 0$. This is obviously true. Therefore,

$$\sum_{\text{cyc}} \frac{a}{b^3 + 4} \geq \sum_{\text{cyc}} a \left( \frac{1}{4} - \frac{b}{12} \right) = \frac{1}{4}(a+b+c+d) - \frac{1}{12}(ab+bc+cd+ca) = 1 - \frac{1}{12}(a+c)(b+d) \geq \boxed{\frac{2}{3}}.$$ ■

### Monotonicity (Calculus)

To prove that $f(x) \geq f(y)$ for all $x \geq y$, we can prove that $f$ is monotonically increasing. We can similarly prove the opposite inequality is $f$ is monotonically decreasing.

**Example 13**

Prove that $e^x \geq x + 1$ for nonnegative $x$.

## Proof

Taking the $\ln$ of both sides gives $x - \ln(x+1) \geq 0$. The derivative of the LHS with respect to $x$ is negative, so this monotonically decreases. $x = 0$ gives that $1 \geq 1$, which is true, completing our proof. ■

## Extra Inequalities

Here are a few inequalities that are less important but can still be useful in some problems. You do not need to memorize all of these, but it is good to know at least some of them.

**Theorem 17** (Rearrangement Inequality)

Let $\sigma$ be a permutation of $(1, 2, \ldots, n)$, and let $x_1, x_2, \ldots, x_n$ and $y_1, y_2, \ldots, y_n$ be sequences such that

$$x_1 \geq x_2 \geq \cdots \geq x_n$$

and

$$y_1 \geq y_2 \geq \cdots \geq y_n.$$

Then,

$$x_1y_1 + x_2y_2 + \cdots + x_ny_n \geq x_1y_{\sigma(1)} + x_2y_{\sigma(2)} + \cdots + x_ny_{\sigma(n)} \geq x_1y_n + x_2y_{n-1} + \cdots + x_ny_1.$$

**Theorem 18** (Nesbitt's Inequality)

For positive $a$, $b$, $c$,

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}.$$

**Theorem 19** (Bernoulli's Inequality)

For all $r \geq 1$ and $x \geq -1$,

$$(1+x)^r \geq 1 + xr.$$

**Theorem 20** (Weighted Chebyshev's Inequality)

Let $\sigma$ be a permutation of $(1, 2, \ldots, n)$, and let $x_1, x_2, \ldots, x_n$ and $y_1, y_2, \ldots, y_n$ be sequences such that

$$x_1 \geq x_2 \geq \cdots \geq x_n$$

and

$$y_1 \geq y_2 \geq \cdots \geq y_n.$$

Furthermore, let $\omega_1, \omega_2, \ldots, \omega_n$ be positive weights such that $\omega_1 + \omega_2 + \cdots + \omega_n = 1$. Then,

$$(\omega_1 x_1 y_1 + \cdots + \omega_n x_n y_n) \geq (\omega_1 x_1 + \cdots + \omega_n x_n)(\omega_1 y_1 + \cdots + \omega_n y_n)$$

**Theorem 21** (Generalization of Schur's Inequality)

Let $f\colon \mathbb{R} \to \mathbb{R}_0^+$ be a function such that $f$ is either convex or monotonic. If $a$, $b$, $c$, $x$, $y$, and $z$ are real numbers such that $a \geq b \geq c$ and either $x \geq y \geq z$ or $z \geq y \geq x$, and if $k$ is a positive integer, then

$$f(x)(a-b)^k(a-c)^k + f(y)(b-a)^k(b-c)^k + f(z)(c-a)^k(c-b)^k \geq 0.$$

## Extra Problems

**Example 14**

Prove that $xy + yz + zx \leq x^2 + y^2 + z^2$ in as many ways as possible.

## Proof

We can use the following different methods to prove this extremely simple inequality.

- Rearrangement Inequality with $3$ variables
- Muirhead's Inequality
- Cauchy-Schwarz Inequality

- AM-GM Inequality
- Trivial Inequality

Try to prove it using all of these ways or as many as possible. All of the proofs are very straightforward.

**Example 15** (Nesbitt's Inequality)

Prove Theorem 18.

## Proof

By Titu's Lemma,

$$\sum_{\text{cyc}} \frac{a^2}{ab+ac} \geq \frac{(a+b+c)^2}{2(ab+bc+ca)} = 1 + \frac{1}{2} \cdot \frac{a^2+b^2+c^2}{ab+bc+ca} \geq \frac{3}{2}. \blacksquare$$

**Example 16**

Prove the Cauchy-Schwarz using vectors.

This is equivalent to stating that, if $\mathbf{a}$ and $\mathbf{b}$ are vectors, then

$$\|\mathbf{a}\| \cdot \|\mathbf{b}\| \geq |\mathbf{a} \cdot \mathbf{b}|.$$

This is true because $\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \cdot \|\mathbf{b}\| \cos(\theta)$, where $\theta$ is the angle between the two vectors. $\blacksquare$

**Example 17** (2012 Balkan Math Olympiad Problem 2)

Prove that

$$\sum_{\text{cyc}} (x+y)\sqrt{(z+x)(z+y)} \geq 4(xy+yz+zx),$$

for all positive real numbers $x$, $y$, and $z$.

## Proof

Let $a = \sqrt{z+y}$, and define $b$ and $c$ similarly. We can see that

$$xy+yz+zx = -\tfrac{1}{4}\left(a^4+b^4+c^4\right) + \tfrac{1}{2}\left(a^2b^2+b^2c^2+c^2a^2\right),$$

so the inequality now becomes

$$a^2bc + ab^2c + abc^2 \geq -a^4 - b^4 - c^4 + 2a^2b^2 + 2b^2c^2 + 2c^2a^2,$$

or
$$a^4+b^4+c^4+a^2bc+ab^2c+abc^2-2a^2b^2-2b^2c^2-2c^2a^2 \geq 0.$$
The polynomial on the LHS can be factored as
$$(a+b+c)(a^3+b^3+c^3+3abc-a^2b-ab^2-a^2c-ac^2-b^2c-bc^2).$$
The problem now reduces to proving that this is positive. Because $a$, $b$,, and $c$ are positive,
$$a+b+c \geq 0,$$
so we just need to prove that
$$a^3+b^3+c^3+3abc-a^2b-ab^2-a^2c-ac^2-b^2c-bc^2 \geq 0,$$
or
$$a^3+b^3+c^3+3abc \geq a^2b+ab^2+a^2c+ac^2+b^2c+bc^2.$$
However, this is just Schur's inequality (for $t=1$), which completes the proof. ■

**Example 18** (1998 IMO Shortlist A3)

Let $x$, $y$, $z$ be positive real numbers such that $xyz=1$. Prove that
$$\frac{x^3}{(1+y)(1+z)}+\frac{y^3}{(1+x)(1+z)}+\frac{z^3}{(1+x)(1+y)} \geq \frac{3}{4}.$$

## Proof

We can write the left-hand side as
$$\sum_{\text{cyc}} \frac{(x^2)^2}{x(1+y)(1+z)}.$$
Therefore, by T2's lemma, the problem is reduced to proving
$$\frac{\left(\sum_{\text{cyc}} x^2\right)^2}{\sum_{\text{cyc}} x+xy+xz+xyz} \geq \frac{3}{4}.$$
After expanding the numerator and simplifying the denominator, we get that the problem reduces to proving
$$\frac{\sum_{\text{cyc}} x^4+2\sum_{\text{cyc}} x^2y^2}{\sum_{\text{cyc}} x+2\sum_{\text{cyc}} xy+3} \geq \frac{3}{4},$$
or
$$4\sum_{\text{cyc}} x^4+8\sum_{\text{cyc}} x^2y^2 \geq 3\sum_{\text{cyc}} x+6\sum_{\text{cyc}} xy+9.$$
We can use the fact that $xyz=1$ to homogenize this and get that we must prove
$$4\sum_{\text{cyc}} x^4+8\sum_{\text{cyc}} x^2y^2 \geq 3\sum_{\text{cyc}} x^2yz+6\sum_{\text{cyc}} x^{\frac{5}{3}}y^{\frac{5}{3}}z^{\frac{2}{3}}+9x^{\frac{4}{3}}y^{\frac{4}{3}}z^{\frac{4}{3}}.$$

Using the fact that

$$x^2y^2 + y^2z^2 \geq 2xy^2z$$

by AM-GM's, cyclicly summing this, and multiplying by $\frac{3}{2}$ gives

$$3\sum_{\text{cyc}} x^2y^2 \geq 3\sum_{\text{cyc}} x^2yz.$$

Now, all we have to prove is

$$4\sum_{\text{cyc}} x^4 + 5\sum_{\text{cyc}} x^2y^2 \geq 6\sum_{\text{cyc}} x^{\frac{5}{3}}y^{\frac{5}{3}}z^{\frac{2}{3}} + 9x^{\frac{4}{3}}y^{\frac{4}{3}}x^{\frac{4}{3}}.$$

By AM-GM, we have that

$$3\sum_{\text{cyc}} x^2y^2 \geq 9x^{\frac{4}{3}}y^{\frac{4}{3}}z^{\frac{4}{3}},$$

meaning it is sufficient to prove

$$4\sum_{\text{cyc}} x^4 + 2\sum_{\text{cyc}} x^2y^2 \geq 6\sum_{\text{cyc}} x^{\frac{5}{3}}y^{\frac{5}{3}}z^{\frac{2}{3}}.$$

Because $(2,2,0) \succ \left(\frac{5}{3}, \frac{5}{3}, \frac{2}{3}\right)$, by Muirhead,

$$2\sum_{\text{cyc}} x^2y^2 \geq 2\sum_{\text{cyc}} x^{\frac{5}{3}}y^{\frac{5}{3}}z^{\frac{2}{3}},$$

meaning we only need to prove that

$$4\sum_{\text{cyc}} x^4 \geq 4\sum_{\text{cyc}} x^{\frac{5}{3}}y^{\frac{5}{3}}z^{\frac{2}{3}}$$

or

$$\sum_{\text{cyc}} x^4 \geq \sum_{\text{cyc}} x^{\frac{5}{3}}y^{\frac{5}{3}}z^{\frac{2}{3}}.$$

However, because $(4,0,0) \succ \left(\frac{5}{3}, \frac{5}{3}, \frac{2}{3}\right)$, this is simply true by Muirhead's inequality. ■

**Example 19** (2018 USAJMO Problem 2)

Let $a$, $b$, $c$ be positive real numbers such that $a+b+c = 4\sqrt[3]{abc}$. Prove that

$$2(ab+bc+ca) + 4\min(a^2, b^2, c^2) \geq a^2 + b^2 + c^2.$$

## Proof

We can use ordering. Without loss of generality, let $a \geq b \geq c$. Our inequality to prove becomes

$$2(ab+bc+ca) + 4c^2 \geq a^2 + b^2 + c^2.$$

Because both the equation we are given and the inequality we must prove are homogeneous, we can let $c = 1$. Now, we are trying to prove that

$$2ab + 2a + 2b + 4 \geq a^2 + b^2 + 1,$$

and the given equation becomes

$$a+b=4\sqrt[3]{ab}-1$$

Now, let $m=ab$ and $n=a+b$. We are now trying to prove that

$$2m+2n+4\geq n^2-2m+1,$$

which can be converted to

$$4m+4\geq n^2-2n+1=(n-1)^2.$$

The equation given in the problem is now

$$n=4\sqrt[3]{m}-1.$$

Substituting this gives that we must prove

$$4m+4\geq(4m^{\frac{1}{3}}-2)^2=16m^{\frac{2}{3}}-16m^{\frac{1}{3}}+4,$$

and after simplifying, we get that this is equivalent to proving

$$m\geq 4m^{\frac{2}{3}}-4m^{\frac{1}{3}},$$

or

$$m^{\frac{1}{3}}\left(m^{\frac{2}{3}}-4m^{\frac{1}{3}}+4\right)\geq 0.$$

However, after dividing by $m^{\frac{1}{3}}$ and simplifying, we can see that we must prove

$$\left(m^{\frac{1}{3}}-2\right)^2\geq 0,$$

which is true by the Trivial Inequality. ■

## Resources

- A Brief Introduction to Olympiad Inequalities by Evan Chen
- Mildorf Inequalities

**Try the problems**

**https://mathdash.com/contest/inequalities-p1**
**https://mathdash.com/contest/inequalities-p2**
**https://mathdash.com/contest/inequalities-p3**
**https://mathdash.com/contest/inequalities-p4**
**https://mathdash.com/contest/inequalities-p5**
**https://mathdash.com/contest/optimization-contest**

https://mathdash.com/maps/map/am-gm

# Monovariants and Invariants

*by Krithik, testsolved by Math645*

**Definition 1**

An **invariant** is a value that does not change after doing a certain process or applying a certain function. For example, assume we are playing a game where we can replace the numbers $x$ and $y$ with $x + y$. In this game, the sum of the numbers never changes, so it is a invariant.

**Definition 2**

A **monovariant** is something that either always increases or always decreases. For example, if we replaced $x$ and $y$ with $2\sqrt{xy}$, the sum would always decrease by the AM-GM inequality, so this would be a monovariant.

**Example 1** (2024 AMC 10B Problem 16)

Jerry likes to play with numbers. One day, he wrote all the integers from $1$ to $2024$ on the whiteboard. Then he repeatedly chose four numbers on the whiteboard, erased them, and replaced them by either their sum or their product. (For example, Jerry's first step might have been to erase $1$, $2$, $3$, and $5$, and then write either $11$, their sum, or $30$, their product, on the whiteboard.) After repeatedly performing this operation, Jerry noticed that all the remaining numbers on the whiteboard were odd. What is the maximum possible number of integers on the whiteboard at that time?

**(A)** $1010$ **(B)** $1011$ **(C)** $1012$ **(D)** $1013$ **(E)** $1014$

**Solution:** Because the problem mentions finding the maximum possible value of something after some process is performed, monovariants might work. After observing this operation be done on some integers, we can get that the number of odd numbers always either stays constant or decreases, making it a monovariant. Therefore, there can be at most 1012 odd numbers. However, we can see that the number of numbers modulo 3 is an invariant, so the final answer must be $2$ mod $3$, giving $\boxed{\textbf{(A) } 1010}$.

**Example 2** (NIMO Winter 2014/3)

The numbers $1, 2, \ldots, 10$ are written on a board. Every minute, one can select three numbers $a, b, c$ on the board, erase them, and write $\sqrt{a^2 + b^2 + c^2}$ in their place. This process continues until no more numbers can be erased. What is the largest possible number that can remain on the board at this point?

**Solution:** The moment we see this problem, we can clearly see that invariants or monovariants can be used. Furthermore, the square roots prompt us that there must be something squared in the invariant. Looking closer, we can see that the sum of the squares of the terms is an invariant because

$$a^2 + b^2 + c^2 = (\sqrt{a^2 + b^2 + c^2})^2.$$

Therefore, the sum of squares must always be

$$1^2 + 2^2 + \cdots + 10^2 = 385.$$

However, we can also see that the number of numbers modulo 2 is also invariant, so there must be $2$ numbers when this process ends. To maximize the larger number, we must minimize the smaller one. This can be done by making the smaller one $1$, so the largest number remaining is $\sqrt{384} = \boxed{8\sqrt{6}}$.

Many monovariant problems require some basic inequalities, so we will present some of the common, simple ones that often appear in monovariant problems.

**Theorem 1** (AM-GM)

For any positive integer $n$ and positive real numbers $a_1, a_2, \ldots, a_n$,

$$a_1 + a_2 + \cdots + a_n \geq n\sqrt[n]{x_1 x_2 \ldots x_n}$$

**Theorem 2** (Power-Mean Inequality)

For any positive integer $n$, positive real numbers $a_1, a_2, \ldots, a_n$, and real numbers $k_1 > k_2$,

$$\left(\frac{1}{n}\sum_{i=1}^{n} a_i^{k_1}\right)^{\frac{1}{k_1}} \geq \left(\frac{1}{n}\sum_{i=1}^{n} a_i^{k_2}\right)^{\frac{1}{k_2}}.$$

**Theorem 3** (Cauchy-Schwarz Inequality)

For any positive integer $n$ and positive real numbers $a_1, a_2, \ldots, a_n$ and $b_1, b_2, \ldots, b_n$

$$(a_1^2 + a_2^2 + \cdots + a_n^2)(b_1^2 + b_2^2 + \cdots + b_n^2) \geq (a_1 b_1 + a_2 b_2 + \cdots + a_n b_n)^2.$$

**Example 3**

The number $1$ is written $n$ times on a whiteboard, and on each turn, Krithik erases $x$ and $y$ and replaces them with $4x+4y$. Prove that the number at the end of this process is at least $n^3$.

**Solution:** Let $x$ be the last number and $S$ be the sum of the cube roots of the numbers. We can see that $(\frac{1}{2}(\sqrt[3]{x}+\sqrt[3]{y}))^3 \leq \frac{1}{2}(x+y)$ by the Power Mean inequality, which is equivalent to $\sqrt[3]{x}+\sqrt[3]{y} \leq \sqrt[3]{4x+4y}$. Therefore, $S$ is always non-decreasing, so $n \leq \sqrt[3]{x}$, and we are done. ■

Monovariants are also commonly used to prove that some process terminates. This works when there are a finite number of possible values of some quantity, and this quantity always increases or decreases.

**Example 4**

There are some rooms in a building with some people in them. Every hour, one person leaves one room and enters another room with more people. Prove that at some point, this process must terminate.

**Solution:** Consider the sum of the squares of the number of people in each room. We can see that this is always increasing, but it has a finite number of possible values, so this process must end. ■

**Try some Monovariants/Invariants Practice Contests**

Invariants Problem 1 (1500)

Invariants Problem 2 (1500)

Invariants Problem 3 (2000)

Invariants Problem 4 (1900)

Invariants Problem 5 (2000)

Invariants Problem 6 (2000)

Invariants Problem 7 (2200)

Invariants Problem 8 (2500)

# Quadratic Residues

*by MathDash Team*

https://mathdash.com/maps/map/quadratic-residues

**Definition 1**

We define $a$ to be a quadratic residue in mod $m$ if there exists an integer $x$ such that $x^2 \equiv a \pmod{m}$

**Theorem 1**

For every odd prime $p$, exactly half of $\{1, 2, \cdots, p-1\}$ are quadratic residues and exactly half are quadratic non-residues.

**Solution:** Consider all the nonzero residues modulo $p$, namely $1$, $2$, $\ldots$, $p-1$, and square them. If two of the squares are equal, we have $x_1^2 \equiv x_2^2 \pmod{p}$, or $x_1^2 - x_2^2 = (x_1 + x_2)(x_1 - x_2) \equiv 0 \pmod{p}$. This means $(x_1 + x_2)(x_1 - x_2)$ is divisible by $p$, so one of the terms has to be divisible by $p$! So, $x_1$ is either $x_2$ or $-x_2$ modulo $p$.

What does this mean? Well, in our set of squares $1^2$, $2^2$, $\ldots$, $(p-1)^2$, this result means that two of them are congruent modulo $p$ if and only if they are the same square or they are in a $x^2, (p-x)^2$ pair. Thus our set of squares must consist of $\frac{p-1}{2}$ distinct values, each occuring twice.

So, these $\frac{p-1}{2}$ distinct values are quadratic residues, and the other $\frac{p-1}{2}$ nonzero values are quadratic nonresidues!

Now let's describe some notation that will come in handy as we work a lot with quadratic residues and nonresidues

**Definition 2** (Legrende)

If $p$ is an odd prime, $a > 0$, and $\gcd(a, p) = 1$, the Legrende symbol $\left(\frac{a}{p}\right)$ is defined as

$$\left(\frac{a}{p}\right) = \begin{cases} 1, & \text{if a is a quadratic residue mod p,} \\ -1, & \text{if a is a quadratic nonresidue mod p,} \\ 0, & \text{if a is divisible by p.} \end{cases}$$

What an arbitrary definition right? Why would we define this.. Well, this symbol has some cool properties.

**Theorem 2** (Euler's Criterion)

For $p$ an odd prime, $\left(\frac{a}{p}\right) \equiv a^{\frac{p-1}{2}} \pmod p$

**Proof:** Well, we know that Fermat's Little Theorem says that $\left(a^{\frac{p-1}{2}}\right)^2 \equiv a^{p-1} \equiv 1 \pmod p$. So, $a^{\frac{p-1}{2}}$ is either $1$ or $-1$ modulo $p$ (because if $x^2 \equiv 1$, then $(x+1)(x-1) \equiv 0$ meaning $x$ is $1$ or $-1$ modulo $p$).

Which one is it? Well, suppose $a$ is a quadratic residue. Then, $a \equiv k^2 \pmod p$, so $a^{\frac{p-1}{2}} \equiv k^{p-1} \equiv 1$. So, $a$ being a quadratic residue means that $a^{\frac{p-1}{2}} \equiv 1$ as desired!

Now, what if $a$ is not a quadratic residue? This is a lot trickier to prove — I'm going to present two proofs to this — one with some heavy machinery!

**Theorem 3** (Lagrange's Theorem)

If $p$ is a prime, and $f(x)$ is a polynomial of degree $d$ with not all coefficients divisible by $p$, then $f(x) \equiv 0 \pmod p$ has at most $d$ integer solutions in $\{0, 1, 2, \cdots, p-1\}$

A quick sketch of the proof of Lagrange's Theorem is to repeatedly use the division algorithm in modulo $p$ to divide out $(x-r)$ for each solution $r$ to this.

In any case, we can use Lagrange's Theorem here because $a^{\frac{p-1}{2}} - 1 \equiv 0$ already has $\frac{p-1}{2}$ roots, namely all of the quadratic residues, so it must not have any other solution! So, $a^{\frac{p-1}{2}}$ must not be $1$ for $a$ any quadratic nonresidue, meaning it is $-1$. This proves the result!

Here's another proof: Consider the set of residues $\{1, 2, \cdots, p-1\}$. Let us pair each residue $r$ with $a \cdot r^{-1}$. Notice since $a$ is not a quadratic residue, each residue is paired with a different nonzero residue. Additionally, the product of each pair of residues is $a$, so the total product is $a^{(p-1)/2}$. However, the product is also $(p-1)! \pmod p$. So, we just need to show now that $(p-1)! \equiv -1 \pmod p$, and this is exactly the statement of Wilson's Theorem.

Now that we've shown $a^{\frac{p-1}{2}} \equiv 1 \pmod p$ for quadratic residues $a$, and $-1 \pmod p$ for quadratic nonresidues $a$, we have proved Euler's Criterion.

One nice property of the Legrende symbol follows immediately from Euler's Criterion

**Theorem 4**

For $p$ an odd prime, and $\gcd(a, p) = \gcd(b, p) = 1$,

$$\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right)\left(\frac{b}{p}\right)$$

**Proof:** This is equivalent to $(ab)^{\frac{(p-1)}{2}} \equiv a^{\frac{(p-1)}{2}} b^{\frac{(p-1)}{2}} \pmod p$ — using Euler's Criterion makes this theorem immediate!

**Theorem 5** (The Law of Quadratic Reciprocity)

If p,q are odd primes, then

$$\left(\frac{q}{p}\right)\left(\frac{p}{q}\right) = (-1)^{\frac{p-1}{2}\cdot\frac{q-1}{2}}$$

**Theorem 6**

If $p$ is an odd prime, then

$$\left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}}$$

The proof of these statements is not simple — perhaps we will add a proof into here in the future — but for now, let's begin using these to solve a lot of number theory questions!

**Example 1**

Show that for every prime number $p$ there exists integers $a$, $b$ such that $a^2 + b^2 + 1$ is divisible by $p$

**Solution:** Notice the set of possible values of $a^2$ has size $\frac{p+1}{2}$ in modulo $p$ (namely, the $\frac{p-1}{2}$ quadratic residues, and $0$). Also, the set of possible values of $-b^2 - 1$ has size $\frac{p+1}{2}$ in modulo $p$. Since there are only $p$ residues modulo $p$, these two sets must share a residue in common, which means there exists a residue $r$ and $a$, $b$ such that $r \equiv a^2 \equiv -b^2 - 1 \pmod{p}$ so $a^2 + b^2 + 1 \equiv 0 \pmod{p}$.

All we needed was an understanding of the number of possible values $a^2$ and $b^2$ could be to do this one! Let's do another.

**Example 2**

Find all integers $x$ such that $x^2 - 31x + 34$ is divisible by $37$.

**Solution:** Notice $x^2 - 31x + 34 \equiv x^2 + 6x - 3 \equiv (x+3)^2 - 12 \pmod{37}$. We can check that $12$ is a quadratic residue modulo $37$ (namely, $7^2 \equiv 12 \pmod{37}$). Thus we see $x + 3 \equiv \pm 7 \pmod{37}$ and $x \equiv 4, 27 \pmod{37}$.

**Example 3**

Does there exist a positive integer $x$ such that $x^2$ leaves a remainder of $74$ when divided by $131$?

**Solution:** The Law of Quadratic Reciprocity is going to come in handy here. We compute as follows:

$$\begin{aligned}\left(\frac{74}{131}\right) &= \left(\frac{2}{131}\right)\left(\frac{37}{131}\right)\\ &= (-1)\cdot(-1)^{18\cdot 65}\left(\frac{131}{37}\right)\\ &= (-1)\left(\frac{20}{37}\right)\\ &= (-1)\left(\frac{5}{37}\right)\\ &= (-1)(-1)^{18\cdot 2}\left(\frac{37}{5}\right)\\ &= (-1)\left(\frac{2}{5}\right)\\ &= (-1)(-1)\\ &= 1\end{aligned}$$

So, indeed there does exist a positive integer $x$ that satisfies this.

**Example 4**

Find the sum of all primes $5 \leq p \leq 30$ for which there exist an $x$ such that $x^2+3$ is divisible by $p$.

**Solution:** We want to find all primes where $-3$ is a quadratic residue. Let's use Quadratic Reciprocity again! We want $\left(\frac{-3}{p}\right) = 1$. Notice $\left(\frac{-3}{p}\right) = \left(\frac{-1}{p}\right)\left(\frac{3}{p}\right) = (-1)^{(p-1)/2}\cdot(-1)^{(p-1)/2}\cdot\left(\frac{p}{3}\right) = \left(\frac{p}{3}\right)$. Furthermore, $\left(\frac{p}{3}\right)$ is $1$ only if $p$ is $1$ modulo $3$ (we know this from checking the two possible residues modulo $3$ to see if they are quadratic residues). Thus, only primes that are $1 \pmod 3$ have $-3$ as a quadratic residue. This gives us an answer of $7+13+19 = \boxed{39}$.

**Example 5**

If $p = 2^n+1$, $n \geq 2$, $p$ is prime, then find the remainder when $3^{(p-1)/2}$ is divided by $p$.

**Solution:** Notice since $p$ is not divisible by $3$, $n$ must be even. Let $n = 2k$. Also notice $p \equiv 1 \pmod 4$ and $p \equiv 4^k+1 \equiv 2 \pmod 3$ so $\left(\frac{p}{3}\right) = -1$. Also, $\left(\frac{3}{p}\right)\left(\frac{p}{3}\right) = (-1)^{(p-1)/2} = 1$. Therefore

$\left(\frac{3}{p}\right) = -1$ so $3^{(p-1)/2} + 1 \equiv 0 \pmod{p}$, and our answer is $\boxed{p-1}$.

**Try some Quadratic Residue Problems**

https://mathdash.com/maps/map/quadratic-residues